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***Electrochemical Monitoring for Molten Salt Reactors:
Status Review***

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Prepared in response to Task 1a in User Need Request NRR-2022-002 by:

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April 2025

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Executive Summary

This report evaluates the potential application of electrochemical methods for monitoring the reduction-oxidation (RedOx) potential of molten salts in molten salt reactors. Electrochemical methods indirectly monitor materials performance in the reactor, since corrosion in molten salts is largely driven by the RedOx condition of the salt. , This report will aid the NRC in its understanding of the capabilities of electrochemical techniques.

Electrochemical methods have been widely used for understanding the thermodynamics and kinetics of molten salts, as well as the chemistry and RedOx potentials of a molten salt system. Electrochemical methods are one such approach for measuring salt chemistry; inductively coupled argon-plasma mass spectroscopy (ICP-MS) is another technique. Two notable electrochemical methods are potentiometry and voltammetry. Potentiometry is based on the measurement of the open circuit potential (OCP) between a sensor electrode (SE) and a reference electrode (RE). For accurate RedOx potential measurements using potentiometry, the potential of the SE is ideally a function of the RedOx potential at a specific moment (determined by salt chemistry) and the RE is independent of the composition changes during the OCP measurement. In voltammetry, typically the current is measured when a voltage is applied to a working electrode (WE) of the electrochemical cell. Note that, in voltammetry, there are typically three multiple electrodes: WE, RE, and counter electrode (CE). There are different voltammetry techniques depending on how the voltage is controlled during measurements, the most common being cyclic voltammetry (CV).

Some electrochemical technologies can be used for multiple purposes. For example, depending on the experimental setup and design, potentiometry can be used for impurity and RedOx potential measurements, as well as to measure specific constituents in the salt being studied. Voltammetry with various potential ramping modes is also a capable analytical technique that can be used for different purposes, such as thermodynamic properties, diffusion coefficients, electrochemical mechanisms, etc. Note that the general principles and theory behind these electrochemical methods are well established, but for each molten salt measurement application, the requirements, accuracy, and sensitivity may be different.

RedOx potential monitoring and control were demonstrated in the Molten Salt Reactor Experiment (MSRE), which used a fluoride salt fuel system. Much operational data and experience were gained from the MSRE. One such observation was for RedOx potential in fluoride salts, the disproportionation of UF_3 to UF_4 and U metal at higher temperatures is a potential issue, meaning that a higher UF_4/UF_3 ratio is needed. For a chloride salt MSR system, however, the available data and best methodology for RedOx potential monitoring and control is very limited. UCl_3 is expected to be more stable than UF_3 , and UCl_4 is known to be very corrosive. Consequently, the presence of UCl_4 should be minimized whenever possible.

It is noted that “RedOx potential” is a widely used and important concept in MSR salt chemistry, though the term “RedOx potential” is not well-defined. “RedOx potential” of molten salt is generally used to describe the salt chemistry and the corrosivity of molten salt to structural alloys. It is noted that this report does not make any conclusions about the conformance of any specific electrochemical technology or design with regulations that would apply to MSRs. The RedOx potential is also a measurable term and can be defined as the potential difference between a working electrode and standard reference electrode. Typically, the more positive the RedOx potential, the more corrosive the salt chemistry to the structural

alloys. In molten salt chemistry, the term can also be applied to the fluorine chemical potential in a fluoride salt, or to the chlorine chemical potential in a chloride salt.

This assessment found that REs for fluorides that typically use graphite and boron nitride containers for reference salts have not been demonstrated for long-term operation. The REs with boron nitride containers were found to be generally stable for up to 12 days. Work is ongoing to develop a reliable RE for long-term operation in fluoride and chloride salts. .

In-line or in-process measurements would provide real time monitoring of salt chemistry. Voltammetry and potentiometry may offer convenient in-line or in-process measurement capabilities. Like the RedOx potential measurements, concentrations of corrosion products, oxide, and moisture impurities can also be monitored using in-line electrochemical methods, and these measurements could be designed and set up in an automated manner. The in-line measurement can provide information that may not be achieved by discrete sampling methods. However, assessment of the performance of cables for electrochemical methods under high radiation environments is not fully understood at this time.

In even the simplest real-world applications, the chemical conditions that develop within a molten salt system can be complex. The salt system is never completely isolated. An inventory of salt is constantly circulated through the reactor systems. Some MSR applications subject the salt to multiple extreme conditions that occur at the same time, including high temperatures, contact with structural materials, contact with gas phases, exposure to different sources of contamination, temperature gradients, pressure gradients, high-velocity flows, entrainment of solids, exposure to radiation, composition changes due to fission processes, and exposure to tritium (even when it is produced at a different location in the reactor).

In salt-fueled MSRs, many of the fission products remain dissolved in the molten salts, at least up to their solubility limits. However, noble metal fission products that accumulate in the molten salt may ultimately deposit on the materials of construction, while noble gases are recovered in the cover gas system via gas sparging. The fission process is mildly oxidizing toward the dissolved fissile elements (e.g., U[IV]) in the fuel salt, and gradually makes the fuel salt more oxidizing if no measures are in place to prevent this from occurring. The salt composition and RedOx potential of the fuel salt is therefore constantly changing, though slowly. This accelerates structural material corrosion by the fuel salt at high temperatures. The introduction of oxides and moisture can also alter the RedOx potential and possibly cause other issues. Salt chemistry in an MSR is critical for reliable operation and minimization of structural material corrosion by fuel salt and should therefore be maintained within acceptable ranges.

The last two decades have seen significant advancements in molten salt electrochemistry. Electrochemical methods are moving toward enabling quantitative measurement of select impurities and RedOx potentials, with potentiometry and linear sweep voltammetry (LSV) having been demonstrated even as far back as the MSRE. In-line salt chemistry monitoring is likely to be seen in the development and deployment of test or commercial MSRs, and electrochemical methods seem well suited for this purpose.

This report provides an overview of RedOx potentials of molten salts relevant to MSRs and the status of multiple electrochemical methods for potential MSR applications. It also provides an overview of electrochemical measurement technologies for molten salts. Electrochemical sensor technologies for molten salts seek to observe at least one of the various signatures related to the chemical conditions of the molten salts. Generally, electrochemical sensors are intended to determine whether the chemical conditions of the molten salt are within acceptable limits of operation. The chemical conditions of concern may be related to corrosion, or perhaps to aspects such as core neutronics, fissile inventory, and phase behaviors, etc. It is important to understand what signature the sensor is designed to observe, and what that signature says about the condition of the salt.

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ACRONYMS

A	ampere
ACU	Abilene Christian University
ARE	Aircraft Reactor Experiment
°C	degrees centigrade
CE	counter electrode
CV	cyclic voltammetry
d	day
DNPV	differential normal pulse voltammetry
DOE	Department of Energy
DPV	differential pulse voltammetry
DRE	dynamic reference electrode
e ⁻	electron
E	electrical potential
EMF	electromotive force
F	Faraday constant
FHR	fluoride salt-cooled high-temperature reactor
FLiBe	LiF-BeF ₂
FLiNaK	LiF-NaF-KF
g	gram
G	Gibbs free energy
HALEU	high-assay low-enriched uranium
HTGR	high-temperature gas-cooled reactor
ICP-MS	inductively coupled argon-plasma mass spectroscopy
IUPAC	International Union of Pure and Applied Chemistry
kg	kilogram
kmol	kilomole
K	degrees Kelvin
ln	natural logarithm
log	logarithm base-10
L	liter

LFTR	liquid fluoride thorium reactor
LSV	linear sweep voltammetry
LWR	light-water reactor
mA	milliampere
MCFR	molten chloride-salt fast reactor
min	minute
mm	millimeter
mol	mole
mol%	mole percent
MSBR	molten salt breeder reactor
MSR	molten salt reactor
MSRE	Molten Salt Reactor Experiment
MT	metric ton
mV	millivolt
MW	mega watt
MWe	megawatt electrical
MWt	megawatt thermal
NPV	normal pulse voltammetry
NRC	Nuclear Regulatory Commission
OCP	open circuit potential
ORNL	Oak Ridge National Laboratory
P	pressure or partial pressure
p	chemical potential
ppm	part per million
PV	pulse voltammetry
PWR	pressurized water reactor
QRE	quasi-reference electrode
R	gas constant
RE	reference electrode
RedOx	reduction-oxidation
s	second
SCE	standard chlorine electrode
SE	sense electrode

SFBR	sodium fast breeder reactor
SHE	standard hydrogen electrode
SWV	square wave voltammetry
$T_{1/2}$	half-life
T	temperature
TRISO	tristructural isotropic
UCB	University of California Berkeley
V	volt
wt%	weight percent
WE	working electrode
yr	year
Z	atomic number

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Electrochemical Monitoring for Molten Salt Reactors: Status Review

1. INTRODUCTION

Molten salt reactors (MSRs), as a family of advanced Generation IV-type reactors, have received a tremendous amount of interest in recent years and MSRs are currently being developed around the world [1, 2, 3, 4]. A salt-fueled MSR uses molten salt as a liquid fuel in the primary heat exchange system and a non-fueled coolant salt in the secondary heat exchange system, as shown schematically in Figure 1. A salt-cooled MSR uses solid fuels and molten salt coolants in both the primary and secondary heat exchange systems. The fluoride salt-cooled high-temperature reactor (FHR) is an example of the latter. Both salt-fueled and salt-cooled MSRs share similarities in inherent reactor safety features since both concepts use molten salt, but there are also significant differences. The U.S. Nuclear Regulatory Commission (NRC) and Department of Energy (DOE) include both salt-fueled and salt-cooled MSRs within the term MSRs.

The MSR concept started with the Aircraft Reactor Experiment (ARE) program in 1954. The ARE used NaF-ZrF₄-UF₄ as fuel salt, and the peak temperature of the salt reached 860°C. Though the duration of the ARE's reactor operation was quite short, it demonstrated the feasibility of the molten salt fuel based on its chemical stabilities, good heat transfer properties, and the high solubility of uranium. It also provided important information for the development of the Molten Salt Reactor Experiment (MSRE) which operated from 1965 to 1969. In the MSRE, the fuel salt composition at the early stage was LiF-BeF₂-ZrF₄-UF₄ (61-29.1-5.0-0.9 mol%), with the uranium consisting of 33% ²³⁵U and 67% ²³⁸U. The coolant salt was LiF-BeF₂. At a later stage, the fissile material was changed to ²³³U. The moderator was graphite and the cover gas for the molten salt was helium [5, 6].

The unique and most important feature of a salt-fueled-MSR compared to traditional light-water reactors (LWRs) is its inherent safety. The molten salts selected for MSRs feature good chemical stability at high-temperature, low volatility, a low melting point and a high boiling point, and the reactor operates at near atmospheric pressure. The negative reactivity coefficient is a significant safety feature for MSRs. When the molten salt temperature increases, the volume increases and fissile materials density decreases, which slows down the chain reaction of the fission process. Unlike solid fuels that contain all the fission products, most of the fission products in MSRs are dissolved in molten salts. The UF₄ salt constituent in the MSRE example was dissolved in the LiF-BeF₂-ZrF₄ carrier salt. The metal fission products accumulate in the salt or deposit on the reactor's construction materials, while the noble gas fission products are recovered in the cover gas by gas sparging. For varying the quantity of fuel salt in the reactor, salt can be added to the reactor vessel and salt can be drained from the vessel.

Based on the MSRE, the average total valence of the fission products of U⁴⁺ is slightly lower than four, leading to a net result of some free fluorine gas being formed during operation. A simplified fission process can be expressed as $UF_4 + n \rightarrow Kr(g) + LnF_3 + 1/2F_2(g)$ [7]. The fluorine gas is very corrosive at high temperatures and may react with UF₃ and form UF₄ (according to $1/2F_2(g) + UF_3 \rightarrow UF_4$) or react with the structural alloys of the reactor vessel. The fission process will gradually make the fuel salt increasingly oxidizing, slowly changing the composition and corrosivity of the fuel salt. To maintain the salt in a non-corrosive state, beryllium metal (encased in a perforated capsule of nickel) was introduced through the sampling system and suspended in the flowing salt stream in the MSRE pump bowl. The reaction with UF₄ by $Be + 2UF_4 = BeF_2 + 2UF_3$ produced UF₃ at a rate of about 600 g per 8-hour treatment.

With the above background, all fuel salts at high temperature can cause corrosion; therefore, corrosion of structural materials is a concern in salt-fueled and salt-cooled MSR. The material corrosion is strongly dependent on the salt chemistry. All corrosion processes are accelerated by the increased corrosivity of salt, including general corrosion, impurity-driven corrosion, dissimilar metal-driven and temperature gradient-driven corrosion, and environmentally assisted cracking from fission product exposure. To ensure years of safe operation, the salt chemistry needs to be maintained so corrosivity stays within an acceptable range.

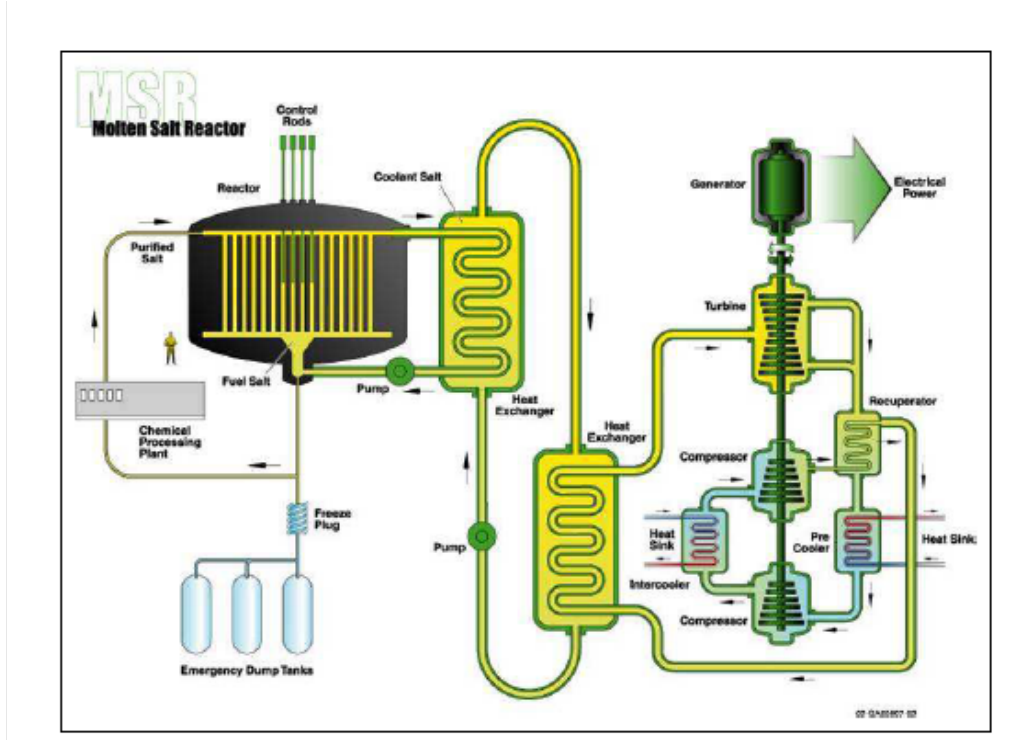


Figure 1. A typical molten salt-fueled MSR concept [8].

The concept of a salt-cooled MSR that uses solid nuclear fuel and liquid salt as its coolant is shown in Figure 2 using the fluoride salt-cooled high-temperature reactor (FHR) as an example. The primary coolant salt is LiF-BeF₂ (66-34 mol%) (FLiBe) with enriched ⁷Li to minimize both the loss of fission neutrons to the salt and the production of tritium. LiF-NaF-KF (46.5-11.5-42 mol%) (FLiNaK) or KF-ZrF₄ (58-42 mol%) was selected as a candidate for the intermediate loop. These reactor concepts will use tristructural isotropic (TRISO) fuel particles (similar to those developed for the high-temperature gas-cooled reactor, [HTGR]) immersed in the primary coolant salt.

Generally, a purified fuel salt is not very corrosive to the structural materials. The FLiBe coolant salt loop during the MSRE operated for over 24,000 hours without any significant corrosion of structures in the loop, suggesting low corrosion rates of the purified salts in the MSRE intermediate loop. The main salt chemistry issue arises from the primary coolant salt system. The FHR primary coolant salt chemistry is simplified in comparison to the MSRE fuel salt chemistry, which contained actinides and fission products, but it can continuously pick up impurities during operations. Neutron irradiation of the ${}^6\text{Li}$ creates tritium and thus converts LiF into ${}^3\text{HF}$, which is corrosive. Refueling operations with new solid fuel could add impurities to the salt. While new solid fuel will be cleaned, the heat-up and cooldown of the fuel over time may allow impurities in the graphite moderator in the reactor core to diffuse out. Therefore, salt chemistry monitoring and control are also needed to ensure corrosion rates will be acceptably low for FHR salt coolants. A detailed review on the salt corrosion and chemistry control for FHRs can be found in the literature [9].

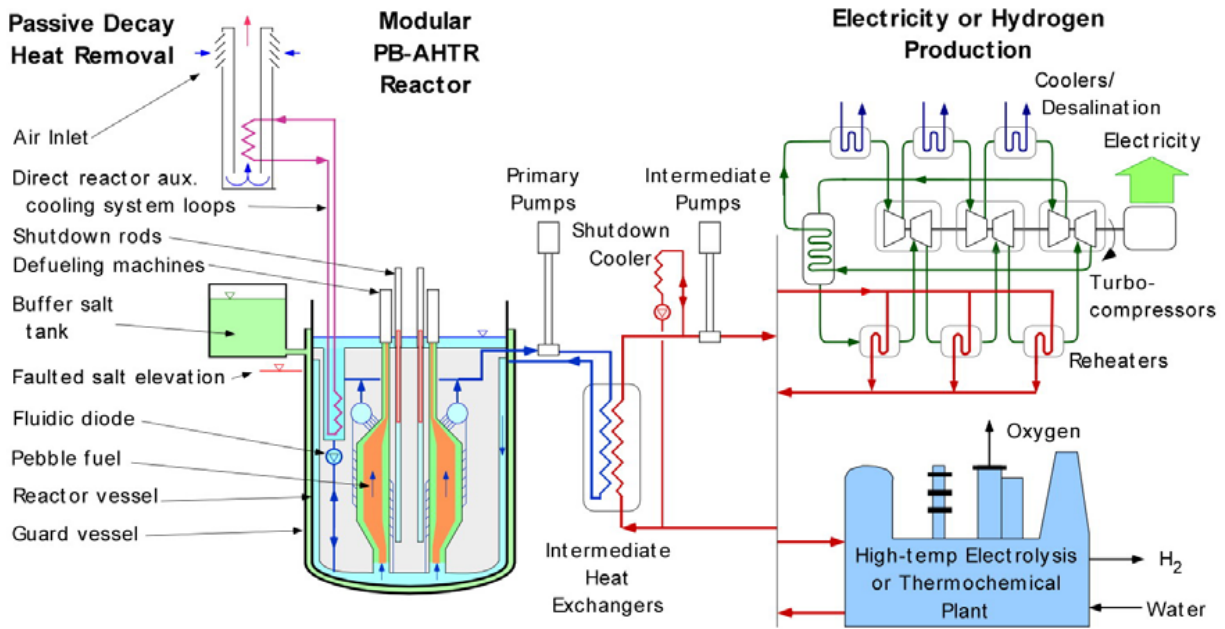


Figure 2. A University of California Berkeley (UCB) FHR reactor system [9].

Extensive research and development activities of different MSR concepts have been carried out around the world. Table 1 lists the typical MSR concepts in publicly available literature or reports. All the MSRs for U.S. commercial deployment will need to be approved and licensed by the NRC. Of the reactor concepts in Table 1, thus far, the solid-fueled fluoride-salt-cooled high-temperature reactor concept has received a construction permit to build the reactor, named the Hermes Demonstration Reactor, in Tennessee [10]. The molten salt research reactor of Abilene Christian University (ACU) is a graphite moderated, fluoride salt-fueled MSR with up to 1 MW (thermal) has also received a construction permit from the NRC [11].

Table 1. Example MSR concepts proposed for development [12].

Company Name	Main Features of MSR	Source
FLiBe Energy	Liquid Fluoride Thorium Reactor (LFTR)– Thorium fuel cycle. Graphite moderated, LiF-BeF ₂ -UF ₄ fuel.	http://flibe-energy.com
Kairos Power	Solid-fueled fluoride-salt-cooled concept with demonstration.	http://kairospower.com
Moltex Energy	NaCl-(Actinide)Cl ₃ fuel salt (<15% ²³⁵ U). NaF-ZrF ₄ coolant salt. Natural circulation of fuel salt within vented fuel pins.	http://www.moltexenergy.com
TerraPower	Molten Chloride-salt Fast Reactor (MCFR) uses chloride-based fuel salt.	http://terrapower.com
Terrestrial Energy USA	Integral MSR concept. Denatured U/Pu fuel cycle and LEU fluoride salt fuel containing constituents such as LiF, NaF, BeF ₂ .	https://aris.iaea.org/pdf/IMS_R400.dpf
Thorcon Power	Fuel salt flowing in the primary loop is a mixture of sodium, beryllium, and uranium fluorides.	http://thorconpower.com
Molten Salt Research Reactor at Abilene Christian University (ACU)	High-assay low-enriched uranium (HALEU) dissolved in FLiBe salt.	https://acu.edu/next

There are multiple proposed MSR designs for utilizing solid fuel and for utilizing liquid fuel. Regardless of the MSR concept, salt chemistry in the primary systems need to be well controlled and monitored with accurate knowledge of fuel salt chemistry to ensure MSR safeguards. Analyzing the salt chemistry by inductively coupled argon-plasma mass spectroscopy (ICP-MS) involves sampling the fuel salt and transferring the sample to an analytical facility capable of receiving high-radiation emitting samples. During the MSRE, electrochemical methods including potentiometry and linear sweep voltammetry (LSV) were used along with traditional chemical analysis for monitoring the reduction-oxidation (RedOx) potential of the fuel salt. More details are provided later. The electrochemical methods involve measuring the open circuit potential (OCP) of a potentiometry sensor immersed in the salt and recording the signals due to the RedOx reactions induced by the voltage sweep applied by the potentiostat. Electrochemical methods can be used to measure the RedOx potential of the salt sampled from fuel salt, or some in-line sensors can be installed in the reactor to provide real-time RedOx potential monitoring.

The primary objective of this report is to evaluate potential applications of electrochemical methods to monitor and/or control molten salt RedOx in MSRs, including both salt-fueled and salt-cooled MSRs. These evaluation results are expected to aid the NRC in its understanding of the capabilities of this technique. This report will help inform NRC reviews of potential license applications with reactor monitoring and control, quality assurance, maintenance, and replacement of electrochemical monitoring systems.

The remainder of this report is organized as follows. Section 2 explains the connection between the chemical potential and RedOx potential of molten salts for MSRs and information on the calculation of chemical potentials in molten salts. Section 3 describes some basic concepts of MSR applications. Section 4 gives background information on operational experience of MSRs and their monitoring, Section 5 gives an assessment of electrochemical methods for molten salts. Section 6 gives an assessment of electrochemical methods for MSR applications. Section 7 summarizes considerations for additional research and development activities, and Section 8 gives an overall summary of the report.

TAKEAWAYS: There have only been two molten salt reactors. The Aircraft Reactor Experiment in 1954 and the Molten Salt Reactor Experiment between 1965 and 1969. Both reactors were based on fluoride salt chemistries and operated at the Oak Ridge National Laboratory. Presently, there are multiple designs of molten salt reactors under investigation based on both chloride and fluoride salt chemistries. The RedOx potential of a salt is a measure of the chemical condition of the salt that provides critical information for reactor operations. Electrochemical techniques can be used to indirectly monitor RedOx potentials.

2. CHEMICAL POTENTIAL OF MOLTEN SALTS

2.1. Reduction/Oxidation Potentials

Understanding of the present chemical conditions of molten salts for MSRs is necessary for their continuous operation and is the motivation for discourse on electrochemical monitoring of MSRs. This section explains the concept of chemical potentials within molten halide salt systems. The salt systems described are limited to chloride and fluoride salts under consideration for use in MSRs as fuel salts, blanket salts, and coolant salts. The examples are based on equilibrium thermodynamic models and are illustrative of, though not rigorously representative of, real molten salt systems. They were chosen to explain concepts that give useful insights into the behaviors of real molten salt systems. Rigorous modeling of molten salt chemistries under specific conditions of reactor operations is beyond the scope of this report and is in many regards beyond the ability of the current state of scientific knowledge. Such modeling efforts are highly application specific with respect to reactor neutronics, salt chemistry, and materials of construction. The examples presented here are for illustrative and educational purposes only.

The term “RedOx potential” sometimes causes confusion because it is used to describe a variety of concepts. During RedOx reactions, the oxidation state is an integer indicative of the number of electrons lost (for positive oxidation states) or gained (for negative oxidation states) relative to an element’s pure elemental state. Since chemical bonding includes sharing of electrons rather than complete loss or gain of electrons, the International Union of Pure and Applied Chemistry (IUPAC) definition of the oxidation state reads: “The oxidation state of an atom is the charge of this atom after ionic approximation of its heteronuclear bonds.” Due to this sharing of electrons in chemical bonds, the integer oxidation state does not represent the real charge on an atom, but the concept is useful, and it serves as a qualitative indication of any increase or decrease of charge for a specific element. The oxidation state concept is used in this report. Two different elements of the same oxidation state should not be considered to have identical charges, but for the purposes of this report, a change in oxidation state for a given element can be taken as an indication of the direction of change in charge on that atom to a more positive valence or a more negative valence.

Thus, for a given element, when reduction occurs, the bonding capacity of that atom is said to decrease, and when oxidation occurs, the bonding capacity of that atom is said to increase. For example, unalloyed metallic uranium has a valence of zero (U^0). Uranium in uranium trichloride (UCl_3) and uranium tetrachloride (UCl_4) have valences of U^{3+} and U^{4+} , respectively. Uranium in UCl_4 is more oxidized than uranium in UCl_3 , so uranium in UCl_4 has more bonding capacity than uranium in UCl_3 as reflected in the number of chlorine atoms bonded. Likewise, uranium in U_3O_8 (two as U^{5+} and one as U^{6+}) is more oxidized than uranium in UO_2 (as U^{4+}) [13]. The chemical properties of an element are partially dependent on its oxidation state. In this context, “oxidized” does not imply the presence of oxygen, it only refers to the valence state of a chemical species being more positive, in this case for uranium.

Without contextual knowledge, for electrochemistry, the term “RedOx potential” may be interpreted to imply that there is an electrical potential at which some RedOx reaction occurs. An example of this is seen during electrochemical measurements such as cyclic voltammetry (CV), where the RedOx potentials of select species in molten salts are measured and compared. During a CV measurement, the electrical potential of a working electrode (WE) is cycled between more oxidizing versus more reducing potentials (or vice versa) and this electrical potential is quantitatively controlled and recorded, typically in units of volts. The electrical potential at the working electrode is recorded based on its potential difference to a reference electrode (RE), where the RE serves as a stable potential datum from which to measure the potential difference to the variable potential of the WE. The current is measured throughout the CV potential sweep, such that the electric current passing through the WE is recorded as a function of the potential difference between the WE and RE.

The authors conducted the following experiment, which is used as an illustrative example. A crucible was filled with 500 g of molten LiCl-KCl eutectic salt heated to 500°C. A WE and counter electrode (CE) were made of 1.5-mm-diameter tungsten wires. An Ag/AgCl RE was made from a 6-mm-diameter closed-bottom mullite tube fitted inside with a 1-mm-diameter silver wire, and a small amount of LiCl-KCl eutectic with 1 wt% AgCl was added to the inside of the mullite tube, enough to wet the bottom of the silver wire. The WE was submerged 1-cm, and the CE and RE were each submerged 10-cm in the salt.

A CV for the LiCl-KCl salt is shown in Figure 3. The WE potential was scanned between -2.2 to 1.6 V (relative to the RE) at a rate of 200 mV s⁻¹ by a potentiostat (BioLogic VSP-300). The WE potential was controlled as a function of time as shown in Figure 4, with four complete cycles between the vertex limits displayed. In other words, -2.2 V was the cathodic vertex potential, and 1.6 V was the anodic vertex potential. The potentiostat was programmed to cycle the voltage between these limits four times at a rate of 200 mV s⁻¹. The entire measurement cycle performed by the potentiostat took about 2.5 minutes to complete. Meanwhile, as the voltage cycles progressed, the potentiostat recorded the direct current response of the circuit as current passed between the WE and CE through the potentiostat. The corresponding current response that was recorded is shown in Figure 5. The CV shown in Figure 3 is an

overlay of four CV cycles, and was produced from the voltage data in Figure 4 combined with the current data in Figure 5 .

At the anodic limit, the tungsten WE began to oxidize, shedding tungsten cations into the salt surrounding the WE. At the cathodic limit, lithium metal was reduced onto the WE. Cathodic current is shown as a negative current, and anodic current is shown as a positive current. Over the section of the CV between these two limits, the current was nearly zero, indicating the absence of any other RedOx reactions. The window of stability for this salt was approximately -2.0 to 1.4 V (a window of 3.4 V).

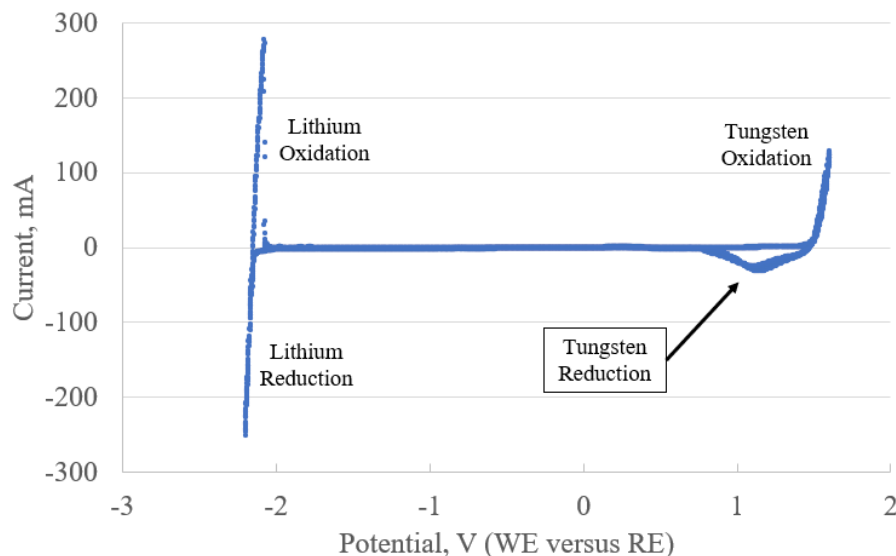


Figure 3. Cyclic voltammogram of LiCl-KCl eutectic at 500°C. Tungsten WE and CE. Ag/AgCl RE.

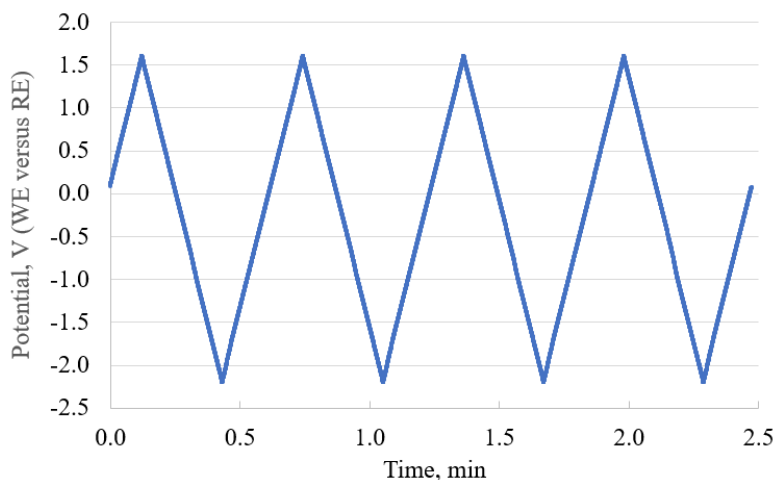


Figure 4. Potential of the WE as controlled by the potentiostat for the cyclic voltammogram of LiCl-KCl eutectic at 500°C. Tungsten WE and CE. Ag/AgCl RE.

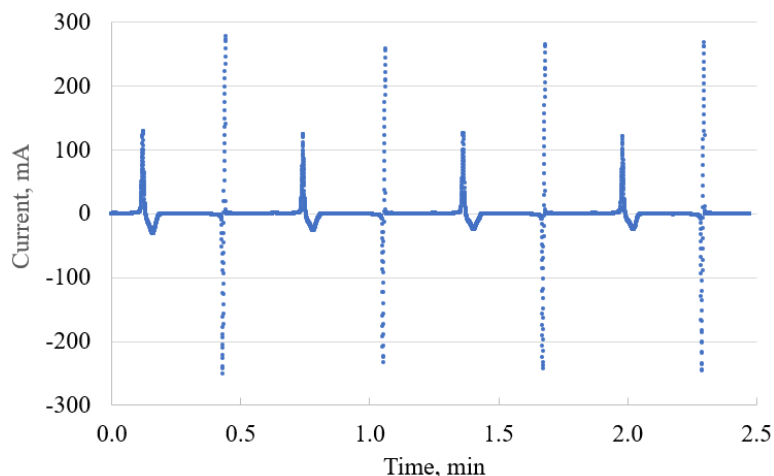


Figure 5. Current response as measured by the potentiostat for the cyclic voltammogram of LiCl-KCl eutectic at 500°C. Tungsten WE and CE. Ag/AgCl RE.

In the second part of the experiment, an additional CV measurement was made with a small amount of MnCl_2 added to the LiCl-KCl in the crucible. A CV for the LiCl-KCl salt with 1 wt% MnCl_2 is shown in Figure 6. The WE potential was scanned between -2.2 to 1.5 V at a rate of 200 mV s^{-1} . At the anodic limit, the tungsten WE began to oxidize. As the WE potential was made more cathodic, manganese metal was reduced onto the WE as the potential approached -0.8 V, and manganese metal continued to be reduced onto the WE beyond this potential. At the cathodic limit, lithium metal was reduced onto the WE along with manganese metal for a brief period. As the WE potential was reversed (i.e., made more anodic), lithium metal was almost immediately oxidized from the WE, and manganese metal was oxidized from the WE as the potential approached -0.8 V.

In both Figure 3 and Figure 6, there are reduction peaks following the relaxation of the potentials after tungsten oxidation. These reduction peaks are interpreted as the reduction of the tungsten cations surrounding the WEs following brief periods of oxidation. However, the reduction of tungsten cations was not sustained once its concentration was depleted. The reduction of manganese cations behaves differently. In Figure 6, the reduction peak for manganese assumes maximum value and then decreases to a steady state value until the onset of lithium reduction. The maximum value was associated with the reduction of the manganese cations immediately surrounding the WE, and the steady value was associated with the reduction of manganese cations from the bulk salt that move to the WE as the result of a steady state concentration gradient.

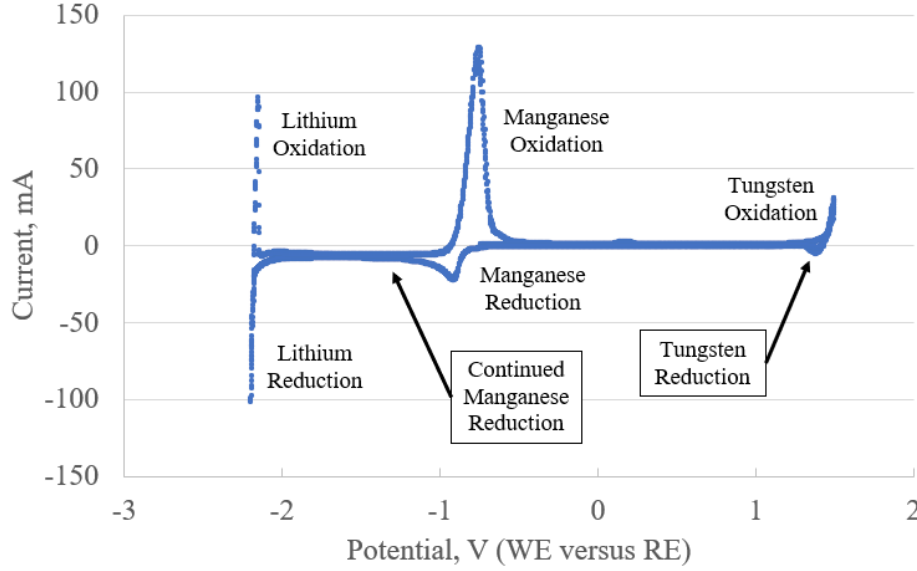


Figure 6. Cyclic voltammogram of LiCl-KCl eutectic with 1 wt% MnCl_2 at 500°C. Tungsten WE and CE. Ag/AgCl RE.

As the WE was cycled to negative potentials by the potentiostat (at potentials more negative than the manganese reduction potential), the net reaction at the WE was the reduction of manganese cations from the salt onto the WE. Manganese reduction continued for as long as the WE potential was less than -0.8 V.



As the WE was cycled back to positive potentials (at potentials more positive than the manganese oxidation potential), the net reaction at the WE was the oxidation of manganese metal from the WE into the salt. Manganese oxidation stopped when all the manganese was oxidized from the WE at potentials greater than -0.8 V.



In this example, the reduction and oxidation potentials were essentially the same potential, namely the RedOx potential of manganese in this salt system as exhibited by the CV. Given this salt system, the RedOx potential of manganese is a function of the MnCl_2 concentration and temperature. This illustrates a valid use of the term “RedOx potential,” but it is not the only meaning of “RedOx potential” in this discussion. RedOx potentials are related to chemical potentials within the salt systems. This interpretation has a long history [14]. Chemical potential could also be used to characterize the RedOx state or RedOx condition of the molten salts. As described by Olander [14], the chemical potential of gas X_2 (e.g., $\text{X} = \text{F}, \text{Cl}$) can be defined in terms of its partial pressure in gas phase, or $\Delta\bar{G}_{\text{X}_2} = RT \ln P_{\text{X}_2}$. A molten salt with known composition at equilibrium fixes the chemical potential of those species within the salt and vice versa. The RedOx potentials observed during electrochemical measurements (potentials as voltages) and RedOx potentials interpreted as chemical potentials (potentials as activities) are not completely unrelated phenomena. In the example shown in Figure 6, if the MnCl_2 concentration was decreased, the electrical reduction potential would shift to more negative values, and if the MnCl_2 concentration was increased, the electrical reduction potential would shift to more positive values. At lower concentrations of a few weight percent or less, electrical reduction potentials are greatly influenced by concentrations. The change of MnCl_2 concentrations will result in the change of chlorine chemical potential.

According to *IUPAC Recommendations 1997*, “RedOx potential is the reduction/oxidation potential of a compound measured under standard conditions against a standard reference half-cell.” [15]. In aqueous electrochemistry, the “standard reference half-cell” is sometimes the “standard hydrogen electrode (SHE)” whose potential is established, by convention, as zero volts. A SHE has a highly specific configuration, which is intended to produce accuracy and reproducibility of measurements. Unfortunately, the SHE is difficult to fabricate and use experimentally. Other reference electrodes are more commonly used because they are more convenient in both of these regards. Often, the data collected using an alternative reference electrode can be corrected to the values that would have been measured if a SHE had been used.

In this context, RedOx potential has a well-defined meaning and the value for a specific half-cell can be determined by the Nernst Equation. In molten salt chemistry and electrochemistry literature, the RedOx potential concept is widely used, but in many cases, it is loosely used. For example, the phrase “RedOx potential of molten salt” does not mean anything itself. It must be given context. In some literature, “RedOx potential of molten salt” refers to the potential difference between a working/indicator electrode and an RE. This measured voltage can provide qualitative information on the chemistry and corrosivity of the salt. Different names like RedOx potential, reduction potential, RedOx condition, RedOx state, and RedOx chemistry have been used interchangeably to describe the chemistry, chemical state, and corrosivity of molten salts.

TAKEAWAYS: Multiple meanings are associated with the term “RedOx potential,” and this can lead to confusion. The terms “reduction” and “oxidation” do not necessarily imply the absence or presence, respectively, of oxygen. Metals exist in their metallic state under sufficiently reducing conditions and form chemical compounds under sufficiently oxidizing conditions. Oxidizing conditions can stem from the presence of oxidizing halide compounds such as $\text{Cl}_2(\text{g})$ and $\text{F}_2(\text{g})$, and from impurities such as oxygen and moisture. A metal in its metallic state has a valency of zero. A metal in a salt compound with halides is a cation and has a positive valency.

2.2. Thermodynamic Stability: Ellingham Diagrams

Molten halide salts are ionic liquids with a net neutral electrical charge, meaning there is a charge balance between cations and anions with respect to the molar stoichiometries of the ions involved. The halides are the anions, either chlorine (Cl^-) or fluorine (F^-), which have only a single anionic valence state of negative one. The metal cations are those selected for the salt system. Examples include lithium (Li^+), beryllium (Be^{2+}), and uranium (U^{3+} or U^{4+}), which may have a single valence state or multiple valence states. For example, consider LiCl , LiF , BeCl_2 , BeF_2 , UCl_3 , UCl_4 , UF_3 , and UF_4 , where lithium and beryllium have single cationic valence states of positive one, while uranium has more than one possible cationic valence state. Molten salts are characterized by their short-range atomic-scale structural order, in contrast to the long-range structural order of crystalline solids. The ions are free to move through the bulk molten salt by the various mass transport processes of diffusion (driven by chemical gradients), convection (driven by density gradients), mechanical mixing (driven by force gradients), and migration (driven by electric field gradients).

Consider the LiCl-KCl eutectic salt system. When molten, the individual cations and anions are free to move about provided that the charge neutrality of the salt is maintained. The association of cations and anions is fluid, and no strong chemical bonding occurs. This visualization of molten salt is convenient, but not necessarily accurate. As the chemical complexity of the salt increases by the addition of more constituents, the formation of complex anionic and cationic molecular species is possible. However, these ionic molecules are difficult to identify and characterize. The analyses presented here ignore this reality and view molten salts as simple mixtures of elemental anions and cations.

Chlorine ($\text{Cl}_2(\text{g})$) and fluorine ($\text{F}_2(\text{g})$) are very reactive and are capable of combining with all the metals on the periodic table to make metal chlorides and fluorides, respectively. Thermodynamics provides a means of ranking the resulting bond strengths between the halides and metals from strongest to weakest. For example, LiCl is much more stable than UCl_3 . Reaction of lithium metal with UCl_3 results in UCl_3 being reduced to uranium metal as lithium metal is oxidized to LiCl as shown in Equation (3).



A formal statement regarding Equation (3) is that the standard state^a Gibbs free energy change (ΔG^0) at 500°C is -336 kJ per mole of UCl_3 . A negative value of ΔG^0 implies that the reaction is favorable for the standard states in the direction written, at that temperature. Temperature is not part of the definition of the standard state (that is, for any given temperature, the standard state of a substance will have a specific energy associated with it at that temperature). Another formal statement regarding Equation (3) is that lithium is more electropositive than uranium, meaning that lithium will tend to lose its electrons to uranium. Conversely, uranium is more electronegative than lithium. A similar thermodynamic construct is to compare the standard state Gibbs free energies of formation (ΔG_f^0) of various chlorides and fluorides. An Ellingham Diagram is a plot of values such as these, as functions of temperature as shown in Figure 7 for chloride salts, and in Figure 8 for fluoride salts.^b The more negative the value of ΔG_f^0 , the more stable the metal chloride. These figures suggest that increasing thermodynamic stability progresses from MoCl_2 to CsCl , and from MoF_3 to BeF_2 . The formation reaction stoichiometries for the data in Figure 7 and Figure 8 are normalized with respect to $\text{Cl}_2(\text{g})$ and $\text{F}_2(\text{g})$, respectively. Formation reactions for UCl_3 and LiCl from their elements are shown in Equation (4) and Equation (5), respectively. These reactions are normalized to one mole of $\text{Cl}_2(\text{g})$.

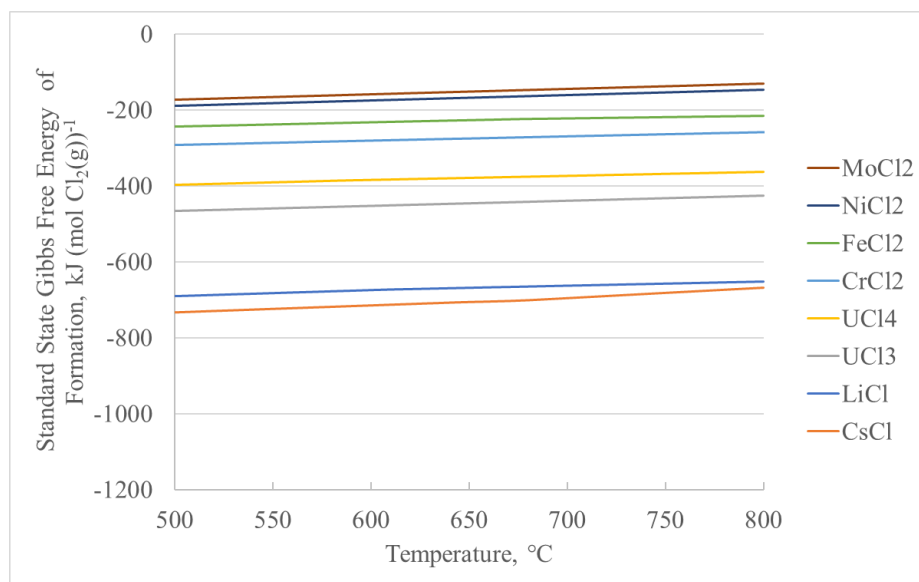


Figure 7. Ellingham diagram for select metal chloride species.

^a Here, standard state is the pure substance at 1 bar pressure, which is the most common selection for standard states encountered in modern literature. However, it is not the only possible selection for a standard state.

^b All standard state thermodynamic data and equilibrium thermodynamic calculations presented in this section were sourced from Outotec HSC Chemistry, Software Version 9.8.1.2.

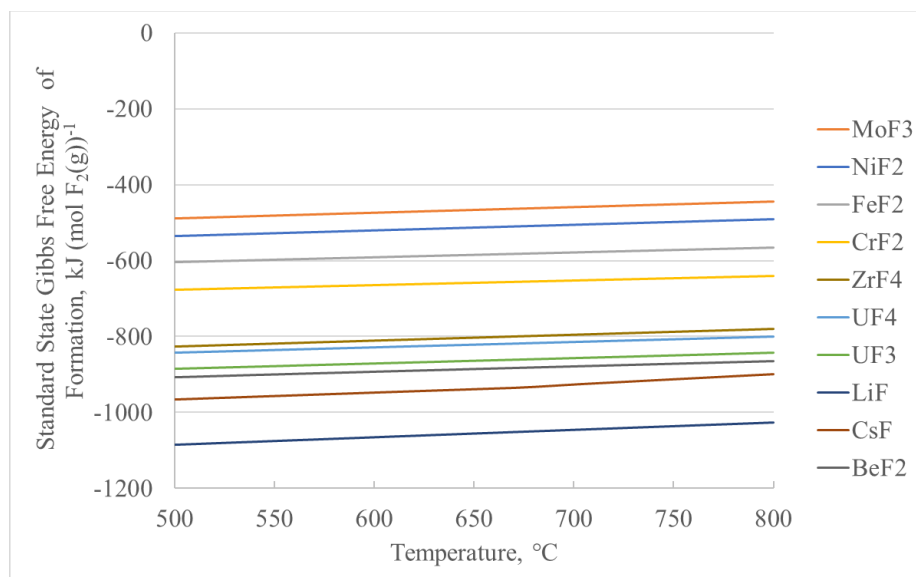


Figure 8. Ellingham diagram for select metal fluoride species.

TAKEAWAYS: Ellingham diagrams provide a way of ranking the standard state thermodynamic stabilities of chemical compounds. Here, Ellingham diagrams are useful to rank the affinities of metals to chemically bond with chlorine and fluorine to form chloride and fluoride compounds, respectively. Metals that have multiple valence states can form a series of Me_xHa_y compounds, where Me is the metal cation, Ha is the halide anion, and x and y are stoichiometric integers. An example of this is UF_3 , UF_4 , and UF_6 .

2.3. Thermodynamic Chemical Potential

In principle, the chemical potential of a species is a measure of its “influence” in a chemical system. Intuitively, the chemical potential of a species would appear to be related to its concentration, which it is. However, the phase of the material (solid, liquid, gas) also influences a species’ chemical potential. For example, two different phases (e.g., solid-solid, solid-liquid, liquid-gas) in equilibrium with each other can contain the same species at the same chemical potential in each phase, but at two different concentrations.

Chemical potential has a strict thermodynamic definition as shown in Equation (6), which states that the chemical potential of species i (μ_i) is defined as the change in the Gibbs free energy (δG) of the system, resulting from a change in the number of moles of species i (δn_i) in the system, keeping constant the temperature (T), pressure (P), and the numbers of moles of all other species in the system ($n_j \dots$).^c The “system” always has a net chemical potential of zero at equilibrium, and the engineering units of chemical potential are energy per quantity, such as kJ mol^{-1} .

$$\mu_i = \left(\frac{\delta G}{\delta n_i} \right)_{T,P,n_j \dots} \quad (6)$$

^c The chemical potential is also known as the partial molar Gibbs energy.

An ideal gas is one in which the molecules of gas do not interact with each other. The gas phase can be comprised of a single species or multiple species. The ideal gas law is shown in Equation (7), where P is system pressure, V is system volume, n is the number of moles of gas molecules, R is the molar gas constant, and T is temperature.

$$PV = nRT \quad (7)$$

A variant of Equation (7) is shown in Equation (8), where P_i is the partial pressure of gas species i , and n_i is the number of moles of gas species i . For a system with multiple species, the system pressure (P) is the sum of the individual partial pressures ($P_i, P_j \dots$) as shown in Equation (9), and the total moles (n) is the sum of the moles of individual species ($n_i, n_j \dots$) as shown in Equation (10).

$$P_i V = n_i R T \quad (8)$$

$$P = P_i + P_j + \dots \quad (9)$$

$$n = n_i + n_j + \dots \quad (10)$$

The compressibility factor (Z) is defined as shown in Equation (11), and has a value of 1 for an ideal gas.

$$Z = \frac{PV}{nRT} = 1 \quad (11)$$

In a real gas, the gas molecules do interact with each other. An example of an equation for the compressibility factor (Z) for a real gas is shown in Equation (12). This is just one example of several possible forms of a *virial equation of state* (i.e., gas pressure as a function of gas density, derived from statistical mechanics). The exact form depends on the physics model used to derive the equation. Here P is pressure, B is the first virial coefficient, C is the second, and D is the third. Equation (12) also expresses that Z is a function of temperature and pressure (T, P), and that the virial coefficients are functions of temperature (T). Notice that as the gas pressure decreases, the latter virial coefficients become less and less significant. Therefore, at sufficiently low pressure, $Z = 1$ is a good approximation, which is to say that the behavior of a real gas approaches the behavior of an ideal gas at low pressure. In many engineering applications at pressures below 1 bar, ideal gas behavior is an acceptable approximation.

$$Z(T, P) = 1 + B(T)P + C(T)P^2 + D(T)P^3 \dots \quad (12)$$

The chemical potential of a real gas species is shown in Equation (13), where the chemical potential of gas species i (μ_i), is defined as the summation of the chemical potential of gas species i under standard state conditions (μ_i^0), as well as a correction term to account for departure from the standard state conditions. The correction term introduces the concept of fugacity (f), where f_i is the fugacity of gas species i , and f_i^0 is the fugacity of gas species i under standard state conditions, where the standard state conditions are hypothetical ideal state conditions. Within this report f_i^0 is defined as having a value of 1 bar. Equation (13) also expresses that μ_i is a function of temperature and pressure (T, P), and μ_i^0 is a function of temperature (T).

$$\mu_i(T, P) = \mu_i^0(T) + RT \ln \left(\frac{f_i}{f_i^0} \right) = \mu_i^0(T) + RT \ln(a_i) \quad (13)$$

In principle, the fugacity of a gas species is a measure of its “effective” partial pressure in that the fugacity value corresponds to the ideal gas absolute pressure value that would produce the same value of $\mu_i(T, P)$. The activity of gas species i is unitless, and is related to its fugacity, given by $a_i = (f_i/f_i^0)$. The introduction of fugacity is also a simplifying mathematical construct because the fugacity value accounts for the nonideal behavior of real gases, and thus enables the use of simpler equations. For example, Equation (13) is a linear expression. Simplification is achieved by using this linear expression during calculations, rather than the nonlinear expression that would otherwise be derived as a function of the gas partial pressure using Equation (12).

The chemical potential of an ideal gas species is shown in Equation (14), which is a corollary to Equation (13). Under conditions where the ideal gas approximation applies, such as at low pressure, partial pressure is equivalent to fugacity. Here, P_i^0 is the standard state pressure of species i which, by convention, is usually chosen as 1 bar. Therefore, for an ideal gas, the partial pressure of gas species i (P_i) becomes a measure of the chemical potential of gas species i (μ_i). Furthermore, the partial pressure of gas species i (P_i) can be calculated from the product of the mole fraction of gas species i (X_i) and total pressure (P_T). Therefore, the concentration gas species i (X_i) can be used as a measure of the chemical potential of gas species i (μ_i).

$$\mu_i(T, P) = \mu_i^0(T) + RT \ln \left(\frac{P_i}{P_i^0} \right) = \mu_i^0(T) + RT \ln \left(\frac{X_i P_T}{P_i^0} \right) \quad (14)$$

Similar concepts apply in the condensed phase, where there is no use for fugacity, but the concept of activity remains applicable. Fugacity is to a gas phase what activity is to a condensed phase. In principle, the activity of species i is a measure of its “effective” concentration, in that the activity describes a unitless relative concentration that would produce the same value of chemical potential for an ideal substance. The activity is a unitless relative concentration because it includes the standard state in the denominator. This is different from fugacity, as fugacity is not a relative concentration, and it retains the units of pressure. The chemical potential of a real condensed species is shown in Equation (15). Here, a_i^0 is the standard state activity of species i which, by convention, is usually chosen as an activity of 1 for pure chemicals in their condensed phase.^d Therefore, the activity of species i (a_i) can be used as a measure of the chemical potential of species i (μ_i). Furthermore, the activity of species i (a_i) is the product of the mole fraction of species i (X_i) and the activity coefficient of species i (γ_i). Therefore, the concentration of species i (X_i) can be used as a measure of the chemical potential of species i (μ_i). In some limited circumstances the activity coefficient can be treated as a constant, but more often it is a function of the state of the system. In an ideal condensed phase, the activity of species i is equal to its concentration, implying that the activity coefficient is equal to 1.

$$\mu_i(T, P) = \mu_i^0(T, P) + RT \ln \left(\frac{a_i}{a_i^0} \right) = \mu_i^0(T, P) + RT \ln \left(\frac{\gamma_i X_i}{a_i^0} \right) \quad (15)$$

^d Standard conditions refer to the selection of a standard pressure and temperature used to describe experimental conditions often involving volume calculations of gases, liquids, and solids. Common standard pressures are 1 atm or 1 bar, and common standard temperatures are 0°C or 25°C. The phrase “standard conditions” refers to an experimental concept, while the phrase “standard states” refers to a thermodynamics concept. “Standard states” refer to the selection of chemical species, phases, concentrations, and pressures for thermodynamic characterizations. Unlike the phrase “standard conditions”, the temperature is not part of the definition of a “standard state”. Instead, a standard state (such as a particular gas species at 1 bar) will be utilized for the temperature at which a thermodynamic calculation is being done. Thus, one might refer to “Xe in its standard state at 500 °C”, which would mean Xe at a pressure of 1 bar at 500 °C. The values of standard conditions and of standard states are chosen for convenience and convention. For example, for solutes that are dissociated ionic species, other conventions are used. For ionic solutes in aqueous solutions, the activity of 1 is not typically set to correspond to pure species, and is instead set to a concentration such as 1 mol/L. Another example is metallurgical slags, where the reference concentration for an activity of 1 is often set to 1 wt% in solution.

The equations presented in this section, Equations (6)–(15), are commonly found in educational materials on chemical thermodynamics [16, 17]. The purpose of presenting these equations here is to illustrate the connections between the formal definition of chemical potential and engineering measurements such as partial pressure and concentration.

TAKEAWAYS: Chemical potential has a strict thermodynamic definition. Nevertheless, engineering quantities such as partial pressure of a gas species, or concentration of a condensed species can be used as indirect measures of chemical potential. In the examples that follow, the partial pressure of chlorine gas will be used as a measure of the chlorine chemical potential, and the partial pressure of fluorine gas will be used as a measure of the fluorine chemical potential.

2.4. Thermodynamic Stability: Phase Stability Diagrams

The equilibrium phase stability diagram for the uranium, chlorine, fluorine system at 500°C is shown in Figure 9. This diagram is based on the selection of three elements (U, Cl, and F) for illustration, and the resultant stable compounds (U, UCl₃, UCl₄, UF₃, UF₄, UF₆, and UClF₃). The partial pressure of Cl₂(g) was selected to represent the Cl chemical potential along the ordinate, and the partial pressure of F₂(g) was selected to represent the F chemical potential along the abscissa. The Cl and F chemical potentials increase as the partial pressures of Cl₂(g) and F₂(g) increase, respectively. On both axes, the pressure scales were limited to 1 bar and made sufficiently low to encompass a portion of the uranium metal stability domain.

Uranium metal is stable only at very low Cl and F chemical potentials. As the Cl chemical potential is increased, the phase boundaries between U and UCl₃, and UCl₃ and UCl₄ are encountered. Likewise, as the F chemical potential is increased the phase boundaries between U and UF₃, UF₃ and UF₄, and UF₄ and UF₆ are encountered. The stability domain for UClF₃ requires both Cl and F chemical potentials to exist. However, Figure 9 gives no information about the compositions of the equilibrium gas phase above the condensed stability domains. Such information can be added by overlaying the partial pressure isobars for select gas species, or the isobars for total system pressure [18].

It can be concluded from Figure 9 that, with respect to uranium, fluorine is much more reactive than chlorine as indicated by the positions of the phase boundaries along the axes. Fluorine begins to react with uranium at much lower partial pressures than chlorine, which gives some insight into why fluoride and chloride salts behave differently regarding corrosion.

Consider the phase boundary between U and UCl₃, which occurs at approximately -31.5 on the ordinate. At this boundary the partial pressure of Cl₂(g) is 3.2×10^{-32} bar, which is equivalent to a concentration of 3.2×10^{-26} parts per million (ppm) (by volume in the gas phase) if the total system pressure is 1 bar. Similarly, the phase boundary between U and UF₃ occurs at a F₂(g) concentration of 1×10^{-54} ppm (by volume in the gas phase). These concentrations are far below the detection limits of chemical analytical instrumentation. Nevertheless, small thermodynamic values such as these are both real and useful.

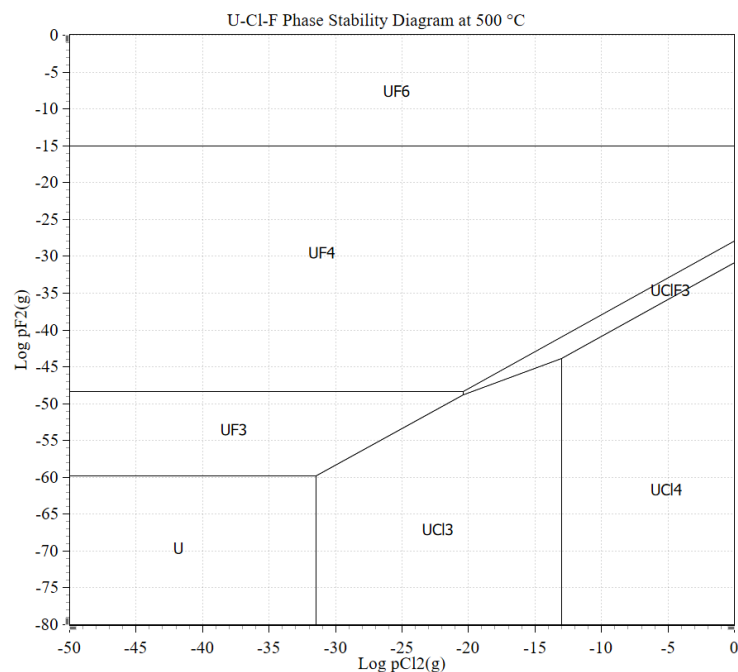


Figure 9. Equilibrium phase stability diagram for the U, Cl, F system at 500°C.

The equilibrium phase stability diagram for the lithium, chlorine, fluorine system at 500°C is shown in Figure 10. This diagram is based on the selection of three elements (Li, Cl, and F), and the resultant stable compounds (Li, LiCl, and LiF). Notice that the lithium metal stability domain requires much lower partial pressures of $\text{Cl}_2(\text{g})$ and $\text{F}_2(\text{g})$ than that of the uranium metal stability domain in Figure 9. This behavior is in agreement with the earlier observations that LiCl and LiF are much more stable than UCl_3 and UF_3 , from Figure 7 and Figure 8, respectively. In Figure 9 and Figure 10, $\text{F}_2(\text{g})$ is reactive with the metals at lower partial pressures than $\text{Cl}_2(\text{g})$.

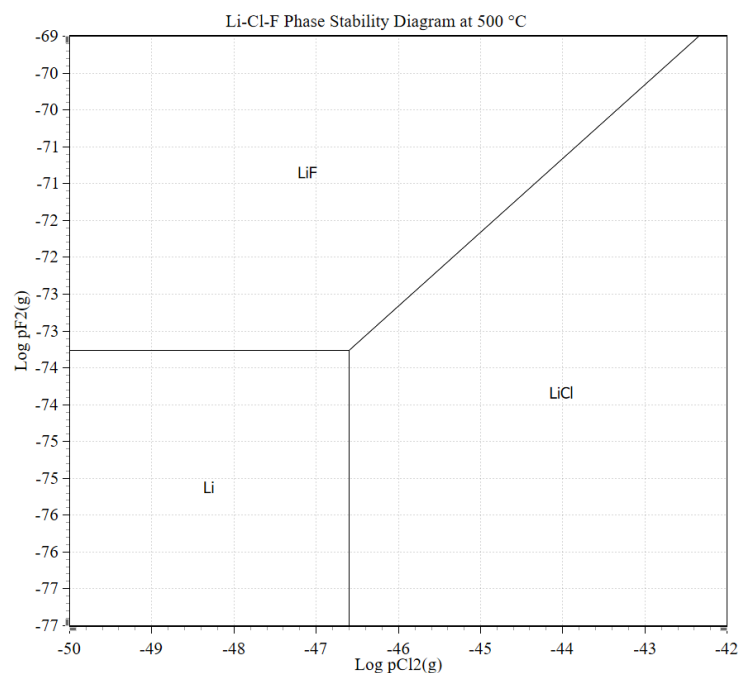


Figure 10. Equilibrium phase stability diagram for the Li, Cl, F system at 500°C.

Of the two diagrams, Figure 9 is more complex in that it has seven stability domains and 10 phase boundaries. These phase boundaries are expressed by the following equations.

$$UF_3 + \frac{3}{2} Cl_2(g) = UCl_3 + \frac{3}{2} F_2(g) \quad (16)$$

$$UClF_3 + Cl_2(g) = UCl_3 + \frac{3}{2} F_2(g) \quad (17)$$

$$UClF_3 + \frac{3}{2} Cl_2(g) = UCl_4 + \frac{3}{2} F_2(g) \quad (18)$$

$$UClF_3 + \frac{1}{2} F_2(g) = UF_4 + \frac{1}{2} Cl_2(g) \quad (19)$$

$$U + \frac{3}{2} F_2(g) = UF_3 \quad (20)$$

$$UF_3 + \frac{1}{2} F_2(g) = UF_4 \quad (21)$$

$$UF_4 + F_2(g) = UF_6 \quad (22)$$

$$U + \frac{3}{2} Cl_2(g) = UCl_3 \quad (23)$$

$$UCl_3 + \frac{1}{2} Cl_2(g) = UCl_4 \quad (24)$$

$$UF_3 + \frac{1}{2} Cl_2(g) = UClF_3 \quad (25)$$

Notice that the phase boundaries in Figure 9 depicted by Equation (16) to Equation (19) have positive slopes (as they involve $Cl_2(g)$ and $F_2(g)$), those depicted by Equation (20) to Equation (22) are horizontal (as they involve only $F_2(g)$), and those depicted by Equation (23) to Equation (25) are vertical (as they involve only $Cl_2(g)$).

Consider the boundary between UF_3 and UCl_3 as expressed in Equation (16). The thermodynamic expressions leading to the equilibrium condition are shown in Equation (26) to Equation (28), where ΔG is the Gibbs free energy change, ΔG^0 is the standard state Gibbs free energy change, R is the gas constant, T is temperature, Q is the reaction quotient, and K_{eq} is the equilibrium reaction quotient.

$$\Delta G = \Delta G^0 + RT \ln(Q) \quad (26)$$

When the chemical system is at equilibrium, the Gibbs free energy change is at a minimum, and the reaction quotient is the equilibrium reaction quotient.

$$\Delta G = 0 \text{ and } Q = K_{eq} \quad (27)$$

Equation (26) and Equation (27) are combined to give Equation (28).

$$\Delta G^0 = -RT \ln(K_{eq}) \quad (28)$$

Equation (16) and Equation (28) are combined to give Equation (29), where a is the chemical activity of the species indicated.

$$\Delta G^0 = -RT \ln \left(\frac{a_{UCl_3} a_{F_2(g)}^{1.5}}{a_{UF_3} a_{Cl_2(g)}^{1.5}} \right) \quad (29)$$

Because UCl_3 and UF_3 are pure condensed phase substances, their chemical activities are equal to 1, as indicated in Equation (30).

$$a_{UCl_3} = 1 \text{ and } a_{UF_3} = 1 \quad (30)$$

Equation (29) and Equation (30) are combined to give Equation (31).

$$\Delta G^0 = -RT \ln \left(\frac{a_{F_2(g)}^{1.5}}{a_{Cl_2(g)}^{1.5}} \right) \quad (31)$$

Equation (31) is rearranged to give Equation (32).

$$\Delta G^0 = -\frac{3}{2} RT \ln \left(\frac{a_{F_2(g)}}{a_{Cl_2(g)}} \right) \quad (32)$$

Equation (32) is rearranged to give Equation (33).

$$\frac{2}{3} \frac{\Delta G^0}{RT} = \ln(a_{Cl_2(g)}) - \ln(a_{F_2(g)}) \quad (33)$$

For low pressures where the gases are sufficiently ideal, the chemical activities of $Cl_2(g)$ and $F_2(g)$ are numerically equal to their partial pressures divided by their standard state pressures, which are both 1 bar.

$$a_{Cl_2(g)} = \frac{P_{Cl_2(g)}}{P_{Cl_2(g)}^0} \text{ and } a_{F_2(g)} = \frac{P_{F_2(g)}}{P_{F_2(g)}^0} \quad (34)$$

Equation (33) and Equation (34) are combined to give Equation (35). For convenience, the terms for the standard pressures have been incorporated into the pressure terms in the following equations, with this incorporation designated by an asterisk, such that $P^*_{Cl_2(g)} = P_{Cl_2(g)} / P_{Cl_2(g)}^0$. Accordingly, the natural log terms remain unitless.

$$\frac{2}{3} \frac{\Delta G^0}{RT} = \ln(P^*_{Cl_2(g)}) - \ln(P^*_{F_2(g)}) \quad (35)$$

Equation (36) is a logarithmic identity.

$$\ln(b) = \ln(10) \cdot \log(b) \quad (36)$$

Equation (35) and Equation (36) are combined to give Equation (37).

$$\frac{2}{3} \frac{\Delta G^0}{RT} = \ln(10) \log(P^*_{Cl_2(g)}) - \ln(10) \log(P^*_{F_2(g)}) \quad (37)$$

Equation (37) is rearranged to give Equation (38).

$$\frac{2}{3} \frac{\Delta G^0}{\ln(10)RT} = \log(P^*_{Cl_2(g)}) - \log(P^*_{F_2(g)}) \quad (38)$$

Equation (38) is rearranged to give Equation (39).

$$\log(P^*_{F_2(g)}) = -\frac{2}{3} \frac{\Delta G^0}{\ln(10)RT} + \log(P^*_{Cl_2(g)}) \quad (39)$$

Once values are assigned to the constants and parameters in Equation (39), and fixing the temperature constant, the expression takes the following form.

$$\log(P^*_{F_2(g)}) = b + m \log(P^*_{Cl_2(g)}) \quad (40)$$

Where the intercept b has the following form.

$$b = -\frac{2}{3} \frac{\Delta G^0}{\ln(10)RT} \quad (41)$$

And the slope m has the following form.

$$m = 1 \quad (42)$$

Similar paths can be used to derive expressions for the other phase boundary equations in Equation (16) to Equation (25). Small steps were taken from Equation (29) to Equation (39) to clearly present the logical progression that led to Equation (39).

TAKEAWAYS: Phase stability diagrams provide a way of illustrating the stability domains of chemical systems as functions of chemical potential and temperature. The partial pressures of $\text{Cl}_2(\text{g})$ and $\text{F}_2(\text{g})$ are often convenient measures of Cl and F chemical potentials, respectively. Some presence of Cl and F chemical potentials are required to form chlorides and fluorides, respectively. The more electropositive the metal, the lower the chemical potential needed to form chlorides and fluorides. $\text{F}_2(\text{g})$ is more chemically reactive than $\text{Cl}_2(\text{g})$ because fluorine is more prone than chlorine to capture an electron and fill its outer electron orbital. As a corollary to the higher elemental reactivity of fluorine relative to chlorine, most fluorine compounds are thus more stable than chlorine compounds, due to these same factors.

2.5. Effect of Fission on Salt Composition

A reactor operating at 1000 MW thermal will consume approximately 1 kg d⁻¹ of fissile metals such as ²³⁵U. A uranium nucleus contains 92 protons (Z = 92). The moment this nucleus splits into two daughters (primary fission fragments), there remain 92 protons between the daughters. The possible combinations of two daughters are listed in Table 2 along each row. These 36 daughters represent about 39% of the elements of the periodic table up to uranium, as shown in Figure 11.

However, there are 383 possible daughter nuclei as each daughter can be born as one of several isotopes as shown in Figure 12 [19]. This figure shows isotopes of zinc to samarium; three isotopes ⁷⁵Cu, ⁹⁶Eu, and ⁹⁷Eu are not shown. Many of these isotopes are ephemeral as the processes of radioactive decay begin immediately upon formation. There are many different modes of radioactive decay, some of the major modes are shown below.

Alpha Decay



Beta Electron Decay



Beta Positron Decay



Electron Capture



Gamma Emission



Spontaneous Fission



Here, Z is the number of protons, A is the atomic number, α is a helium nucleus, β is an electron (-) or a positron (+), ν is a neutrino, ν' is an antineutrino, x-ray and γ -ray are forms of high energy electromagnetic radiation, and n is a neutron.

Table 2. Primary fission fragment pairs of uranium.

Daughter A	Z	Daughter B	Z
Copper (Cu)	29	Europium (Eu)	63
Zinc (Zn)	30	Samarium (Sm)	62
Gallium (Ga)	31	Promethium (Pm)	61
Germanium (Ge)	32	Neodymium (Nd)	60
Arsenic (Ar)	33	Praseodymium (Pr)	59
Selenium (Se)	34	Cerium (Ce)	58
Bromine (Br)	35	Lanthanum (La)	57
Krypton (Kr)	36	Barium (Ba)	56
Rubidium (Rb)	37	Cesium (Cs)	55
Strontium (Sr)	38	Xenon (Xe)	54
Yttrium (Y)	39	Iodine (I)	53
Zirconium (Zr)	40	Tellurium (Te)	52
Niobium (Nb)	41	Antimony (Sb)	51
Molybdenum (Mo)	42	Tin (Sn)	50
Technetium (Tc)	43	Indium (In)	49
Ruthenium (Ru)	44	Cadmium (Cd)	48
Rhodium (Rh)	45	Silver (Ag)	47
Palladium (Pd)	46	Palladium (Pd)	46

H																	He
Li	Be											B	C	N	O	F	Ne
Na	Mg											Al	Si	P	S	Cl	Ar
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
Cs	Ba	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
Fr	Ra	Ac															
			Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu	
			Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr	

Figure 11. Primary fission fragment pairs (green) presented as the Periodic Table.

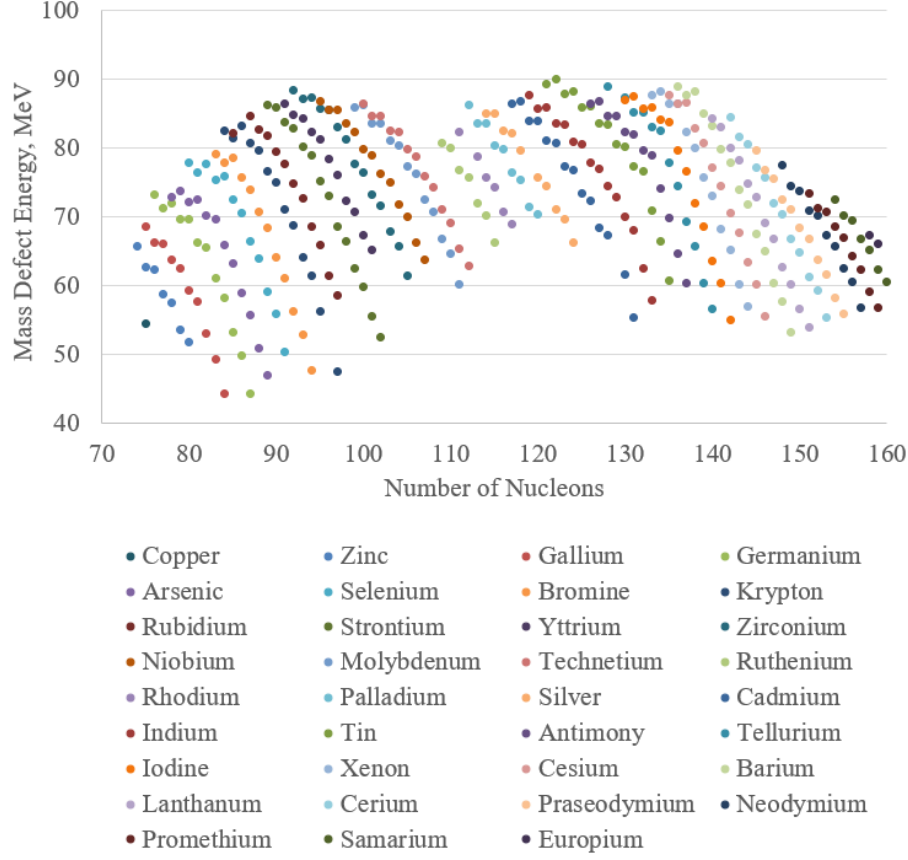
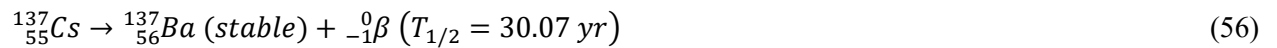


Figure 12. Possible daughter nuclei from the fission of uranium [19].

As an example, uranium can fission into ^{140}Xe and ^{94}Sr , and decay according to the following modes for ^{140}Xe (Equation (49) to Equation (52)) and for ^{94}Sr (Equation (53) and Equation (54)). $T_{1/2}$ is the half-life.



Alternately, uranium can fission into ^{137}Xe and ^{95}Sr , and decay according to the following modes for ^{137}Xe (Equation (55) and Equation (56)) and for ^{95}Sr (Equation (57) to Equation (60)).





Notice that the decay chains of ${}^{140}\text{Xe}$ and ${}^{94}\text{Sr}$ lead to stable Ce and Zr, while those of ${}^{137}\text{Xe}$ and ${}^{95}\text{Sr}$ lead to stable Ba and Mo. Each of these decay paths are occurring simultaneously along with all other possible fission product pairs and isotopes, and their respective decay paths. Computer modeling of these phenomena for a specific reactor application can be done using “burn-up code,” which predicts how the composition of the fuel changes over time, based on the various parameters associated with reactor operations.

In the case of an MSR operating with a fuel salt, the effect of fission processes is that the composition of the fuel salt is always changing. The composition never reaches equilibrium or steady state during operation. To complicate matters even more, the fuel salt is in contact with the construction materials used to contain the salt in the reactor system such as tanks, piping, pumps, heat exchangers, and the reactor core. In a broader sense, the materials of construction currently favored are primarily metal alloys and graphite due to their technological readiness level. The favored metal alloys for MSR applications are compositions of ferro-alloys and nickel-alloys based largely on transition metals Fe, Ni, Cr, Mo, Mn, and Co. Chemical interactions between a fuel salt and the construction materials are inevitable.

The complete description of fission processes is much more complex than those presented here. For example, certain isotopes can fission into three fragments (one typically being helium), and fertile isotopes can transmute into fissile daughters. Two important examples of the latter are the transmutations of ${}^{232}\text{Th}$ to ${}^{233}\text{U}$ (Equation (61) to Equation (63), and ${}^{238}\text{U}$ to ${}^{239}\text{Pu}$ (Equation (64) to Equation (66)), by neutron capture and decay.



The half-life of ${}^{233}\text{U}$ is 159,000 y.



The half-life of ${}^{239}\text{Pu}$ is 24,110 y.

TAKEAWAYS: Radiation effects can result in significant chemical changes to a salt system. The primary nuclear mechanisms responsible for changing the salt chemistry are fission, transmutation, and radioactive decay. A simple fuel salt system that begins with only a few metal halides, can change to a complex salt system with a large portion of the periodic table present as metal halides, and each metal can appear as one of several different isotopes. The composition of the salt system is not stable once removed from the reactor system due to the continuation of radioactive decay.

2.6. Dynamic Behaviors of Select Molten Salt Systems

Salt composition can change dynamically as a function of changes to the chemical potentials of the system resulting from internal processes and external influences. To illustrate the effects of compositional changes on salt chemistries, several examples are given in the following sections, and more in Appendix A. Each example begins with a defined chemical system, to which one or more components are added. These examples are meant only to illustrate different behaviors under different scenarios. These examples are greatly simplified and have no basis in actual performance.

Four examples follow in the body of the text. These include:

- Dynamic Fission of U to Ba and Mo: at the expense of U, Ba and Mo are added to a system containing NaCl and UCl_3 to show the effect on the chlorine chemical potential.
- Effects of Oxygen and Moisture: Phase stability diagrams are shown for the U, Cl, O and U, F, O systems.
- Chloride Salt Exposed to Oxygen and Moisture: $\text{O}_2(\text{g})$ and $\text{H}_2\text{O}(\text{g})$ are added to a system containing LiCl, KCl, and UCl_3 to show the effects on the chlorine and oxygen chemical potentials.
- Fluoride Salt Exposed to Oxygen and Moisture: $\text{O}_2(\text{g})$ and $\text{H}_2\text{O}(\text{g})$ are added to a system containing LiF, BeF_2 , ZrF_4 , and UF_4 to show the effects on the fluorine and oxygen chemical potentials.

The following examples are presented in Appendix A. Some of these examples may be conceptually simpler than those presented in the body of the text.

- Dynamic Behavior of the Cl, Cr, Fe, Ni System: $\text{Cl}_2(\text{g})$ is added to a system of Cr, Fe, and Ni to show the sequential formations of CrCl_2 , FeCl_2 , and NiCl_2 .
- Dynamic Behavior of the F, Cr, Fe, Ni System: $\text{F}_2(\text{g})$ is added to a system of Cr, Fe, and Ni to show the sequential formations of CrF_2 , FeF_2 , and NiF_2 .
- Dynamic Behavior of the Cl, U System: $\text{Cl}_2(\text{g})$ is added to a system containing U to show the sequential formations of UCl_3 and UCl_4 .
- Dynamic Behavior of the Cl, Na, U System: $\text{Cl}_2(\text{g})$ is added to a system containing Na and U to show the sequential formations of NaCl, UCl_3 , and UCl_4 .
- Dynamic Behavior of the F, U System: $\text{F}_2(\text{g})$ is added to a system containing U to show the sequential formations of UF_3 and UF_4 .
- Dynamic Behavior of the F, Li, Be, U, Zr System (Version 1): $\text{F}_2(\text{g})$ is added to a system containing Li, Be, U, and Zr to show the sequential formations of LiF, BeF_2 , UF_3 , ZrF_4 , and UF_4 .
- Dynamic Behavior of the F, Li, Be, U, Zr System (Version 2): Be is added to a system containing LiF, BeF_2 , UF_4 , and ZrF_4 to show the sequential reductions of UF_4 to UF_3 , and ZrF_4 to BeF_2 . In these examples, the chlorine chemical potential is related to the partial pressure of $\text{Cl}_2(\text{g})$, and the fluorine chemical potential to the partial pressure of $\text{F}_2(\text{g})$. These are valid thermodynamic constructs as derived in Sections 2.3 and 2.4.

2.6.1. Dynamic Fission of U to Ba and Mo

The purpose of this model is to show how the fission of uranium can result in dynamic chemical potentials and can involve the precipitation of molybdenum metal from the fuel salt. The example provided is greatly simplified to the point of being unrealistic in that it presents the fission of uranium (an actinide metal) into barium (an alkaline earth metal) and molybdenum (a transition metal), according to Equation (67). As described earlier, the fission process may involve multiple fission pathways that, in aggregate, produce a large array of fission product elements and isotopes. In this example, barium was chosen from Equation (56) and molybdenum from Equation (60) as a possible stable fission product pair produced after secondary nuclear reactions. Fission products from uranium are known to cause the formation of metallic transition metal fission products within molten salts, that can precipitate and/or chemically interact with the materials containing the molten salt, such as piping, valves, and pumps.



The model begins with an initial amount of NaCl (6.6 kmol), UCl_3 (1.7 kmol), and UCl_4 (1.7 kmol), in contact with $Ar(g)$ (1.0 kmol). Constant total pressure of 1 bar. The UCl_4 is incrementally and stoichiometrically converted to $BaCl_2$ and $MoCl_2$ to simulate fission. The model ends when all the initial UCl_4 is gone. The dynamic behavior of this simplified Cl, Na, U system at 500°C is shown in Figure 13.

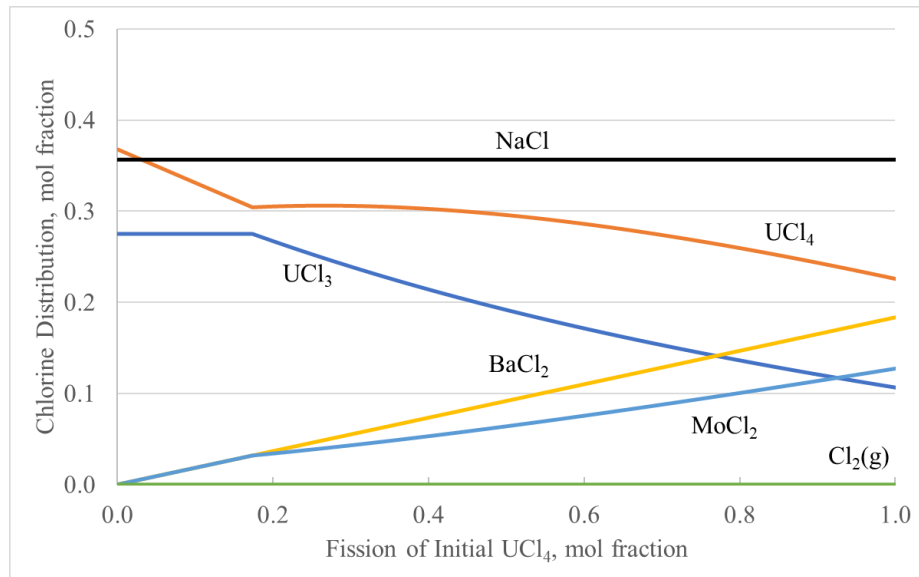


Figure 13. Chlorine distribution as a function of the fission of UCl_4 .

The redistribution of uranium between UCl_3 and UCl_4 as a function of the fission of the initial UCl_4 is shown in Figure 14. There is some change in the Cl chemical potential with progress of the fission process as shown in Figure 15. The primary factors affecting the Cl chemical potential are the production of $BaCl_2$, $MoCl_2$, and Mo in the system.

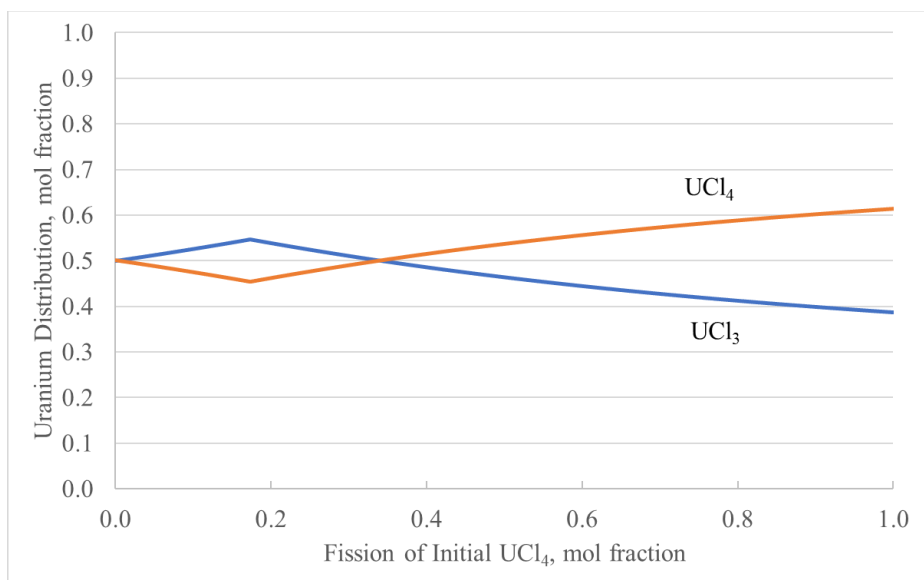


Figure 14. Uranium distribution as a function of the fission of UCl₄.

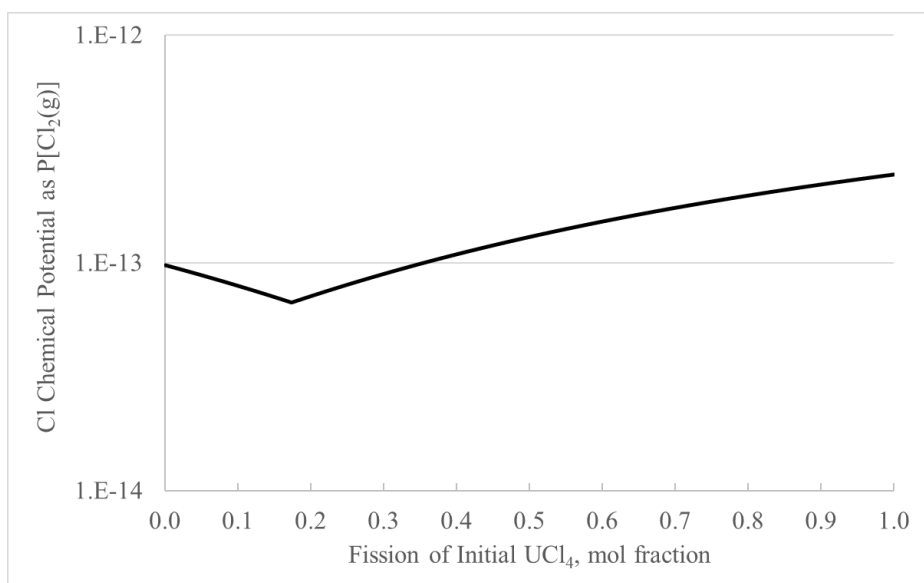


Figure 15. Chlorine chemical potential as a function of the fission of UCl₄.

The redistribution of molybdenum between MoCl₂ and Mo as a function of the fission of the initial UCl₄ is shown in Figure 16. Metallic Mo is formed when the MoCl₂ reacts with UCl₃ according to Equation (68).



Notice that MoCl₂ is the least stable metal chloride in Figure 16, meaning it is capable of chlorinating Ni, Fe, and Cr. In a chloride MSR application, the salt may be contained in a steel alloy. Formation of MoCl₂ could result in the degradation of the steel by converting the Ni, Fe, and Cr into metal chlorides, and by alloying of the Mo with the steel.

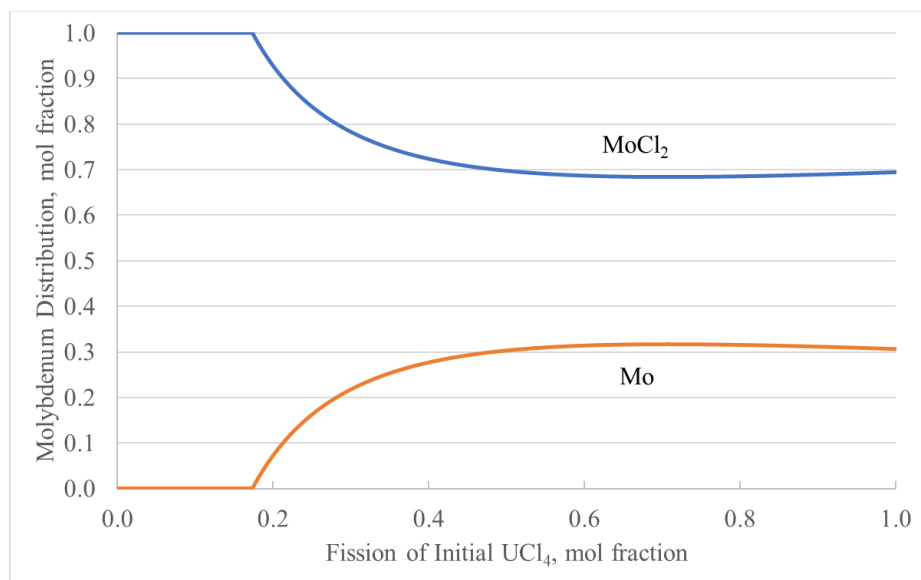


Figure 16. Molybdenum distribution as a function of the fission of UCl_4 .

TAKEAWAYS: Fission products include transition metals. As the transition metals appear as metal cations in the halide salt, the metal cations can be reduced to a metallic state by chemical interactions with other metals that are more electropositive. As transition metal fission products are formed, some of them may alloy with the materials of construction and cause degradation of components. During the Oak Ridge National Laboratory, Molten Salt Reactor Experiment, transition metal fission products that were particularly prone to reduction and plating onto the Hastelloy-N components were Nb, Mo, Tc, Ru, Rh, Pd, Ag, Sb, and Te.

2.6.2. Effects of Oxygen and Moisture

The equilibrium phase stability diagram for the uranium, chlorine, and oxygen system at 500°C is shown in Figure 17. This diagram is based on the selection of three elements (U, Cl, and O), and the resultant stable compounds (U, UCl_3 , UCl_4 , UO_2 , U_3O_8 , and UOCl_2). The partial pressure of $\text{O}_2(\text{g})$ was selected to represent the O chemical potential along the ordinate, and the partial pressure of $\text{Cl}_2(\text{g})$ was selected to represent the Cl chemical potential along the abscissa. The O and Cl chemical potentials increase as the partial pressures of $\text{O}_2(\text{g})$ and $\text{Cl}_2(\text{g})$ increase, respectively. The pressure scales were limited to 1 bar on the high sides and encompassed the uranium metal stability domain on the low sides. Atmospheric pressure is approximately 1 bar. This high side limit was reasonable because MSR salt systems are not expected to be operated at pressures much above 1 bar.

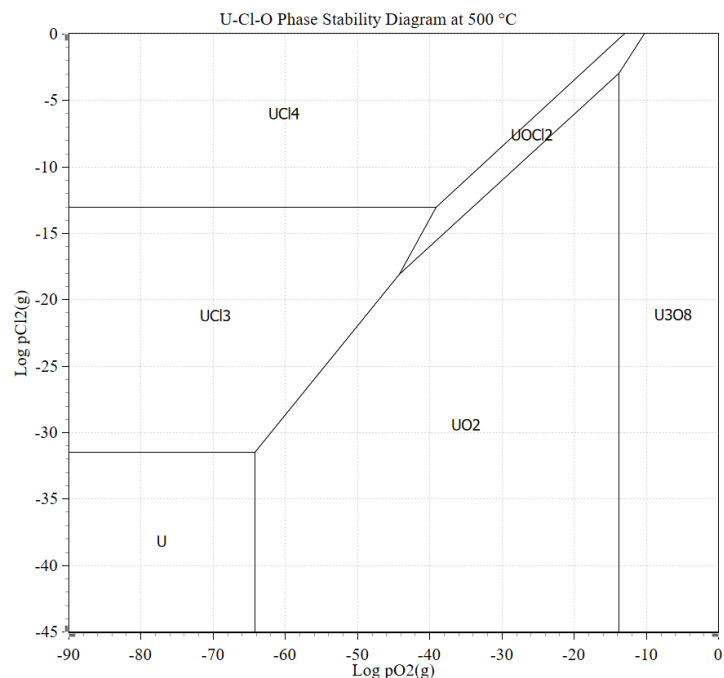


Figure 17. Equilibrium phase stability diagram for the U, Cl, O system at 500°C.

Similarly, the equilibrium phase stability diagram for the uranium, fluorine, and oxygen system at 500°C is shown in Figure 18. This diagram is based on the selection of three elements (U, F, and O), and the resultant stable compounds (U, UF₃, UF₄, UF₆, UO₂, U₃O₈, and UO₂F₂). The partial pressure of O₂(g) was selected to represent the O chemical potential along the ordinate, and the partial pressure of F₂(g) was selected to represent the F chemical potential along the abscissa. The O and F chemical potentials increase as the partial pressures of O₂(g) and F₂(g) increase, respectively. The pressure scales were limited to 1 bar on the high sides and encompassed the uranium metal stability domain on the low sides.

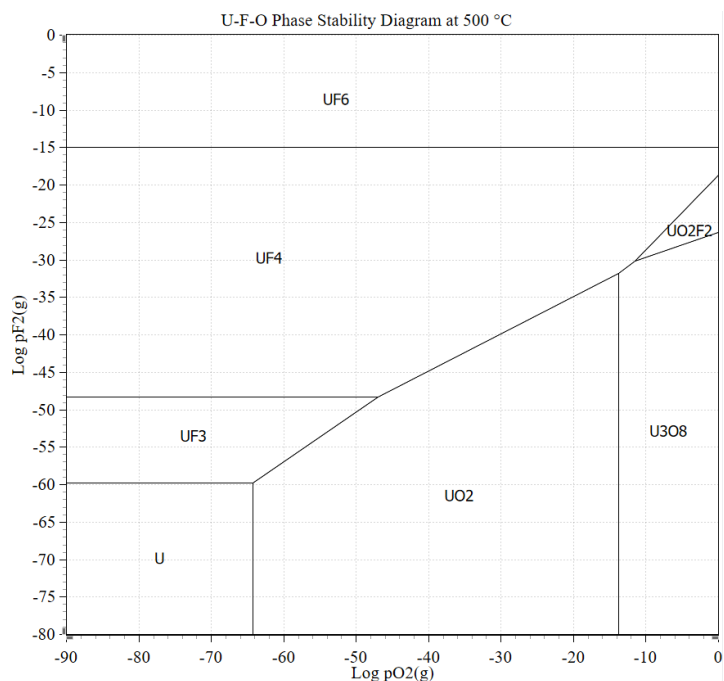


Figure 18. Equilibrium phase stability diagram for the U, F, O system at 500°C.

Figure 17 and Figure 18 show that the formation of the oxychloride, UOCl_2 , requires both O and Cl chemical potentials, and the formation of the oxyfluoride, UO_2F_2 , requires both O and F chemical potentials. However, as before, they give no information about the compositions of the equilibrium gas phase above the condensed stability domains.

The ingress of oxygen and/or moisture into a molten salt introduces an O chemical potential that then competes with either the Cl chemical potential (in a chloride salt) or the F chemical potential (in a fluoride salt). Generally, introduction of oxygen will form oxides, oxychlorides, or oxyfluorides in the salt; and introduction of moisture will form hydroxides in the salt or be liberated as $\text{H}_2(\text{g})$, $\text{HCl}(\text{g})$, or $\text{HF}(\text{g})$.

TAKEAWAYS: The exposure of a molten salt to oxygen and moisture can alter the chemical composition of the salt. In a chloride salt, the oxygen can alter the chlorine chemical potential and allow oxychloride species to form. In a fluoride salt, the oxygen can alter the fluorine chemical potential and allow oxyfluorides to form. Once contaminated, recovery of the salt chemistry may be challenging because simple heating under a vacuum may be insufficient to expel the impurities. It may be necessary to provide chlorine or fluorine chemical potentials to purify the salts.

2.6.3. Chloride salt exposed to oxygen and moisture

The following model illustrates the behavior of a chloride salt exposed to incremental additions of oxygen and moisture. The salt consists of LiCl (5.82 kmol) and KCl (4.18 kmol) as the LiCl - KCl eutectic composition with UCl_3 (0.5 kmol) and U (0.1 kmol). Constant total pressure is 1 bar. As a representation of humid air, every mole of $\text{O}_2(\text{g})$ introduced to the system is accompanied by 0.2 mole of $\text{H}_2\text{O}(\text{g})$ in $\text{Ar}(\text{g})$ (1 kmol). $\text{O}_2(\text{g})$ was added up to 0.5 kmol, and $\text{H}_2\text{O}(\text{g})$ up to 0.1 kmol. The gas phase species included $\text{Ar}(\text{g})$, $\text{O}_2(\text{g})$, $\text{H}_2(\text{g})$, $\text{H}_2\text{O}(\text{g})$, $\text{Cl}_2(\text{g})$, and $\text{HCl}(\text{g})$. The salt phase included LiCl , KCl , UCl_3 , and UCl_4 . The oxide phase included UO_2 and UOCl_2 .

The distribution of uranium between the uranium-containing species as a function of the $\text{O}_2(\text{g})/\text{H}_2\text{O}(\text{g})$ addition is shown in Figure 19. Notice that U metal is the first to be consumed, followed by UCl_3 and UCl_4 . The distribution of oxygen between the oxygen-containing species as a function of the $\text{O}_2(\text{g})/\text{H}_2\text{O}(\text{g})$ addition is shown in Figure 20. The distributions to $\text{H}_2\text{O}(\text{g})$ and $\text{O}_2(\text{g})$ are minimal. The composition of the gas phase as a function of the $\text{O}_2(\text{g})/\text{H}_2\text{O}(\text{g})$ addition is shown in Figure 21. Three regions of buffering (plateau-like regions) are seen in this figure. The first is while U remains, the second is while UCl_3 remains, and the third is while UCl_4 remains.

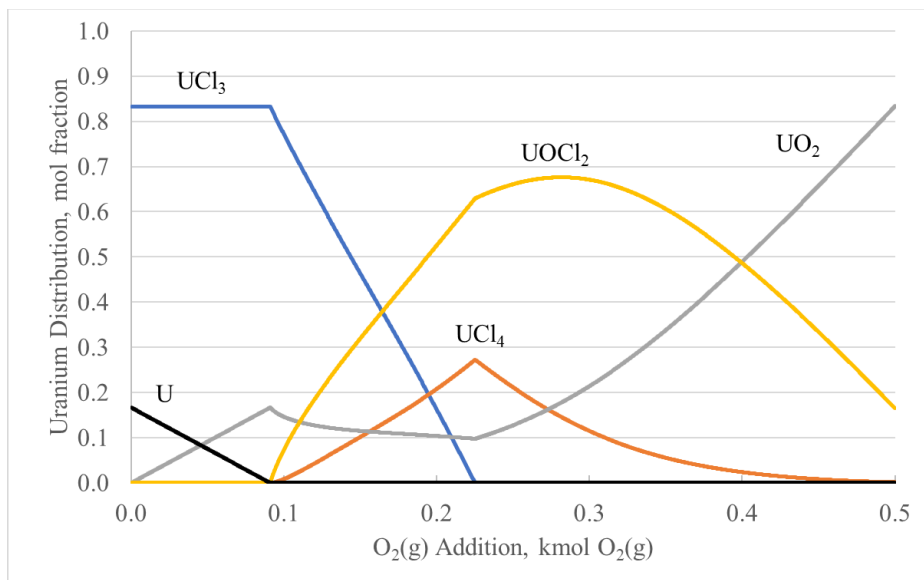


Figure 19. Uranium distribution as a function of the $O_2(g)/H_2O(g)$ addition.

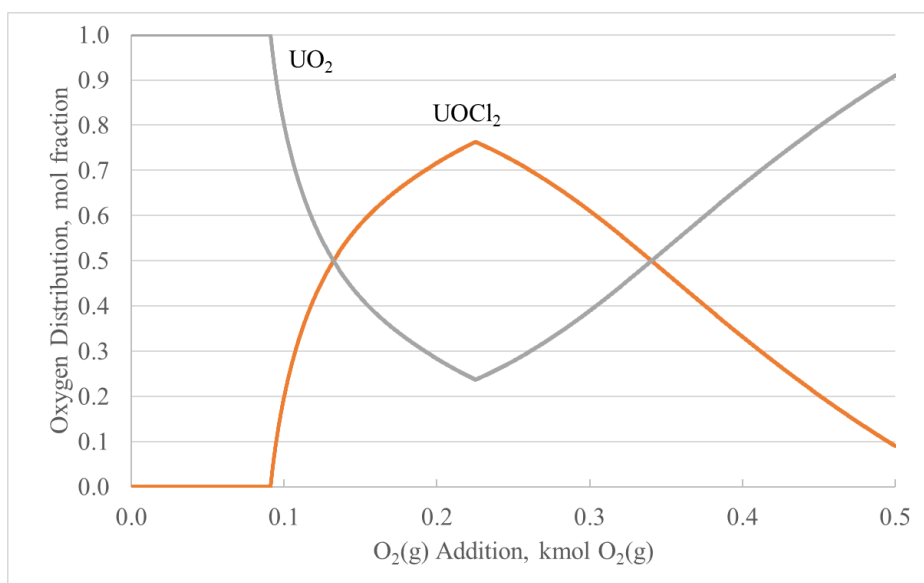


Figure 20. Oxygen distribution as a function of the $O_2(g)/H_2O(g)$ addition.

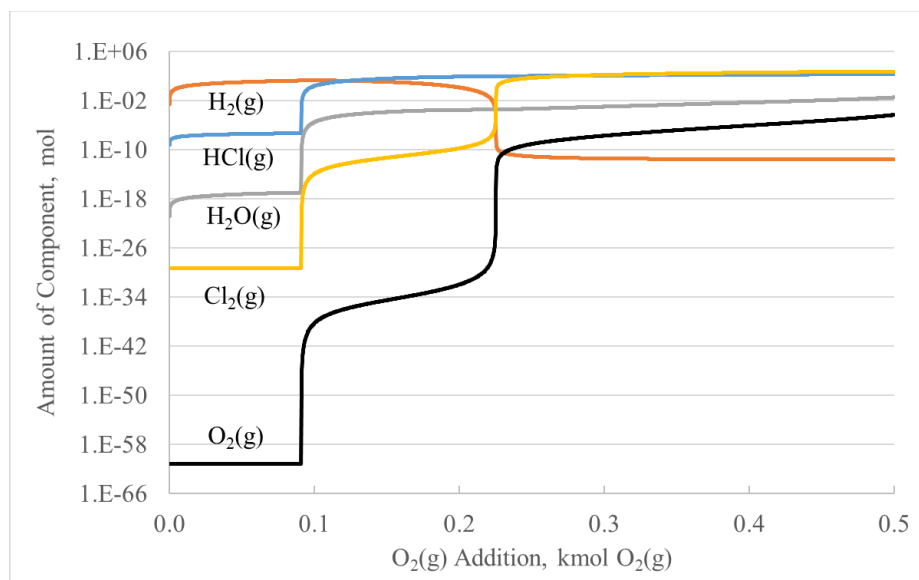


Figure 21. Equilibrium composition of the gas phase as a function of the $\text{O}_2(\text{g})/\text{H}_2\text{O}(\text{g})$ addition.

The Cl and O chemical potentials as a function of the $\text{O}_2(\text{g})/\text{H}_2\text{O}(\text{g})$ addition are shown in Figure 22. In this example, the Cl and O chemical potentials increase as $\text{O}_2(\text{g})$ and $\text{H}_2\text{O}(\text{g})$ are introduced into the system. The three buffered plateaus are related to the U/UCl_3 , $\text{UCl}_3/\text{UCl}_4$, and $\text{UCl}_4/\text{UOCl}_2/\text{UO}_2$ phase equilibria. Essentially, chlorine is chemically liberated as uranium is oxidized.

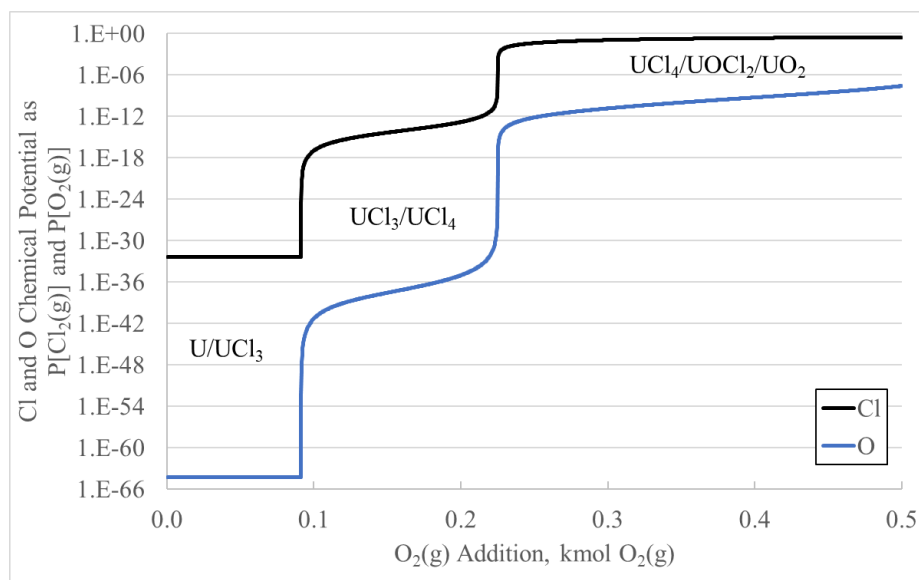


Figure 22. Cl and O Chemical Potentials as a function of the $\text{O}_2(\text{g})/\text{H}_2\text{O}(\text{g})$ addition. Unit of pressure: bar

TAKEAWAYS: The addition of oxygen and moisture into a uranium-containing chloride salt resulted in increasing both the chlorine and oxygen chemical potentials. These effects would cause the salt to become more corrosive to materials of construction. The salt would be made less corrosive by minimizing the oxygen and chlorine chemical potentials.

2.6.4. Fluoride salt exposed to oxygen and moisture

The following model illustrates the behavior of a fluoride salt exposed to incremental additions of oxygen and moisture. The salt consists of LiF (6.5 kmol), BeF₂ (2.92 kmol), ZrF₄ (0.5 kmol), and UF₄ (0.08 kmol). Constant total pressure of 1 bar. As a representation of humid air, every mole of O₂(g) introduced to the system is accompanied by 0.2 mole of H₂O(g) in Ar(g) (1 kmol). O₂(g) was added up to 0.5 kmol, and H₂O(g) up to 0.1 kmol. The gas phase species included Ar(g), O₂(g), H₂(g), H₂O(g), F₂(g), and HF(g). The salt phase included LiF, BeF₂, ZrF₄, UF₃, and UF₄. The oxide phase included ZrO₂, UO₂, and UO₂F₂.

The distribution of uranium between the uranium-containing species as a function of the O₂(g)/H₂O(g) addition is shown in Figure 23. Notice the rapid development of UO₂F₂. The most significant effect of introducing oxygen and moisture was the conversion of UF₄ to UO₂F₂.

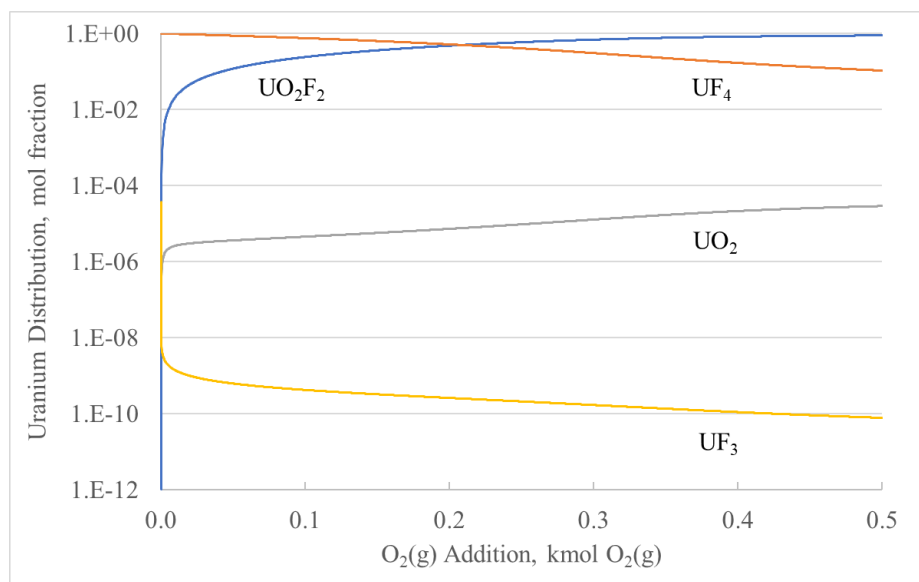


Figure 23. Uranium distribution as a function of the O₂(g)/H₂O(g) addition.

The distribution of zirconium between the zirconium-containing species as a function of the O₂(g)/H₂O(g) addition is shown in Figure 24. There was little effect on the distribution of zirconium between ZrF₄ and ZrO₂.

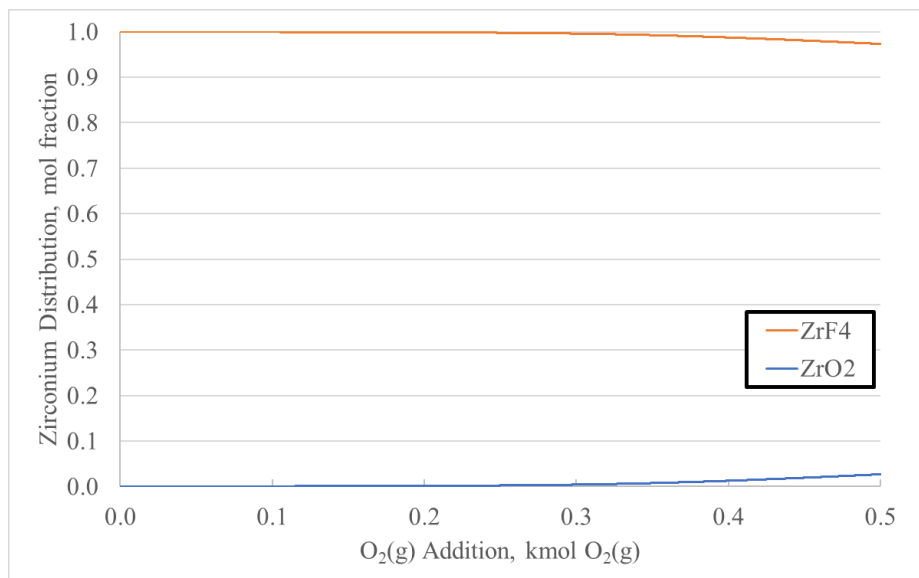


Figure 24. Zirconium distribution as a function of the O₂(g)/H₂O(g) addition.

The distribution of oxygen between the oxygen-containing species as a function of the O₂(g)/H₂O(g) addition is shown in Figure 25.

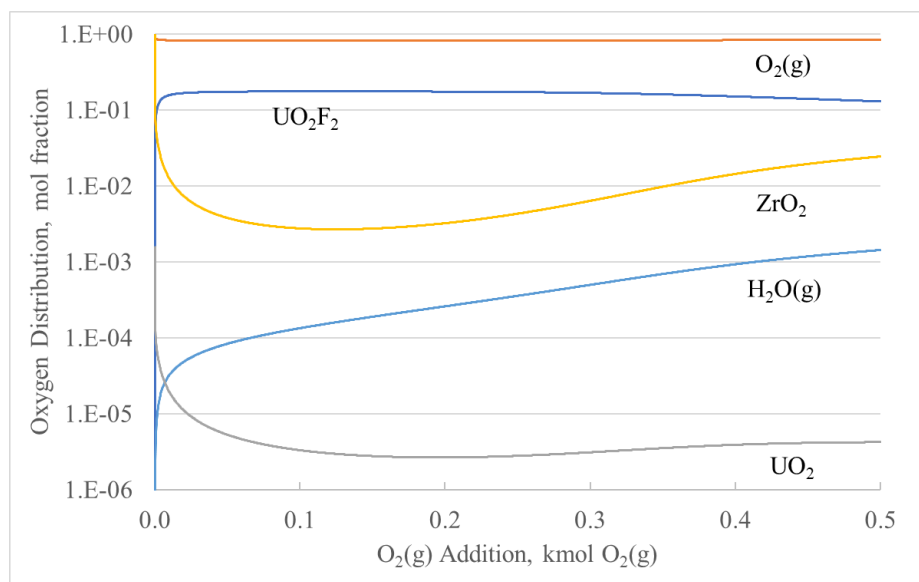
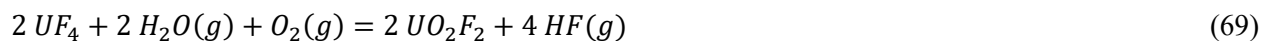


Figure 25. Oxygen distribution as a function of the O₂(g)/H₂O(g) addition.

The composition of the gas phase as a function of the O₂(g)/H₂O(g) addition is shown in Figure 26. The HF(g) is formed from the hydrolysis of the UF₄. A possible reaction mechanism is shown in Equation (69).



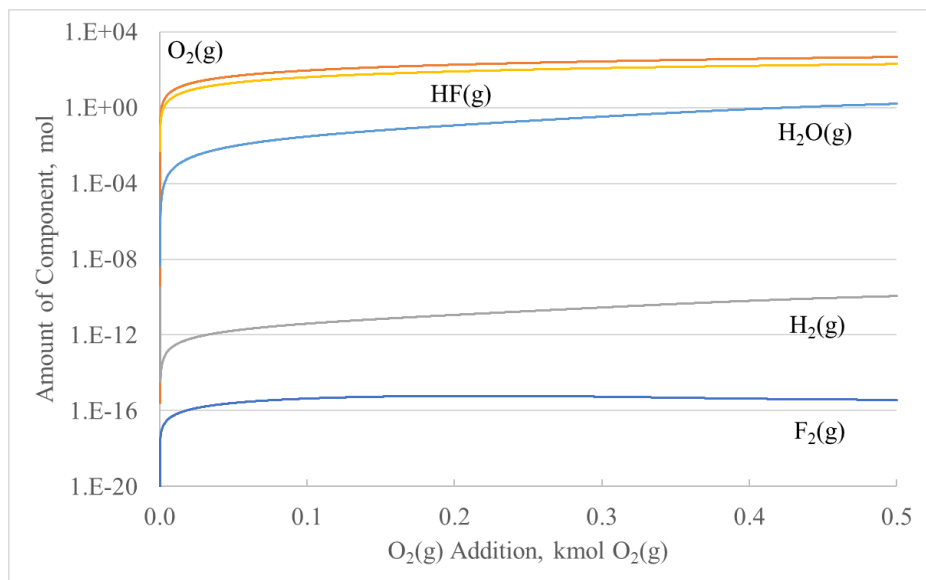


Figure 26. Equilibrium composition of the gas phase as a function of the $\text{O}_2(\text{g})/\text{H}_2\text{O}(\text{g})$ addition.

The F and O chemical potentials as a function of the $\text{O}_2(\text{g})/\text{H}_2\text{O}(\text{g})$ addition are shown in Figure 27.

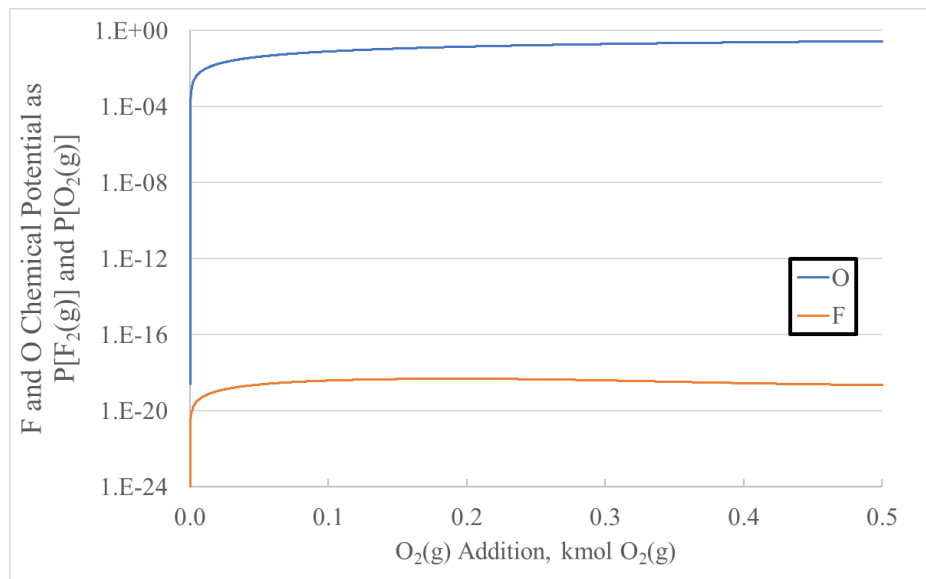


Figure 27. F and O Chemical Potentials as a function of the $\text{O}_2(\text{g})/\text{H}_2\text{O}(\text{g})$ addition. Unit of pressure: bar

TAKEAWAYS: The addition of oxygen and moisture into a uranium-containing fluoride salt resulted in the formation of oxyfluoride (UO_2F_2) and the evolution of hydrogen fluoride ($\text{HF}(\text{g})$). Although there was little effect on the oxygen and fluorine chemical potentials of the molten salt, $\text{HF}(\text{g})$ is potentially very reactive to the alloys in contact with the molten salt.

2.7. Thermodynamic Stability: Littlewood Diagrams

A Pourbaix diagram is a type of predominance area diagram used in aqueous chemistry. Most commonly the axes are pH (a function of the chemical activity of the hydrogen cation) and E_H (potential relative to the standard hydrogen electrode). A Littlewood diagram is an analogous approach used to construct a predominance area diagram of a molten salt system. In this example the two axes are a measure of chemical activity (pO^{2-}) and a measure of potential (potential E relative to the standard chlorine electrode [SCE]). In Littlewood diagrams, the specific measures used are selected to illustrate relevant information for a specific molten salt system under consideration.

This section shows the derivation of a simple Littlewood diagram for a metal, chlorine, oxygen solute in a LiCl-KCl eutectic solvent, as shown in Figure 28. There are five stability domains and eight phase boundaries in the diagram labeled “a” to “h.” These phase boundaries are expressed by the following equations.



Notice that the phase boundaries depicted by Equation (70) to Equation (74) have positive slopes (as they involve e^- and O^{2-}), while those depicted by Equation (75) is horizontal (as it involves only e^-), and those depicted by Equation (76) and Equation (77) are vertical (as they involve only O^{2-}).

There are three additional horizontal lines in Figure 28 (lines “i,” “j,” and “k”). These represent the stability of the solvent salt and are also reduction reactions involving only e^- . Line “i” is the reduction of lithium metal, line “j” is the reduction of potassium metal, and line “k” is the oxidation of chlorine according to the following equations.



Equation (70) to Equation (79) are chemical reactions with unique solutions that rely on the availability of thermodynamic data. The points of intersection are determined from the mathematics of two or more intersecting lines.

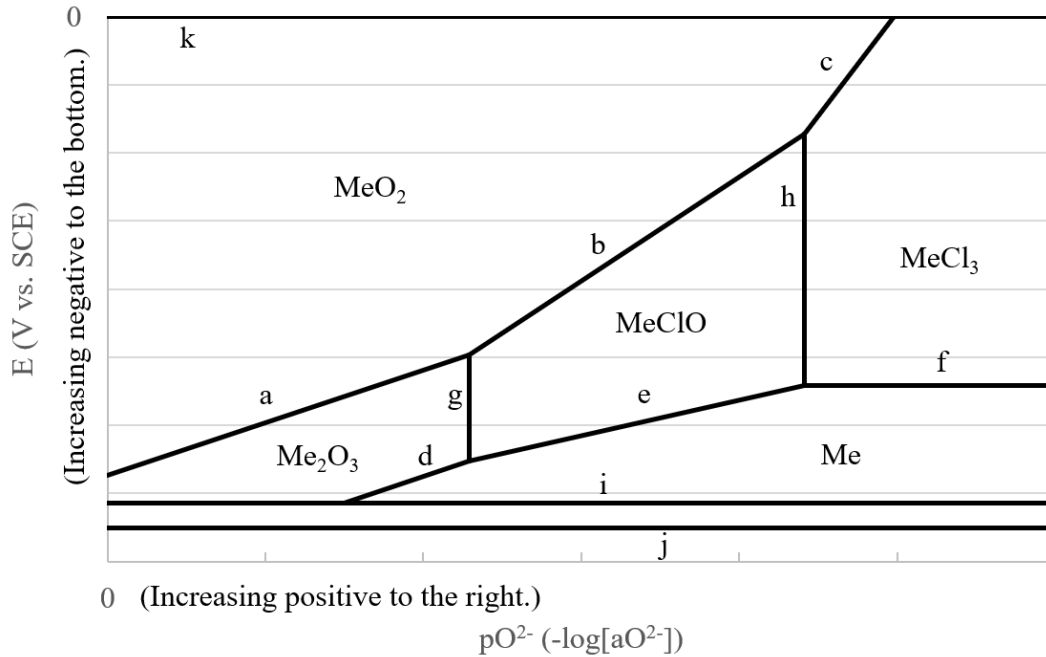


Figure 28. Generic predominance area diagram for a metal, chlorine, oxygen solute in LiCl-KCl eutectic salt solvent at some temperature at which the solvent salt is molten.

Consider the boundary between MeO_2 and MeCl_3 as expressed in Equation (72) and illustrated by line “c” in Figure 28. The thermodynamic expressions leading to the Nernst Equation are shown in Equation (81) to Equation (86), where ΔG is the Gibbs free energy change, ΔG^0 is the standard state Gibbs free energy change, R is the gas constant, T is temperature, Q is the reaction quotient, K_{eq} is the equilibrium reaction quotient, E is the potential relative to the SCE, E^0 is the standard potential, n is the number of electrons transferred, and F is the Faraday constant.

$$\Delta G = \Delta G^0 + RT \ln(Q) \quad (81)$$

$$\Delta G^0 = -RT \ln(K_{eq}) \quad (82)$$

$$\Delta G^0 = -nFE^0 \quad (83)$$

$$E^0 = -\frac{\Delta G^0}{nF} \quad (84)$$

$$E = E^0 - \frac{RT}{nF} \ln Q \quad (85)$$

$$E = -\frac{\Delta G^0}{nF} - \frac{RT}{nF} \ln Q \quad (86)$$

Equation (72) and Equation (86) are combined to give Equation (87), where a is the chemical activity of the species indicated.

$$E = -\frac{\Delta G^0}{F} - \frac{RT}{F} \ln \left(\frac{a_{\text{MeCl}_3} a_{\text{O}^{2-}}^2}{a_{\text{MeO}_2} a_{\text{Cl}^-}^3} \right) \quad (87)$$

Equation (87) is rearranged to give Equation (88).

$$E = -\frac{\Delta G^0}{F} - \frac{RT}{F} \left(\ln \left(\frac{a_{\text{MeCl}_3}}{a_{\text{MeO}_2}} \right) + \ln \left(\frac{a_{\text{O}^{2-}}^2}{a_{\text{Cl}^-}^3} \right) \right) \quad (88)$$

Equation (88) is rearranged to give Equation (89).

$$E = -\frac{\Delta G^0}{F} - \frac{RT}{F} \ln \left(\frac{a_{MCl_3}}{a_{MO_2}} \right) - \frac{RT}{F} \ln \left(\frac{a_{O^{2-}}^2}{a_{Cl^-}^3} \right) \quad (89)$$

The expression can be simplified further, for examining various scenarios, if certain species concentrations are kept constant. Equation (90) fixes chemical potential of the chlorine anion to one.

$$a_{Cl^-} = 1 \quad (90)$$

Equation (89) and Equation (90) are combined to give Equation (91).

$$E = -\frac{\Delta G^0}{F} - \frac{RT}{F} \ln \left(\frac{a_{MCl_3}}{a_{MO_2}} \right) - \frac{RT}{F} \ln(a_{O^{2-}}^2) \quad (91)$$

Equation (91) is rearranged to give Equation (92).

$$E = -\frac{\Delta G^0}{F} - \frac{RT}{F} \ln \left(\frac{a_{MCl_3}}{a_{MO_2}} \right) - \frac{2RT}{F} \ln(a_{O^{2-}}) \quad (92)$$

Equation (93) is a logarithmic identity.

$$\ln(b) = \ln(10) \cdot \log(b) \quad (93)$$

The logarithmic identity in Equation (93) is applied to Equation (92) to give Equation (94).

$$E = -\frac{\Delta G^0}{F} - \frac{RT}{F} \ln \left(\frac{a_{MCl_3}}{a_{MO_2}} \right) - \frac{2RT \ln(10)}{F} \log(a_{O^{2-}}) \quad (94)$$

The expression pO^{2-} is defined in Equation (95). It is analogous to the expression pH, which is similarly related to the activity of the hydrogen anion in aqueous systems.

$$pO^{2-} = -\log(a_{O^{2-}}) \quad (95)$$

The expression in Equation (95) is applied to Equation (94) to give Equation (96).

$$E = -\frac{\Delta G^0}{F} - \frac{RT}{F} \ln \left(\frac{a_{MCl_3}}{a_{MO_2}} \right) + \frac{2RT \ln(10)}{F} pO^{2-} \quad (96)$$

Once values are assigned to the constants and parameters in Equation (96), then the expression takes the following form.

$$E = f(pO^{2-}) = b + m(pO^{2-}) \quad (97)$$

Where the intercept b has the following form.

$$b = -\frac{\Delta G^0}{F} - \frac{RT}{F} \ln \left(\frac{a_{MCl_3}}{a_{MO_2}} \right) \quad (98)$$

And the slope m has the following form.

$$m = \frac{2RT \ln(10)}{F} \quad (99)$$

Similar paths can be used to derive expressions for the other phase boundary equations in Equation (70) to Equation (80). Small steps were shown from Equation (87) to Equation (96) to clearly present logical progression of the derivation.

In Figure 28, the chemical potential of oxygen is expressed along the abscissa as the activity of the oxygen anion, and it decreases from left to right. The chemical potential of chlorine is expressed along the ordinate as the Nernst potential relative to the SCE, and it decreases from top to bottom.

The relationship between oxygen gas and oxygen anions is expressed in Equation (100), which, after following a similar derivation as before, leads to the thermodynamic expression in Equation (106). Equation (106) can be used to plot oxygen gas isobars on Figure 28.

$$O_2(g) + 4 e^- = 2 O^{2-} \quad (100)$$

Equation (100) and Equation (86) are combined to give Equation (101), which is rearranged in a series of steps to give Equation (106). For convenience, the terms for the standard pressures have been incorporated into the pressure terms in the following equations, with this incorporation designated by an asterisk, such that $P^*_{O_2(g)} = P_{O_2(g)}/P_{O_2(g)}^0$. Accordingly, the natural log terms remain unitless.

$$E = -\frac{\Delta G^0}{4F} - \frac{RT}{4F} \ln \left(\frac{a_{O^{2-}}^2}{P^*_{O_2(g)}} \right) \quad (101)$$

$$E = -\frac{\Delta G^0}{4F} - \frac{RT}{4F} (\ln(a_{O^{2-}}^2) - \ln(P^*_{O_2(g)})) \quad (102)$$

$$E = -\frac{\Delta G^0}{4F} - \frac{RT}{4F} \ln(a_{O^{2-}}^2) + \frac{RT}{4F} \ln(P^*_{O_2(g)}) \quad (103)$$

$$E = -\frac{\Delta G^0}{4F} - \frac{RT}{2F} \ln(a_{O^{2-}}) + \frac{RT}{4F} \ln(P^*_{O_2(g)}) \quad (104)$$

$$E = -\frac{\Delta G^0}{4F} - \frac{RT \ln(10)}{2F} \log(a_{O^{2-}}) + \frac{RT}{4F} \ln(P^*_{O_2(g)}) \quad (105)$$

$$E = -\frac{\Delta G^0}{4F} + \frac{RT \ln(10)}{2F} pO^{2-} + \frac{RT}{4F} \ln(P^*_{O_2(g)}) \quad (106)$$

The expression between chlorine gas and chlorine anions is expressed in Equation (107), which, after following a similar derivation as before, leads to the thermodynamic expression in Equation (111). Equation (111) can be used to plot chlorine gas isobars on Figure 28.

$$Cl_2(g) + 2 e^- = 2 Cl^- \quad (107)$$

Equation (107) and Equation (86) are combined to give Equation (108).

$$E = -\frac{\Delta G^0}{2F} - \frac{RT}{2F} \ln \left(\frac{a_{Cl^-}^2}{P^*_{Cl_2(g)}} \right) \quad (108)$$

Because Equation (107) is being treated as the SCE, the Standard State Gibbs free energy is zero as indicated in Equation (109).

$$\Delta G^0 = 0 \quad (109)$$

Equation (90) and Equation (109) are combined to give Equation (110).

$$E = -\frac{RT}{2F} \ln \left(\frac{1}{P^*_{Cl_2(g)}} \right) \quad (110)$$

Equation (110) is rearranged to give Equation (111).

$$E = \frac{RT}{2F} \ln(P^*_{Cl_2(g)}) \quad (111)$$

TAKEAWAYS: Many variants of Littlewood diagrams are possible that depend on the salt system, species of interest (both condensed and gaseous), and the availability of thermodynamic data. In many cases, thermodynamic data that was developed for pure species as solids or liquids are being applied to the interpretation of molten salt systems comprised of mixed salts. The individual salt species are regarded as a solvent or a solute as an approximation when rigorous mixture information is not available. Also in many cases, the thermodynamics of the solvent salt are completely neglected during approximations, as the solvent salt is merely considered the medium in which the solute species interact.

2.8. Thermodynamic Stability: Lewis Acid Analogy

In the Arrhenius theory of acids and bases, an acid increases the hydrogen cation concentration in an aqueous solution, and a base increases the hydroxide anion concentration in an aqueous solution. This theory is intuitive with strong acids such as hydrochloric acid (HCl) and strong bases such as sodium hydroxide (NaOH) that dissociate to produce those ions.

In the Brønsted-Lowry theory of acids and bases, an acid is a proton donor, and a base is a proton acceptor. In this context, a “proton” is a hydrogen cation. The Brønsted-Lowry theory is consistent with the Arrhenius theory and expands the concept of acids and bases to nonaqueous systems.

In the Lewis theory of acids and bases, an acid is an electron pair acceptor, and a base is an electron pair donor. The Lewis theory is consistent with the Arrhenius and Brønsted-Lowry theories and expands the concept of acids and bases to systems not involving hydrogen. It is the Lewis theory that can be applied to the interpretation of halide molten salt systems. The term “electron pair” is often used to describe the Lewis theory, but when applied to molten salts, single or multiple electrons can be involved.

Consider the following equation involving equilibrium within a halide molten salt.



Here Me is a metal cation, X is a halide anion, q is the oxidation state of the metal cation, $q + r$ is the coordination number of the metal cation, and r is the net charge of the complex anion. The species MeX_q is a metal halide solute species dissolved in a metal halide solvent salt, rX^- are the free halide anions provided by metal halide solvent salt, and MeX_{q+r}^{r-} is the resultant complex metal halide anion. Equation (112) has the following form:



Metal cations that are likely to accept a greater number of halide anions are more acidic; these include transition, lanthanide, and actinide metals. Metals that are likely to accept a lesser number of halide anions are less acidic; these include the alkali and alkali earth metals. However, many metal halides may act as an acid in one set of circumstances and as a base in another set of circumstances.

A halide salt is acidic when it has a deficiency (low chemical potential) of free halide anions, and basic when it has an excess (high chemical potential) of free halide anions. Chloroacidity (pCl^-) is defined by Equation (114) for a chloride molten salt, and fluoroacidity (pF^-) is defined by Equation (115) for a fluoride molten salt.

$$pCl^- = -\log(a_{Cl^-}) \quad (114)$$

$$pF^- = -\log(a_{F^-}) \quad (115)$$

These concepts of acidity are directly related to the chemical activity of the halide. For example, as the concentration and chemical activity of the chlorine anion decreases, the measure of pCl^- increases, and the salt is less acidic and less corrosive. This is analogous to how, in aqueous chemistry, a low pH (corresponding to high hydrogen cation concentration) is acidic, and a high pH (corresponding to high hydroxide anion concentration) is basic.

Bulk charge neutrality is a constraint common to every halide molten salt system. There can be no bulk net charge within a molten salt system. The bulk net valence charge of the anions must balance against the bulk net valence charge of the cations. The term “free halide anions” refers not to an excess of anions leading to a net negative charge within the bulk molten salt, but instead to halide anions that are less constrained by association to cations and are thus freer to migrate through the molten salt. By such mechanisms, acidity results in structural effects within the molten salt that can affect physical properties such as density and viscosity.

TAKEAWAYS: The Lewis acid analogy considers the state of the halide anions in the salt. The interactions of the halide anions with the metal cations influences not only the salt chemistry, but also some of the intrinsic physical properties of the bulk salt. Salt chemistry is more complex than a simple mixture of individual metal cations and individual halide anions that are loosely associated by the need for charge neutrality in the bulk salt. Complex ions comprised of groupings of metals and halides and oxides are present, and in many cases, accurate thermodynamic values are not currently known.

2.9. Limitations of Equilibrium Thermodynamic Calculations

The examples presented in this section were meant to illustrate various chemical behaviors encountered in molten halide salt systems. All calculations were based on equilibrium thermodynamic data with many simplifying assumptions. The examples were not intended to be rigorously illustrative of real chemical systems.

One description of chemical equilibrium is a system in which there are no longer any driving forces for bulk change. A system that is in a transient state is not at equilibrium, so it will not have compositions described by equilibrium calculations. Rate of change toward equilibrium is a kinetic consideration and can be fast or slow, and it can be so slow that for all practical purposes the system is not expected to reach equilibrium. However, lack of change does not mean there are no longer any driving forces for change, and lack of change does not mean the system is at equilibrium. A system could thus appear to have a constant set of concentrations that do not match equilibrium concentrations.

One description of steady state is a dynamic system in which there are no longer any spatial bulk changes. As any location in the system accommodates constant mass and energy flow, the conditions at that location are stable with respect to parameters such as temperature, pressure, and composition. Therefore, a system at chemical equilibrium is also at steady state, but a system at steady state is not necessarily at chemical equilibrium. For example, a system defined as fluid being mixed within a tank can be at steady state, but it is not a system at chemical equilibrium. At the very least, the act of mixing is imparting the fluid with heat, which dissipates into the surroundings through the walls of the tank. Likewise, an MSR operating at steady power may be considered to be at steady state (at least in the short term), but the system is far from equilibrium.

Mathematically describing equilibrium thermodynamics relies on the establishment of an internally consistent set of standard states for all species involved, and then accounting for the energetic potential differences as the system is shifted away from the standard state conditions of those species. Temperature is not part of the definition of the standard state (that is, for any given temperature, the standard state of a substance will have a specific energy associated with it at that temperature). The standard state of a substance is often defined as the pure substance at 1 bar pressure and in its natural phase (e.g., solid, liquid, gas), and the states of the species and their potential differences from their standard states are calculated at the temperature of interest. All chemical thermodynamics serve as a means of describing and predicting the behaviors of real chemical systems to varying degrees. Calculations based on equilibrium thermodynamic data are often highly idealized and are, in some cases, good approximations of real systems, and in some cases not.

Real MSR salt systems have chemistries that are difficult to characterize because they (1) are in contact with and may react with materials of construction such as metal alloys, ceramics, and graphite, (2) have headspaces that may contain reactive gases such as air and moisture, and (3) may be under the constant influence of fission and decay processes (particularly in the case of fuel salts). Furthermore, study of thermodynamic properties of ionic species in molten salts does not have complete quantitative consensus, both regarding theoretical interpretations and experimental methods of measurement.

There can be uncertainty about the accuracy of the thermodynamic data. The published thermodynamic data are derived from several sources, including data derived from direct experimental measurements, data extrapolated beyond experimental measurements, and data estimated by empirical and fundamental methods. The level of confidence in the accuracy of thermodynamic data is based on expert technical assessment of the source of that data, and the estimated uncertainty values may vary between individual assessments.

The chemical behaviors of real systems are complicated by many phenomena, which include thermodynamics, kinetics, passivation, diffusion, thermal gradients, pressure gradients, corrosion mechanisms, cavitation, materials strain, radiation effects, and so on. Thus, performance predictions need to consider more than solely equilibrium thermodynamic considerations, though equilibrium thermodynamic considerations are an important factor in performance predictions.

TAKEAWAYS: Predictive modeling of molten salt systems draws upon a wide range of scientific and engineering experience from disciplines such as chemistry, thermodynamics, kinetics, fluid mechanics, heat transfer, and neutronics. As thorough as the modeling effort may be, real molten salt systems can behave in ways that are extremely complex and unique to a particular application. This attribute of real molten salt systems can lead to a misalignment between predicted behaviors and observed behaviors. Predictive modeling provides powerful tools that support the design and operation efforts, but may not be a replacement for process demonstrations.

3. MOLTEN SALT REACTOR SYSTEMS

Three basic types of MSRs are shown below. Figure 29 represents a thermal-spectrum graphite-moderated MSR with a fuel salt and coolant salt. The fuel salt flows through the reactor core and a heat exchanger, and the coolant salt does not flow through the reactor core but only through heat exchangers. Figure 29 is representative of the ORNL MSRE. The fissile inventory of this type of fuel salt tends to be lower than that of a fast-spectrum type, for the same power output.

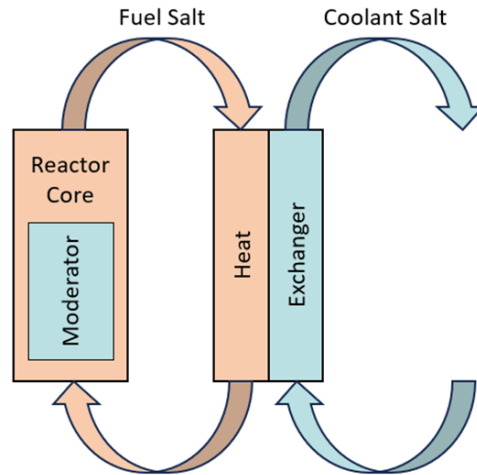


Figure 29. Thermal-spectrum MSR with moderated core and fuel salt.

Figure 30 represents a fast-spectrum MSR with a fuel salt and coolant salt. As before, the fuel salt flows through the reactor core and a heat exchanger. The coolant salt does not flow through the reactor core but only through heat exchangers. The fissile inventory of this type of fuel salt tends to be higher than that of a thermal-spectrum type, for the same power output.

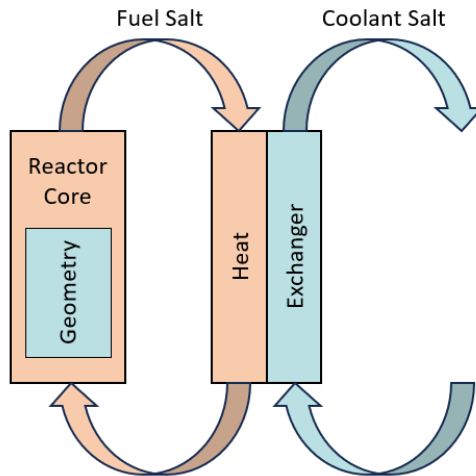


Figure 30. Fast-spectrum MSR with unmoderated core and fuel salt.

Figure 31 represents a salt-cooled MSR. The primary coolant flows through the reactor core and a heat exchanger but does not carry any fuel. The secondary coolant does not flow through the reactor core but only through heat exchangers. The fuel is stationary and resides in the reactor core.

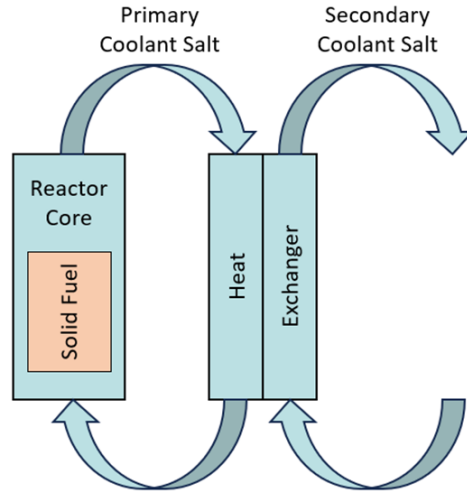


Figure 31. Solid-fueled molten salt cooled reactor.

The basic requirements of the different types of salts are summarized in Table 3. These requirements greatly narrow down the number of candidate fuel salt and coolant salt systems. Thermal-physical properties include intrinsic properties such as density, viscosity, heat capacity, surface tension, and vapor pressure. These are important features from a fluids engineering perspective. All salt systems must be chemically compatible with the “wetted” materials of construction. Wetted materials are those that come into direct contact with the molten salts. Salts that flow through the reactor core have the added requirement of being compatible with the core neutronics. For example, natural lithium is 7.6 wt% ^6Li , with the balance made of ^7Li . The salt used in the ORNL MSRE was enriched with respect to ^7Li to reduce tritium generation from the neutron-induced fission of ^6Li according to the following reaction.



Salts that carry the fissile fuel load must also provide adequate solubilities for actinides and fission products.

Table 3. Basic considerations for the different salt applications in an MSR.

Requirement	Fuel Salt	Primary Coolant Salt	Secondary Coolant Salt
Acceptable thermal-physical properties	Yes	Yes	Yes
Compatible with the materials of construction.	Yes	Yes	Yes
Compatible with neutronics in the reactor core.	Yes	Yes	-
Adequate solubility for actinides.	Yes	-	-
Adequate solubility for fission products.	Yes	-	-

There are excellent discussions on salt chemistry and the selection criteria of salt systems for use in MSRs [20, 21, 22, 23, 24, 25, 26, 27, 28, 29]. There are many different concepts for MSRs, but only two have been operated in the past: the ARE at ORNL from November 3 to 12, 1954, and the MSRE at ORNL from June 1, 1965, to December 12, 1969 with over 13,000 hours of full power operation [30]. There are excellent technical reports from ORNL on the experience with salt chemistry from operation of the MSRE [31, 32, 33, 34, 35].

Regardless of the specific MSR type, heat is extracted from the reactor core by the salt flowing through the reactor core. In an MSR (whether salt-fueled or not), salt is the heat exchange medium, just as water is the heat exchange medium in an LWR and sodium is the heat exchange medium in a sodium-cooled fast reactor (SFR).

The engineering design of an MSR can follow many bifurcations. These include selections of fuels, fuel cycles, salts, materials of construction, power output, fission product management, and so on. As the engineering design options of the MSR are down selected, certain requirements will become more obvious. These requirements include the selection of the fuel salt and/or coolant salt chemistries, as well as the selection of sensor technologies for monitoring the salt chemistries and RedOx potentials.

The design specifications of three different reactors are summarized in Table 4, one of which is an MSR design. These three reactor designs have different coolants: salt, sodium, and water, respectively. Each reactor was designed to produce approximately 1000 MWe. The three designs include:

- ORNL single-fluid molten salt breeder reactor (MSBR) design [36] (which was a design study and not an operational reactor).
- Orano Super-Phenix sodium fast breeder reactor (SFBR) design [37] (which did operate).
- Westinghouse AP1000 pressurized water reactor (PWR) design [38] (which are currently operating).

Notice that water and sodium have similar densities, while the density of salt is much greater. Also, notice that salt and sodium have similar heat capacities, while the heat capacity of water is much greater. Power efficiency differences are attributed to differences in the operating temperatures.

Table 4. Comparison of the design specifications of three different reactor types.

	ORNL Single-Fluid MSBR	Orano Super-Phenix	Westinghouse AP1000
Reactor Type	MSBR	SFBR	PWR
Status	Conceptual	Decommissioned	Operating
Thermal Power, MWt	2,250	3,000	3,415
Electrical Power, MWe	1,000	1,200	1,117
Power Efficiency, %	44.4	40.0	32.7
Liquid Coolant	Molten Fluoride Salt	Sodium	Pressurized Water
Inlet Temperature, °C	566	395	281
Outlet Temperature, °C	705	545	321
Increase in Temperature, °C	139	150	40
Heat Capacity, J g ⁻¹ K ⁻¹	1.357	1.258	6.137
Density, kg L ⁻¹	3.281	0.811	0.917
Liquid Coolant Inventory, MT	160	3,500	178
Fuel Type	Molten Fluoride Salt	Mixed Oxide	Uranium Oxide
Breeder Fuel Cycle	²³² Th/ ²³³ U	²³⁸ U/ ²³⁹ Pu	-
Neutron Spectrum	Thermal	Fast	Thermal

Fuel consumptions based on the thermal (MWt) to electrical (MWe) power conversion efficiencies of the three reactors are compared in Figure 32. The comparison is made based on the fission of 1-kg ²³⁵U producing approximately 1000 MWt of energy, and the design thermal-to-electrical efficiencies of the three reactor types. For example, the MSBR consumes approximately 2.4 kg of ²³⁵U per day to produce 1000 MWe. For all practical purposes, the fission of 2.4 kg of ²³⁵U produces 2.4 kg of fission products that accumulate in the salt.

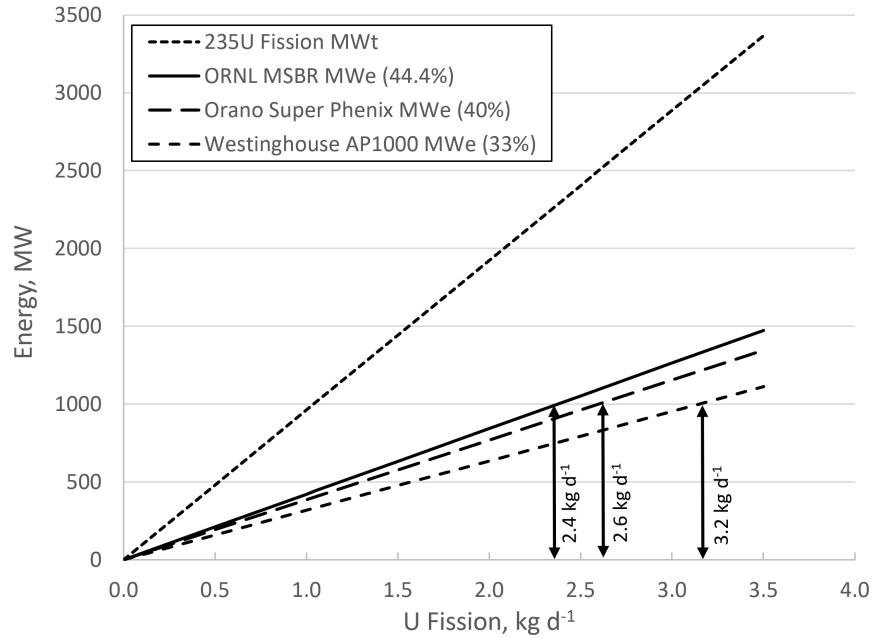


Figure 32. Comparison of approximate fuel consumptions. The vertical axis units reflect the thermal units of the fuel in the top series (units of MWt) and the corresponding electrical power conversion efficiencies (units of MWe) of the three reactor types. The mass flow rates of coolant through the cores are compared in Figure 33. In each example, a range of temperatures is shown (rather than a single temperature) to provide small curves for comparison rather than single points. The three mass flow rates are similar.

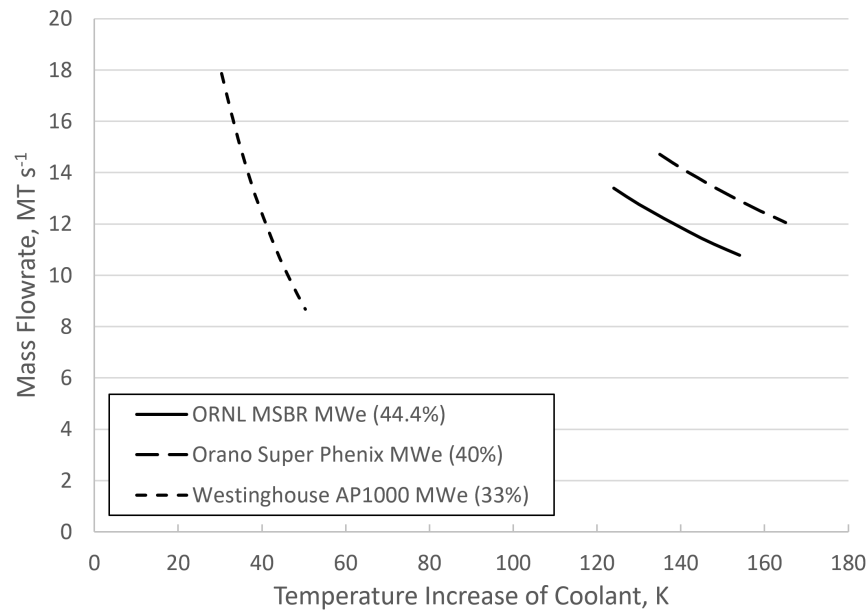


Figure 33. Comparison of mass flowrates of coolant through the core.

The volume flow rates of coolant through the cores are compared in Figure 34. The volume flow rate of the MSBR is much lower than the flow rates of the PWR and the SFBR because the salt is much denser than water or sodium.

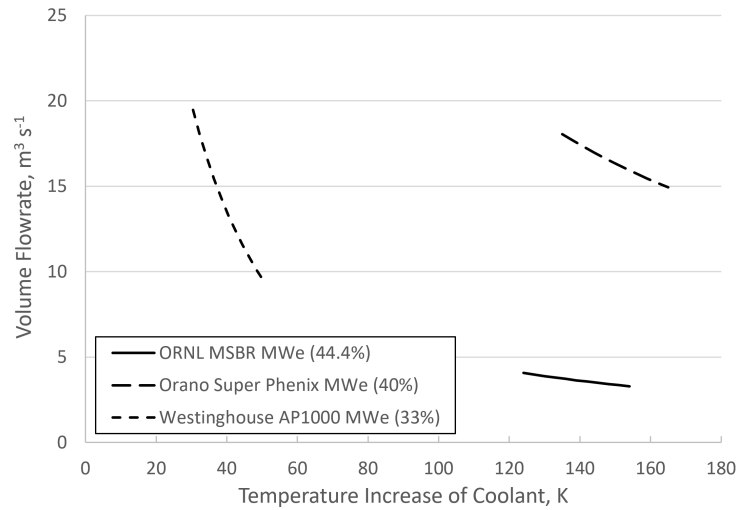


Figure 34. Comparison of volume flowrates of coolant through the cores.

The times to circulate the entire inventory through the cores are compared in Figure 35. The time required for the SFBR is much greater than the other two because the SFBR is a pool-type reactor. As shown in Table 4, the SFBR has a much greater inventory of coolant, approximately 20 times the mass of coolant compared to the other reactors. In the case of the MSBR and PWR, then, entire inventories of coolant are circulated through the cores in approximately 12 to 14 seconds.

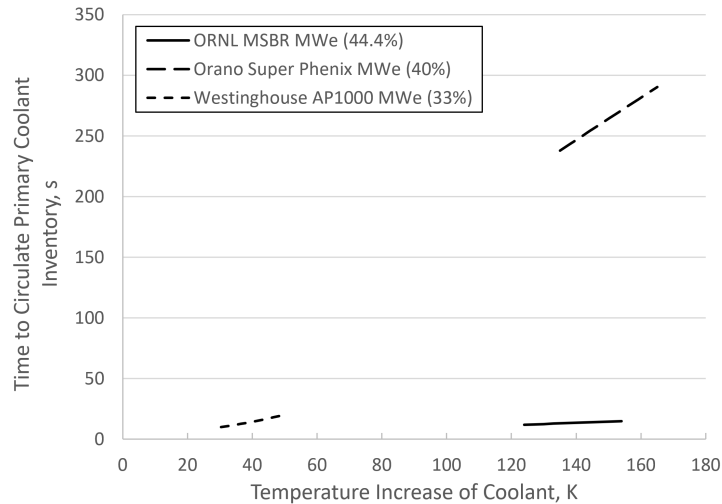


Figure 35. Comparison of the times required to circulate the entire inventories of coolant through the cores.

TAKEAWAYS: The engineering design of an MSR must consider many factors. These include selections of fuel salts, coolant salts, fuel cycles, materials of construction, power output, fission product management, and so on. Down selection of engineering design options of the MSR will place more restrictive requirements on the selection of the fuel and coolant salts. As expected, there are logical relationships between the performance characteristics of MSRs, SFBRs, and PWRs. A reactor core is essentially a heat exchanger. The rate of heat production in the reactor core must be balanced against the rate of heat extraction from the reactor core. Heat extraction is performed by a heat transfer fluid that captures heat from the core based on its heat capacity, flowrate, and temperature increase upon passing through the core. In the case of a salt-fueled MSR, the entire inventory of fuel salt is circulated through the core at a rapid rate.

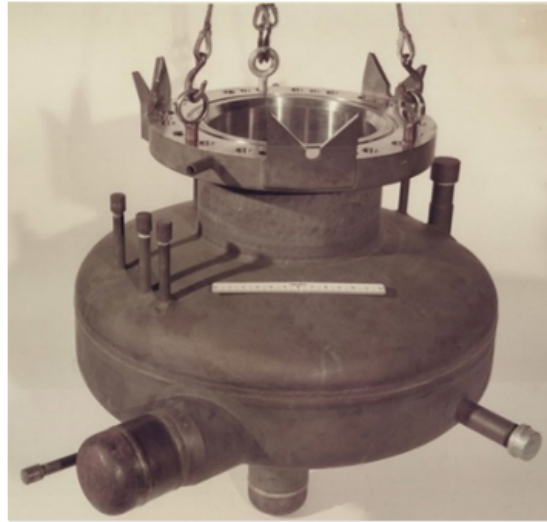
4. OPERATIONAL EXPERIENCE ON SALT CHEMISTRY MONITORING IN THE MSRE

The MSRE was an 8-MWt reactor with fluoride salt-based fuel, which operated at a temperature of around 650°C. Oxidation potential of the fuel salt during operation was an important practical consideration. The fuel composition was LiF-BeF₂-ZrF₄-UF₄. The ZrF₄ was added as an oxygen getter to avoid the inadvertent precipitation of UO₂. If excessive oxygen impurity were introduced to the fuel salt system, it would preferably react with ZrF₄ and form ZrO₂, which has a more negative formation energy and a lower solubility in the salt (600 ppm for ZrO₂ versus 1000 ppm for UO₂). An inert helium cover gas with oxygen and moisture concentrations of less than 10 ppm was used to control the oxide formation by maintaining a clean atmosphere above the salt. The UF₄ was selected because of its relatively higher chemical stability compared with UF₃. Pure UF₃ was found to disproportionate at temperatures higher than 1000°C, via reaction $4\text{UF}_3 = 3\text{UF}_4 + \text{U}$. UF₃ was found to be unstable at lower temperatures in molten fluoride salt mixtures [20]. Excess amounts of UF₄ can avoid the formation of U metal, which may alloy with the structural materials and should be avoided.

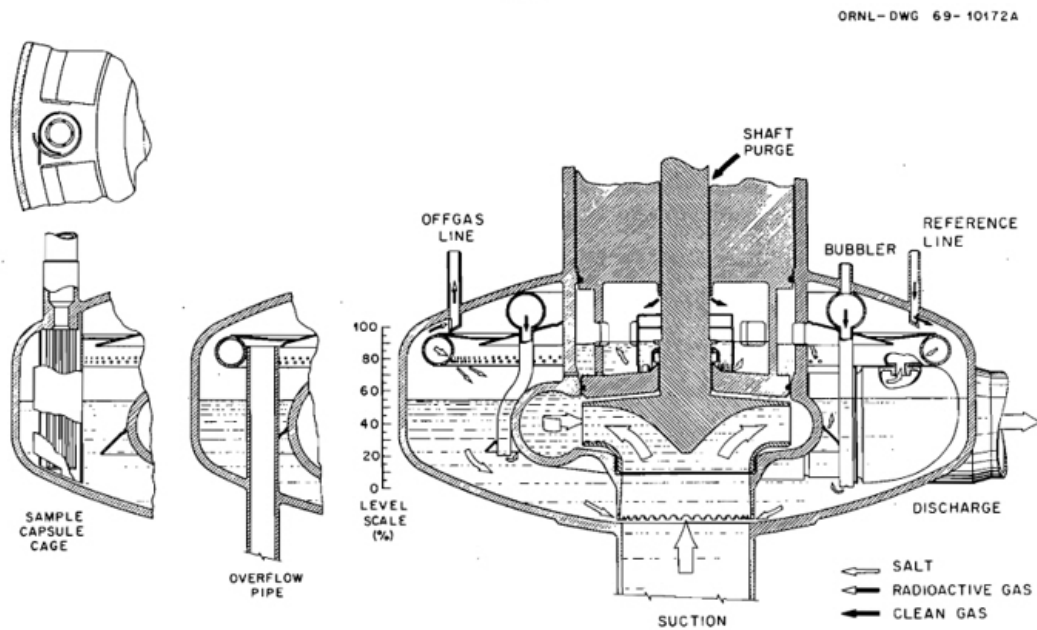
During fission of UF₄ fuel in the MSRE, the average total valence was observed to be slightly lower than four [39], which slowly made the fuel salt more and more oxidizing and accelerated structural material corrosion in the reactor vessel and other components that came into contact with the fuel salt in the reactor. The fuel salt must be controlled within an acceptable salt chemistry so that the salt does not become corrosive to the structural materials and other components in the reactor. When the UF₄/UF₃ ratio was maintained at a range of 100 to 200, the corrosion of the materials was tolerable [40, 41]. Too low UF₄ concentration would result in the formation of metallic uranium, and too high UF₄ concentration would lead to a highly corrosive salt, both of which should be avoided.

In the MSRE, to maintain the salt in a non-corrosive state, when the RedOx potential was higher, as indicated by the UF₄/UF₃ ratio, beryllium metal (encased in a perforated capsule of nickel) was introduced through the salt sampling system and suspended in the flowing salt stream in the fuel salt pump bowl. The reaction with UF₄ by $\text{Be} + 2\text{UF}_4 = \text{BeF}_2 + 2\text{UF}_3$ at a rate of about 600 g of UF₃ was produced during an 8-hour treatment. This may produce some local over-reduction of UF₄ in the Be-treated salt, but equilibrium was achieved with the excess unreacted salt in the pump bowl and reactor circuit [20].

To ensure the salt chemistry stayed within the range of acceptable compositions and impurity levels during the operation of the MSRE, salt samples were periodically removed and analyzed for major constituents, corrosion products, and oxide contamination. The isotopic composition of the salt samples was characterized by the spectroscopy technique. The salt sampling and fuel addition were carried out through the pump bowl. Helium flowed through the gas space in the bowl to sweep xenon and krypton to the off-gas disposal system. Figure 36 shows the photograph of a fuel salt pump bowl and flow patterns in a salt pump in MSRE. Figure 37 shows the process for salt sampling transfer and salt addition.

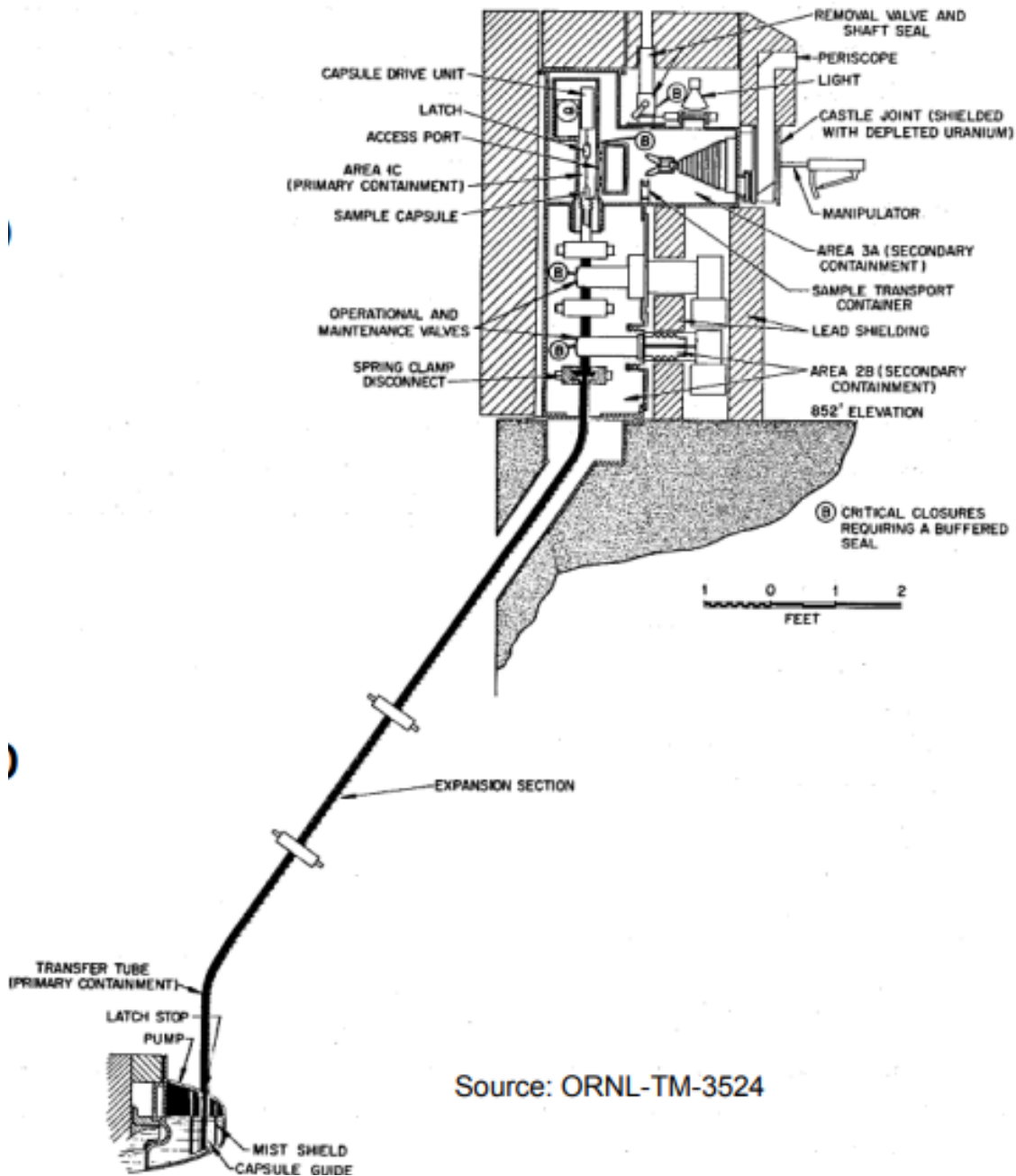


(a)



(b)

Figure 36. (a) Photograph of a fuel salt pump bowl, and (b) flow pattern in the MSRE fuel pump [42].



Source: ORNL-TM-3524

Figure 37. Salt sampling transfer and salt addition [43].

The UF_4/UF_3 ratio of the salt samples was determined by spectroscopy and electrochemical techniques such as potentiometry and voltametric techniques. The potentiometry was based on the measurement of OCP of two electrodes immersed in the fuel salt samples collected from the pump bowl and remelted. The OCP between the platinum indicator electrode and Ni/NiF_2 RE was used to measure the UF_4/UF_3 ratio. The voltametric method used was the linear sweep voltammetry technique. The current measured from the reduction of UF_4 was used to determine the RedOx potential determined by the UF_4/UF_3 ratio. The chemical analysis results were used as a calibration for the potentiometry and voltammetry measurements of the UF_4/UF_3 . (The details about the potentiometry and voltammetry techniques are provided later.)

Figure 38 shows the schematic thermal convection loop with three electrodes installed in the surge tank for electrochemical measurements of RedOx potentials. This setup was able to perform in-line electrochemical measurements. Figure 39 and Figure 40 show how the RedOx potential changes in two different loops made by Hastelloy-N and Inconel alloys, respectively. It clearly shows that in the loop made by Inconel alloy, the RedOx potential indicated by the U^{4+}/U^{3+} ratio decreased more significantly and more quickly than that in the loop made by Hastelloy-N, suggesting that Inconel is corroded more severely than Hastelloy-N. Note that the corrosion of the salt with a U^{4+}/U^{3+} ratio as its RedOx buffer is mainly attributed to the reaction of Cr (the most active element in these alloys) with salt and the reaction will reduce the RedOx potential of the salt.

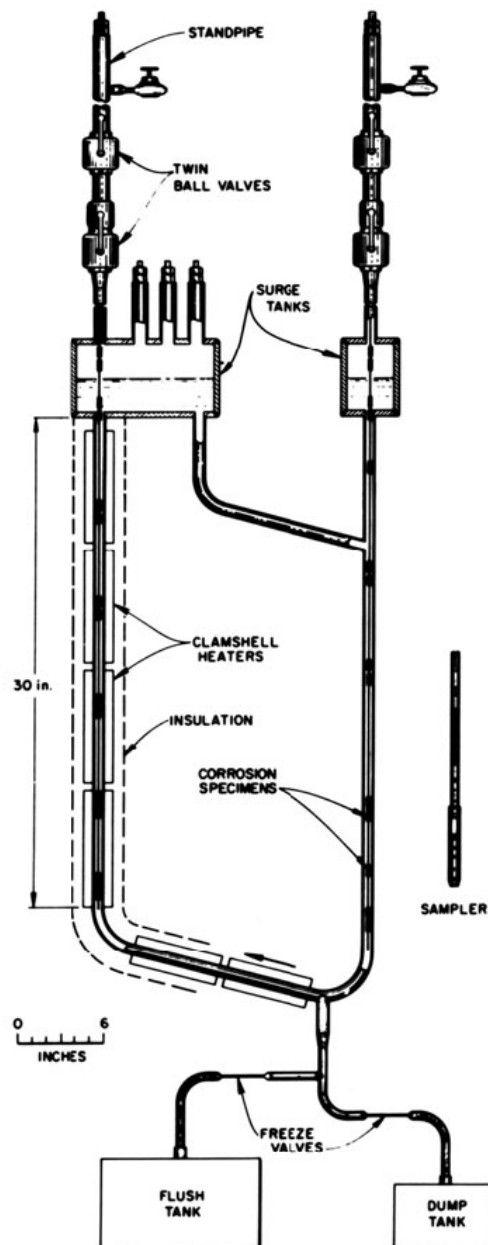


Figure 38. Schematic of a thermal convection loop. Note three electrodes in the surge tank for electrochemical measurements of RedOx potential [44].

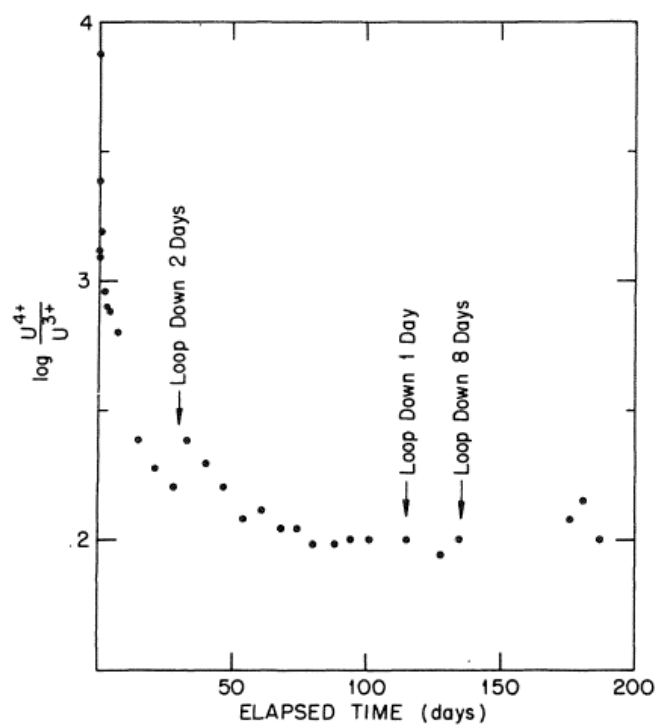


Figure 39. RedOx monitoring for a forced convection Hastelloy-N corrosion loop [44].

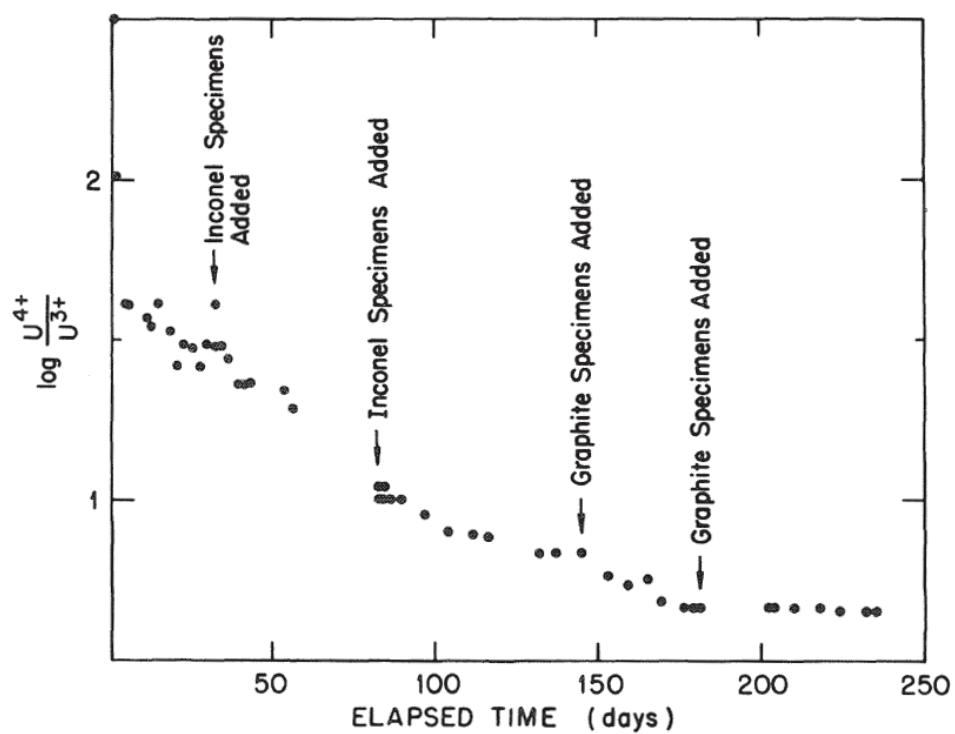


Figure 40. RedOx monitoring via the U(III/IV) couple of an Inconel corrosion loop [44].

Figure 41 shows the RedOx potential indicated by the U^{4+}/U^{3+} ratio during the MSRE operation. Initially the U^{4+}/U^{3+} ratio is around 250, then increased to 630 after an 1,800-hour operation. After beryllium addition, the ratio decreased quite fast. The lowest value during the MSRE operation was around 60. Note that the $U(III)/(\sum U)$ ratios in percentage versus full power hours were reported. The beryllium metal in contact with fuel salt increased the $U(III)/(\sum U)$ ratios, thereby decreasing the oxidizing RedOx potential of the salt by altering the $U(IV)/U(III)$ ratio.

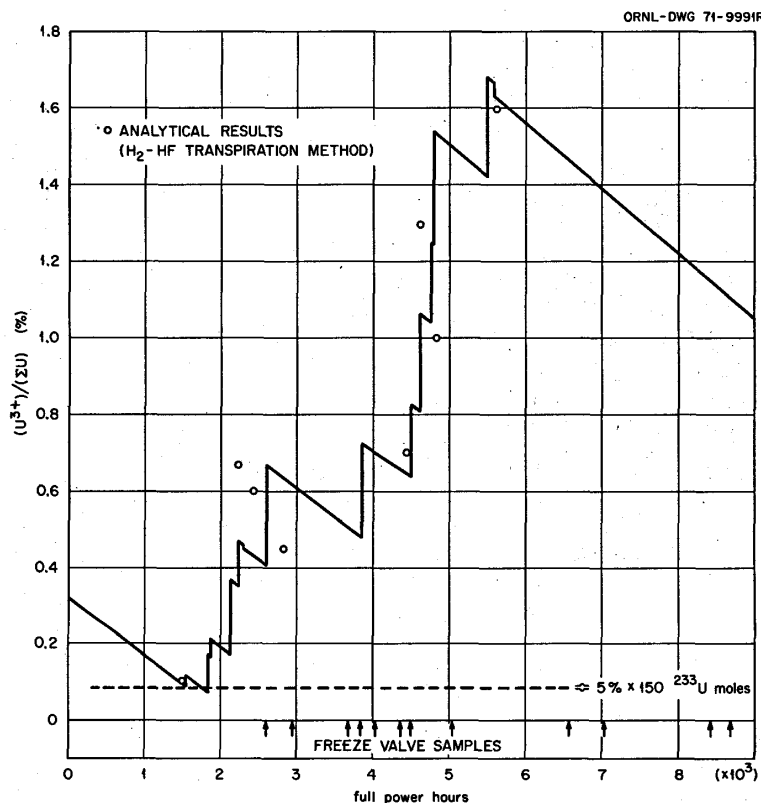


Figure 41. RedOx changes during MSRE operation [33].

The MSRE demonstrated the practicality of the MSR concept, and the safety, reliability, and tractability of simple maintenance methods. The salt chemistry was also maintained adequately by periodic adjustment of the RedOx potential through the contact of beryllium in the fuel salt. But because of the inherent disadvantages, particularly the delayed results of chemistry control associated with the salt sampling, transfer, and analysis, it was concluded that in-line salt chemistry monitoring is required. In the prominent report, Chemical Aspects of MSRE operations, upon its completion, Thoma stated: "A salient conclusion from the chemical studies described in the current report is that development of automated in-line methods for determination of RedOx potential ($U^{3+}/(\text{total } U)$) of fuel salts, for dynamic assessment of corrosion rates, and for measurement of the presence of oxides at low (<50 ppm) concentrations in flowing salt will be required for operation of large reactors." [33]. As will be further described below, oxides can also be monitored using electrochemical methods.

TAKEAWAYS: Measuring and maintaining the $U(IV)/U(III)$ ratio of the salt was of critical importance to the operation of the MSRE. The formation of excess $U(IV)$ was inhibited by the addition of beryllium metal into the salt. The beryllium metal reduced the $U(IV)$ to $U(III)$ while the beryllium metal was oxidized to $Be(II)$.

5. ASSESSMENT OF ELECTROCHEMICAL METHODS FOR MOLTEN SALTS

Electrochemical methods are techniques that can be used for analyzing the species in the molten salts of interest by measuring the potentials and currents under controlled conditions in an electrochemical cell. In some cases, they can be used to assess the composition or the RedOx potential of the molten salt. Two typical categories of electrochemical methods are potentiometry and voltammetry techniques. The potentiometry is based on the measurement of the open circuit potential (OCP) between a sensor electrode (SE) and a reference electrode (RE). The SE is also called an indicator electrode in different contexts. For accurate RedOx potential measurements using potentiometry, the potential of the SE is ideally a function of RedOx potential at a specific moment (determined by salt chemistry) and the RE is independent of the composition changes during the OCP measurement. In the voltammetry technique, typically the current is measured when a voltage is applied to the working electrode (WE) of the electrochemical cell. There are different voltammetry techniques depending on how the voltage is controlled during measurements: linear sweep voltammetry (LSV), cyclic voltammetry (CV), square wave voltammetry (SWV), and pulse voltammetry (PV), such as normal pulse voltammetry (NPV) and differential normal pulse voltammetry (DNPV). In voltammetry, there are typically three multiple electrodes: working electrode (WE), RE, and counter electrode (CE). Electrochemical methods have been widely used for understanding the thermodynamics and kinetics of molten salts, as well as the chemistry and RedOx potentials of a molten salt system. For MSRs, salt chemistry and RedOx potentials are time- and fission-dependent. In this section, we will focus mainly on the overview of salt chemistry monitoring by typical electrochemical methods for fluoride and chloride classes of molten salt systems with MSR-relevant chemical compositions.

5.1. RedOx Potential of a Salt System

The chemical behaviors of a molten salt system are determined by the chemical potentials of its constituents. In practice, a molten salt system does not exist in isolation but is always in contact with solids and gases. Solids include construction materials, solids deposited from the salt onto construction materials, and solids entrained in the molten salt. Gases include vapors formed in headspaces, fission product gases, circulation pumps, and sparging operations.

A standalone expression such as “RedOx potential of the salt” means nothing by itself because it is not a definitive, universally accepted expression. To know what is meant, it is necessary to provide a precise definition of the expression. Generally, a RedOx potential is only well-defined for a single species rather than for a mixture of species in molten salt electrochemistry. On the other hand, chemical attributes of a salt mixture are explained thermodynamically by the concepts of chemical speciation and chemical potentials.

Sensors of all sorts, when they work, can only be correlated to the chemical property they are designed to sense. There are no techniques for the direct measurements of chlorine chemical potential, fluorine chemical potential, U^{3+}/U^{4+} ratio, moisture concentration, or corrosion products. The best that can be attained is an electrochemical sensor that responds to these attributes in a consistent and reproducible way, and which may have theoretical relations that can be used for understanding. In some cases, the response of the sensor can be explained and interpreted by thermodynamic expressions, and in some cases the response can only be interpreted empirically. Empirical interpretations are still useful, provided that their limitations and scope of applicability are understood.

TAKEAWAYS: Typically, the term “RedOx potential” is used to describe the chemical condition, including corrosivity, of the molten salt. However, when the term “RedOx potential” is used quantitatively it must be clearly defined because there is no standardized meaning.

5.2. Reference Electrodes for Molten Salts

Voltage is the electrical potential difference between two locations. Furthermore, the absolute electrical potential of either location has no meaning, and it is only the difference that can be measured. A RE acts as a sufficiently stable electrical potential from which the potential difference to other electrodes can be measured, resulting in the measurement of a voltage. An analogy is the measurement of elevation. When flying an airplane, it is useful to know the elevation above sea level (which is related to air density) and the elevation above the ground (which is related to avoiding obstacles). In this example, the sea level is one useful datum, and the ground level is another useful datum. During electrochemical measurements, it is useful to know the potential differences between the RE and other electrodes in the system. In this case, the RE is the datum. By means of convention, the absolute elevation of sea level can be set to zero, or the absolute elevation of the ground can be set to zero. In a similar fashion the absolute potential of some REs can be set to zero (e.g., the standard hydrogen electrode for aqueous electrochemistry).

An RE is a critical component in potentiometry and voltammetry techniques for molten salt systems. Ideally, the potential of an RE is independent of the bulk salt chemistry. There are typically two types of REs for electrochemical measurements: a thermodynamic RE (in which the half-cell reaction is well characterized) and a quasi-reference electrode (QRE) (in which the half-cell reaction is not kept at a controlled chemical potential, or not well-defined).

TAKEAWAYS: A reference electrode serves as a datum from which to measure the potentials of other electrodes. The reference electrode and the other electrodes must share a common pool of molten salt.

5.2.1. Thermodynamic Reference Electrodes

The potential of the thermodynamic RE is determined by a well-defined chemical RedOx couple, like NiF_2/Ni for fluoride salt and AgCl/Ag for chloride salt. Take the NiF_2/Ni RE for a FLiNaK fluoride salt system as an example. A small amount of NiF_2 was mixed with FLiNaK salt and housed in a container of boron nitride (BN) or graphite. This housing was put in the FLiNaK salt to monitor the chemistry of the FLiNaK. The container for the NiF_2/Ni RE functions as the ionic conductor between the RE salt and FLiNaK salt. The composition of the salt inside the RE container is not affected by the composition of the salt outside the RE container. Two basic requirements for an RE are needed. One is the chemical compatibility of the container material with fluoride salt which is important for long-term operations. The other is to provide the ionic conductivity between the salt inside and outside the RE container. The RE container materials need to provide a mechanism for ionic conductivity either by providing a small amount of porosity that can be filled with salt, or by themselves being ionically conductive. Materials like mullite and yttria stabilized zirconia that are used for chloride salt systems (discussed later) are ionically conductive. Another example of an ionically conductive material is single crystal LaF_3 which was used for fluoride salts [45].

Due to the requirement for sufficient chemical compatibility with molten fluoride salt systems, which exhibit strong fluxing capability and are corrosive for most ceramic materials like oxides, there are limited options in the selection of RE container materials. Graphite and BN were typically used as container materials for fluoride salt systems. Graphite with engineered porosities were used. Graphite is chemically compatible with fluoride salt and is expected to last a long time. BN was also used as a container material. Some researchers reported that BN cannot last long in high-temperature fluoride salt systems. Figure 42 shows a NiF_2/Ni RE used in FLiNaK salt systems. Figure 43 shows an RE that used BN-coated graphite as container. Figure 44 shows a NiF_2/Ni that uses single crystal LaF_3 as container material. Figure 45 shows a schematic of a NiF_2/Ni RE that uses BN as container material

When boron nitride was used as the container materials for the reference salt, a junction potential, usually in the range of a few mV was observed. This junction potential may affect the chemical composition measurement by potentiometry because the composition is exponentially dependent on the OCP. The effect of the junction potential needs to be removed or corrected for accurate RedOx potential measurement. During the MSRE program, the junction potential was measured by measuring the NiF_2/Ni RE with a BN container and the bulk salt with the same NiF_2 content.

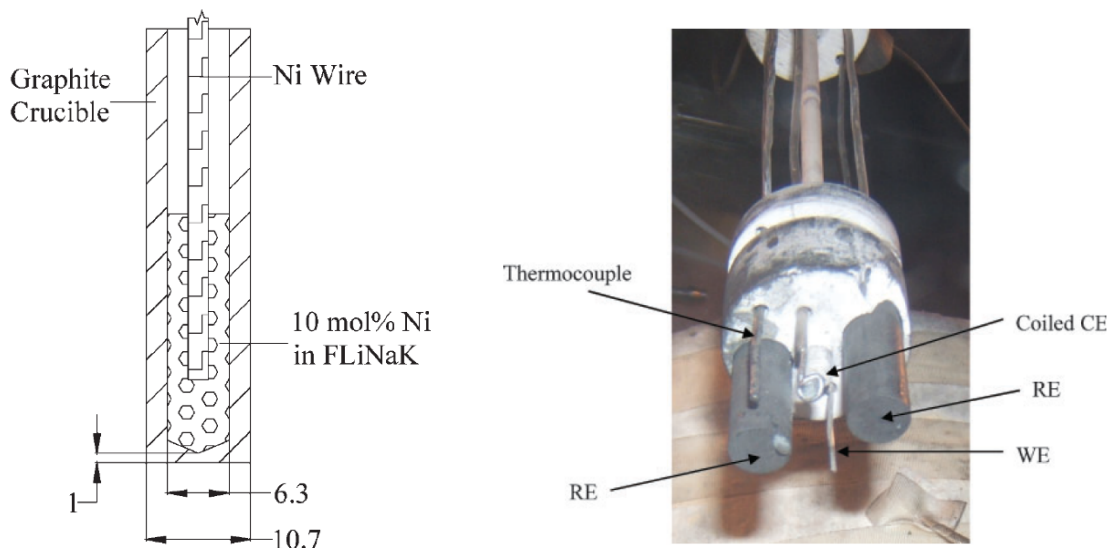


Figure 42. NiF_2/Ni reference electrode for FLiNaK salt [46].

The Ni/NiF_2 RE with graphite container coated with BN was found to exhibit Nernstian response and the wetting time for RE is quite short (in hours scale) for the graphite tube with relatively large pores. The graphite tube can be used for up to 2 days. It is easy to make and inexpensive, making it a potential alternative RE for fluoride salts [47].

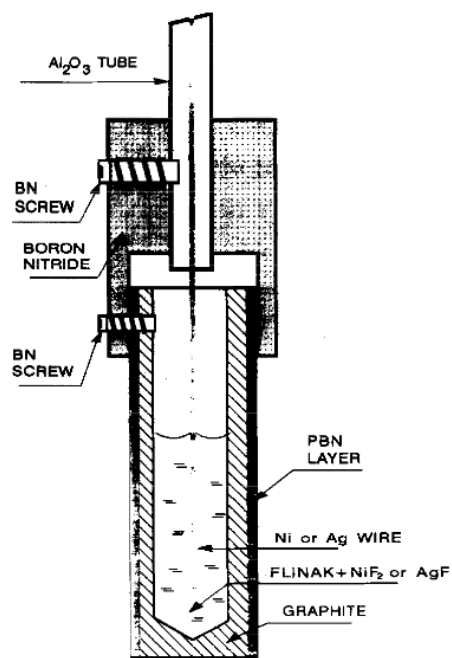


Figure 43 A schematic of an RE used in FLiNaK with a BN-coated graphite container [47].

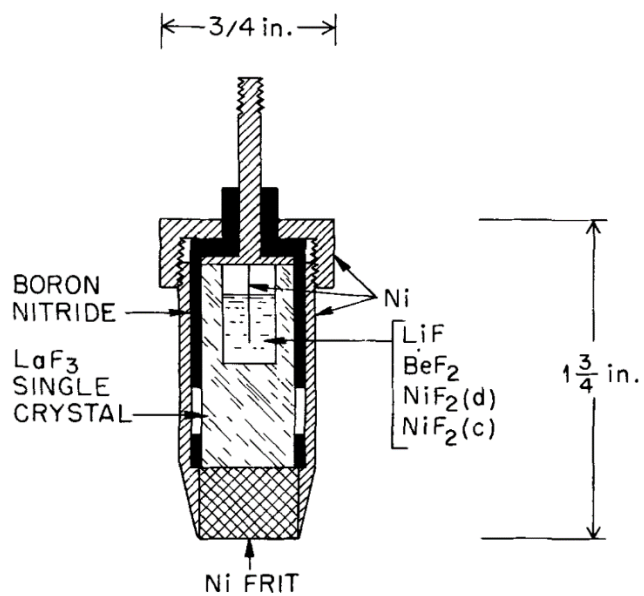


Figure 44. A schematic of a NiF₂/Ni RE using LaF₃ as a container material for fluoride salt system [45].

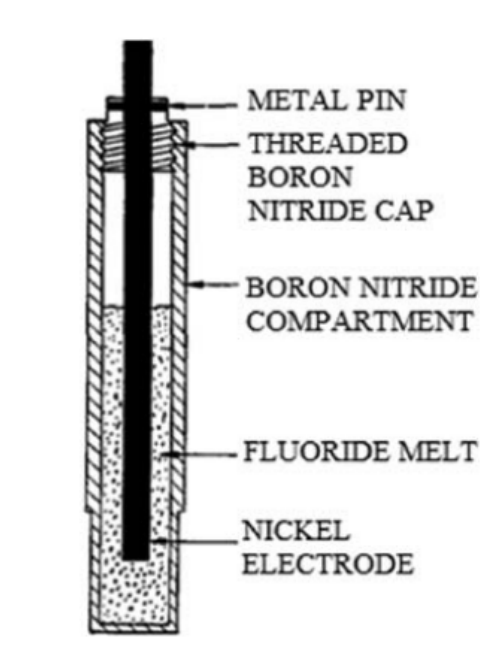


Figure 45. A schematic of a NiF_2/Ni RE that use BN as container material [48].

To achieve reliable and consistent performance, a RE needs to be compatible with the salt system, the electrochemical reaction in the RE should be reversible, and the RE should have adequate mechanical strength, particularly for the extreme environment expected in the engineering applications of MSRs. For example, the RE must be able to survive thermal shock due to the sudden temperature change induced from frequent removal and insertion of the RE in the salt, high velocity salt flow, and corrosion.

Based on the experiences of researchers, for REs made with BN as reference salt containers, the working temperatures are in the range of 500 to 600°C. It may take 24 to 48 hours to stabilize and reach an equilibrium with the molten salt for testing, and measurement variation in the range of ± 6 mV over 12 days in FLiBe salt was observed.

For the REs made with graphite as reference salt containers, the working temperature range reported was 650 to 850°C, and it took about 3 hours to equilibrate with the salt for testing. Stable measurements were reported for 2 days.

For the LaF_3 single crystal as an ionically conductive membrane, the working temperature range was 500 to 600°C, and it took only 5 minutes to equilibrate with the molten salt for testing. No measurements more than 2 days were reported to assess the long-term performance of the LaF_3 membrane in molten salts.

To evaluate how the REs perform, some relative potential tests were performed prior to experiments. In the relative potential tests, the potential difference was measured from two similar REs in the same reference salt. If the potential difference between the two REs was close to zero, it suggested that the two REs had reached an equilibrium or quasi-equilibrium with the molten salts. Some tests were also performed to evaluate if the REs responded to changes of the analytes in the salt, as would be expected from the Nernst Equation. Figure 46 shows a plotting of electromotive force (EMF) between two REs versus time. Figure 47 shows the plots for two Ni^{2+}/Ni REs in two fluoride salts.

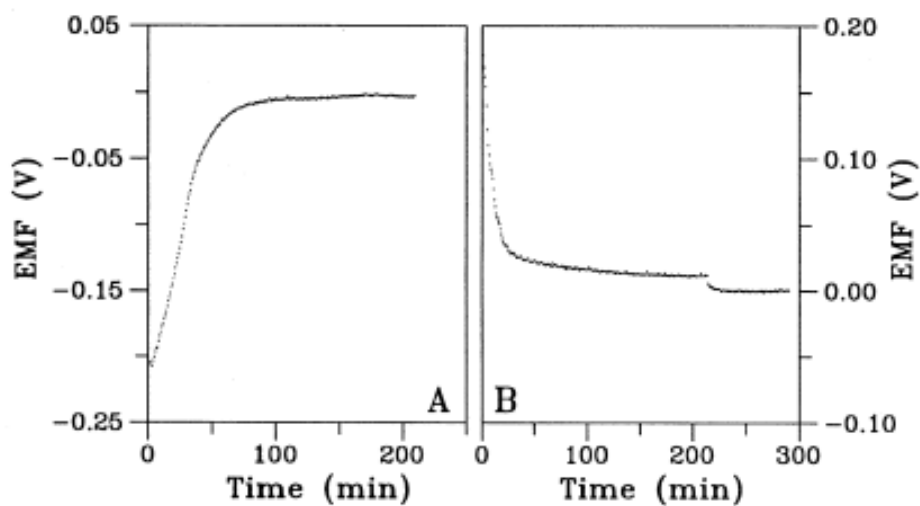


Figure 46. EMF vs. time between two NiF_2/Ni REs. (A) EMF vs. time between two Ni/NiF_2 REs (1 mol% in FLiNaK). (B) EMF vs. time between two Ag/AgF REs (1 mol% in FLiNaK) [47].

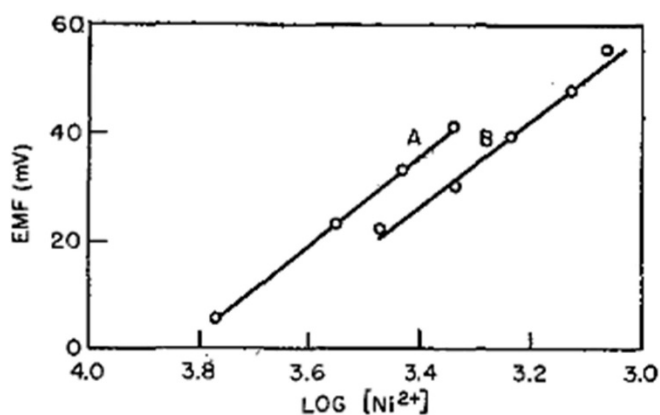


Figure 47. Nernst Equation based on log plots for the $\text{Ni}/\text{Ni(II)}$ couple in molten fluorides. (A) LiF-NaF-KF (517°C) slope 0.080, theory 0.078. (B) LiF-BeF-ZrF_4 (507°C) slope 0.081, theory 0.077 [48].

Because many different oxide ceramic materials are generally compatible with high-temperature molten chloride salts, there are more options in the selection of container materials for building REs for chloride salts than for fluoride salts. Typical container materials reported in literature are Vycor glass, mullite, alumina, and quartz. Similar to the REs for fluoride salts, which usually have a porous frit section for ionic conductivity, the RE designs with frits also worked for chloride salts. However, for long-term monitoring purposes, ultimately there is some mass transfer between the reference salt and the working salt under electrochemical tests, and REs that rely on frits are a poor choice. In a chloride salt system, the main reason for not needing a frit section is because the container materials can provide adequate ionic conductivity. For example, the Vycor glass and mullite are ionic conductors at high temperature in molten salt. The Na^+ ions in the Vycor and mullite can provide ionic conductivity. Figure 48 shows a typical Ag/AgCl RE with some membrane for containing the reference salt. To reduce the resistance of the membrane, the bottom of the membrane was thinned, such that ionic conductivity occurs primarily through the thinner wall section depicted at the bottom of Figure 48, for this design. From the experience of the authors, some mullite tubes (e.g., McDanel Advanced Ceramic Technologies) with a thickness of 1 to 2 mm meet the requirements of an RE quite well. Figure 49 shows an RE design using alumina that is reported in the literature for a chloride salt system.

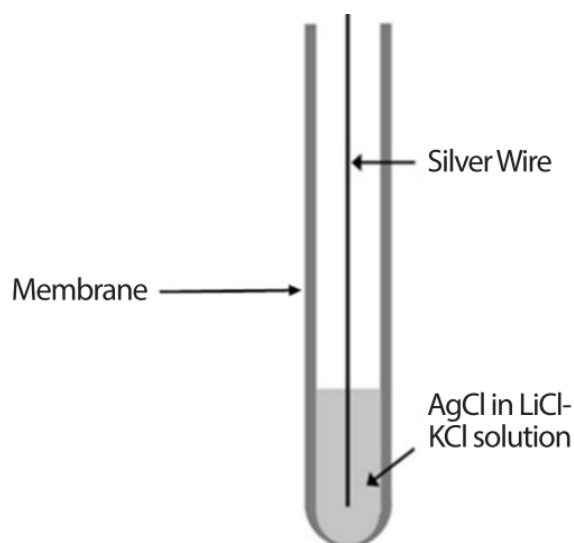


Figure 48. Schematic AgCl/Ag reference electrode for chloride salt [49].

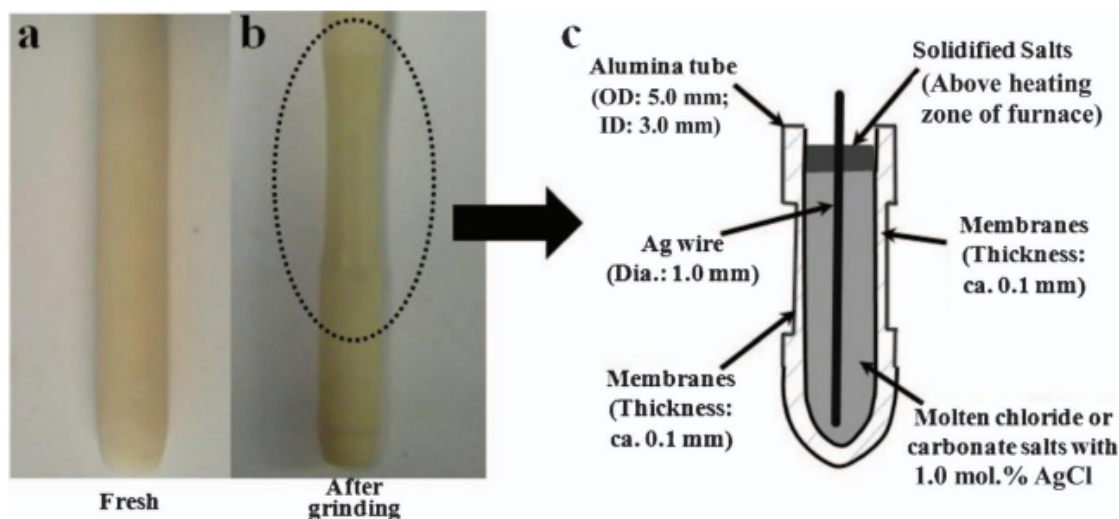


Figure 49. AgCl/Ag reference electrode contained in alumina membrane [50].

In addition to the above mullite and alumina membranes for chloride salts, quartz, and even MgO are also reported as being used for container materials for REs [51, 52]. The selection of a good RE is dependent on the specific salt system.

For all the RE designs with ceramic and glass membranes, it is common to exhibit some junction potential associated with the REs during electrochemical measurements. This junction potential was typically in the range of several mV. The absolute value of a junction potential is often small or negligible for most applications but may considerably affect the concentration measurement if using potentiometry methods. Therefore, it may be necessary to correct for the effects of the junction potential by calibration as was done for fluoride salts.

Because of the variations of the different REs in real tests, sometimes it is difficult to compare the results achieved by different investigators under similar conditions. Some investigators have tried to convert the potential difference of the WE and Ag/AgCl RE to the potential difference between the WE and a $\text{Cl}_2(\text{g})/\text{Cl}^-$ RE. In this way, the difference between the Ag/AgCl RE and $\text{Cl}_2(\text{g})/\text{Cl}^-$ can be measured. Figure 50 shows how the difference between an Ag/AgCl RE and $\text{Cl}_2(\text{g})/\text{Cl}^-$ RE was measured. The data are summarized in Table 5.

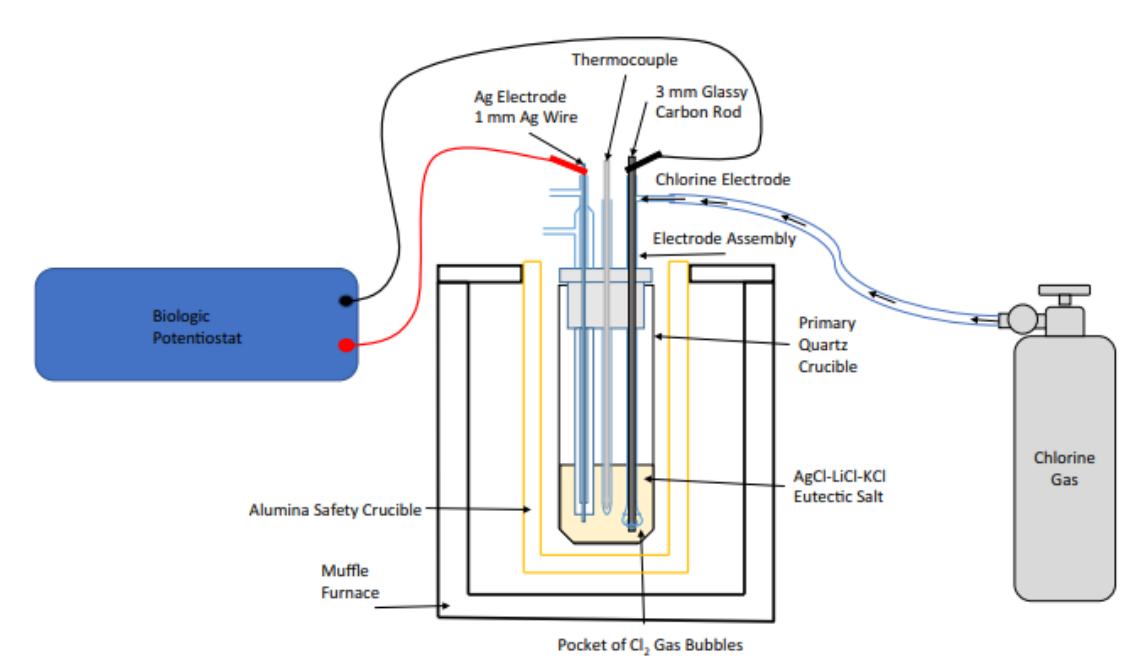


Figure 50. A schematic $\text{Cl}_2(\text{g})/\text{Cl}^-$ RE [53].

Table 5. Compiled comparison of the potential differences of Ag/AgCl REs with 0.00039 to 0.10 molar fraction of AgCl in reference salt and $\text{Cl}_2(\text{g})/\text{Cl}^-$ RE, at temperature range of 723 K to 823 K [53].

Temperature (K)	AgCl concentration (mole fraction)	$E_{\text{Ag}/\text{AgCl}}$ vs. $E_{\text{Cl}_2/\text{Cl}^-}$ (V)	Standard deviation (V)	AgCl concentration (mole fraction)	$E_{\text{Ag}/\text{AgCl}}$ vs. $E_{\text{Cl}_2/\text{Cl}^-}$ (V)	Standard deviation (V)
723	0.00039	-1.322	0.0014	0.0043	-1.195	0.0005
748		-1.328	0.0004		-1.199	0.0028
773		-1.339	0.0009		-1.21	0.0001
798		-1.352	0.0005		-1.206	0.0028
823		-1.361	0.0003		-1.211	0.0021
723	0.010	-1.139	0.0008	0.022	-1.0893	0.0017
748		-1.1393	0.0017		-1.0927	0.0009
773		-1.144	0.0006		-1.0950	0.0014
798		-1.147	0.001		-1.0970	0.0021
823		-1.1507	0.0009		-1.0940	N/A
723	0.050	-1.0330	0.0088	0.10	-0.9817	0.0091
748		-1.0407	0.0012		-0.9907	0.0012
773		-1.0484	0.0013		-0.9873	0.0045
798		-1.0418	0.0027		-0.9873	0.0012
823		-1.044	0.0018		-0.9853	0.0009

For some molten chloride salts containing constituents, like oxides, that are reactive with container materials, the mullite and Vycor glass may not be suitable options. For example, in LiCl-salt with dissolved Li_2O , which is typically used for pyroprocessing of spent nuclear oxide fuels, Li_2O will react with ZrO_2 and mullite. Therefore, the more chemically compatible MgO oxide ceramic can be used in conjunction with a MgO frit section [54, 55, 56]. For fuel salts in MSRs, oxygen content is an impurity that should be maintained as low as achievable. The reaction of oxide impurity with common RE containers like mullite and yttria stabilized zirconia is likely not an issue. More information on the REs for molten salts can be found in literature.

TAKEAWAYS: A thermodynamic reference electrode is a reference electrode that has a well-defined half-cell reaction. Not always, but in certain circumstances, a thermodynamic reference electrode can be used to quantitatively measure the thermodynamic properties of electrochemical reactions relative to standard state conditions. There are many variants of thermodynamic reference electrodes and they are selected to meet the requirements of specific applications.

5.2.2. Quasi-Reference Electrodes

In contrast to thermodynamic REs, quasi-reference electrodes (QREs) have potentials that are determined by salt chemistry in contact with the QREs. Typical metallic QREs include Pt, Ir, W, and Ta. Glassy carbon is also used as a QRE. These QREs were typically used in voltammetric techniques. One basic requirement for QREs is that they are compatible with molten salts. In contrast to thermodynamic REs with well-defined chemical RedOx coupling in the RE container, the potential of the QRE is determined by the salt chemistry (with potentially different QRE electrode chemical RedOx reactions in different salts) and thus is dependent on the composition of the salt melt that is in contact with the electrode. Despite the drawbacks of QREs, because of their simplicity and easy availability, QREs are widely used for voltammetry experiments in molten salts, including chlorides and fluorides [57, 58, 59, 60, 61, 62, 63, 64]. Figure 51 shows a photograph of an electrode assembly, and Figure 52 shows two example experimental setups for voltammetry using QREs.

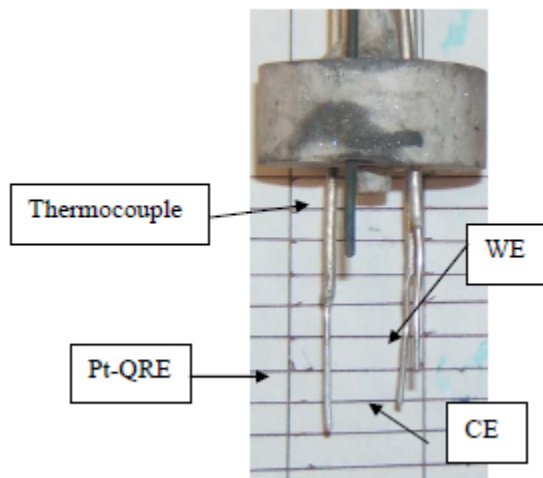


Figure 51. Electrochemical setup using Pt wires for WE, CE, and QRE [63].

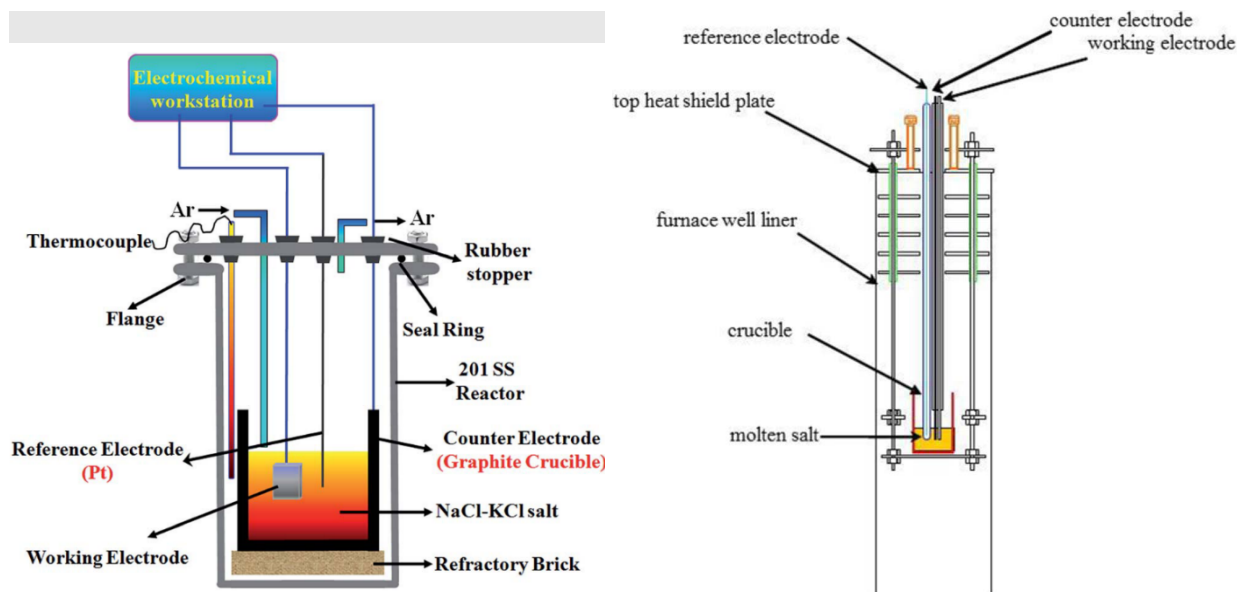


Figure 52. Two example experimental setups for voltammetry studies quasi-reference electrode [62, 64].

TAKEAWAYS: A quasi-reference electrode is a reference electrode that does not have a well-defined half-cell reaction. A quasi-reference electrode can be susceptible to measurement drift because the surface electroactive properties of the quasi-reference electrode are influenced by changes in the chemistry of the bulk salt. Nevertheless, quasi-reference electrodes can be robust and serve useful purposes.

5.2.3. Dynamic Reference Electrode

Dynamic reference electrode (DRE) is a type of RE with the following features: (1) a thin layer of element M (e.g., Be in FLiBe salt) is cathodically deposited on an electrode (e.g., W, Mo, Ir, Pt), and (2) the deposited metal M (e.g., Be in FLiBe again) can be considered for a short time (on the order of seconds) to be the thermodynamic RE, with the potential determined by the RedOx potential of M/M^{z+} (e.g., Be/Be²⁺ in FLiBe salt). Figure 53 shows a cell for RedOx potential measurement using a DRE. The advantage of the DRE is that it does not have the junction potential of REs that use containers because the deposited metal is in direct contact with the molten salt. The disadvantage is that the deposited metal is usually a reactive metal that can be redissolved into the salt, meaning it functions as an RE for only a short time. For example, the DRE in Figure 54 is effective for the duration of the voltage plateau (e.g., ~60 seconds in Figure 54b and ~10 seconds in Figure 54d) in measuring the OCP. Note that a voltage plateau is needed for the DRE to measure the OCP in molten salt systems.

Typically, DREs are used as potentiometry measurements, such as RedOx potential measurements. Figure 54 shows the measurements of RedOx potentials in a FLiBe salt using a DRE, based on a Be/BeF₂ RedOx couple and with varied deposition parameters during the deposition stage. The voltage plateaus in the plots were dependent on multiple factors such as deposition current, deposition time, the metal deposition layer, and impurities.

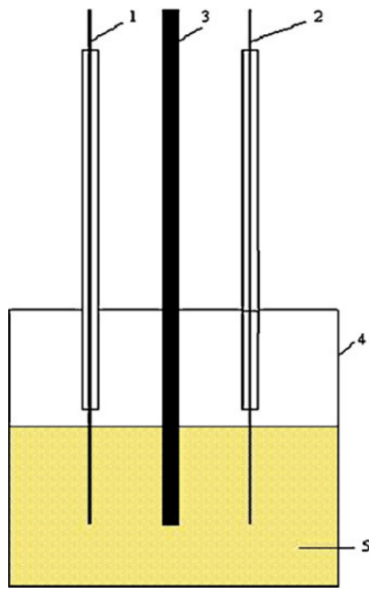


Figure 53. Schematic of a cell for RedOx potential measurement using a dynamic reference electrode (1) indicator electrode; (2) dynamic reference electrode (cathode); (3) glassy carbon anode; (4) crucible; and (5) molten salt [65].

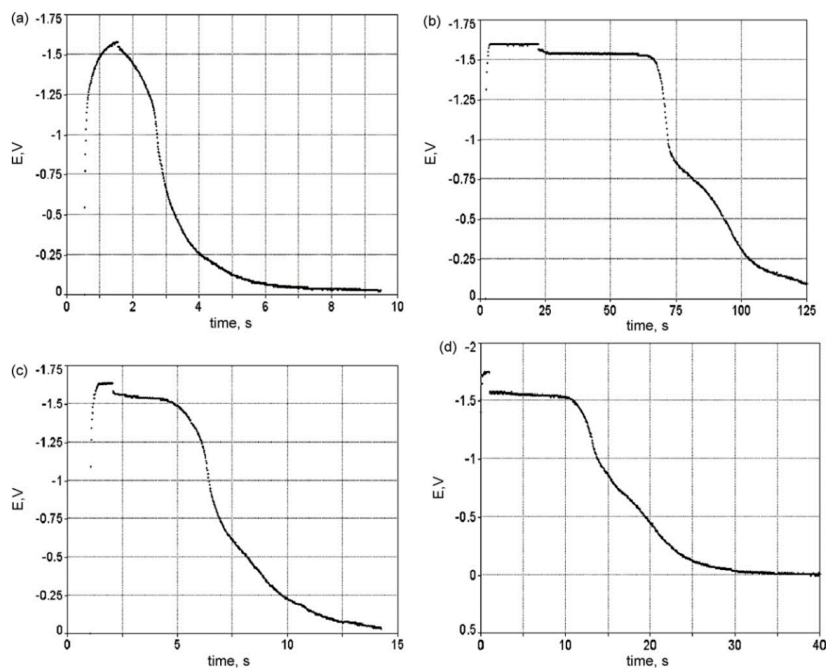


Figure 54. Effect of deposition conditions in preparing dynamic reference electrodes on the EMF relaxation curves at 600°C: (a) $I = 20 \text{ mA}$, $t = 1 \text{ s}$; (b) $I = 20 \text{ mA}$, $t = 20 \text{ s}$; (c) $I = 40 \text{ mA}$, $t = 1 \text{ s}$; and (d) $I = 100 \text{ mA}$, $t = 1 \text{ s}$ [65].

TAKEAWAYS: A dynamic reference electrode can be considered a refinement of a quasi-reference electrode. The surface electroactive properties of the dynamic reference electrode can be manipulated to make it behave like a thermodynamic reference electrode, but the conditions for functioning as a thermodynamic reference electrode last a short time and will drift over time. Nevertheless, dynamic reference electrodes can be robust and serve useful purposes.

5.3. Potentiometry Techniques for Molten Salts

In the MSRE, the uranium constituent in the fuel salt was mainly UF_4 with a small amount of UF_3 . As mentioned previously, UF_3 is unstable at high temperature in FLiBe salt and is susceptible to disproportionation that results in the formation of uranium metal and UF_4 . To avoid this, a significant amount of UF_4 was added to avoid the disproportionation of UF_3 . The ratio of $\text{U(IV)}/\text{U(III)}$ was maintained within a range that avoided the UF_3 disproportionation and to prevent the salt from becoming significantly corrosive to structural materials. The $\text{U(IV)}/\text{U(III)}$ ratio was indicative of the RedOx condition of the salt and is also called a RedOx buffer. The measurement of the $\text{U(IV)}/\text{U(III)}$ ratio is therefore essential to a fundamental understanding of fuel chemistry. The sensor design is shown in Figure 55.

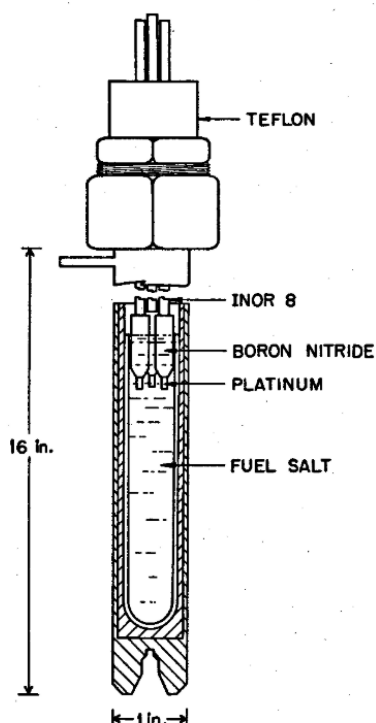


Figure 55. Electrochemical cell for $\text{U(IV)}/\text{U(III)}$ ratio determination in MSRE fuel [66].

Potentiometric determination of this ratio requires the value of the standard electrode potential of the $\text{U(IV)}/\text{U(III)}$ ratio relative to the standard electrode potential of a reference couple. The standard potential of the $\text{U(IV)}/\text{U(III)}$ ratio was calculated with respect to a hypothetical unit mole fraction $\text{Ni}/\text{Ni(II)}$ electrode (arbitrarily set at 0.0 V) from EMF measurements on the galvanic cell $\text{Pt}|\text{U(IV)U(III)}||(\text{Ni(II)})|\text{Ni}$. The cell consisted of a $\text{Ni}/\text{Ni(II)}$ RE contained in a boron nitride compartment and a platinum wire immersed in a melt containing U(III) and U(IV) . The cell potential was measured as a function of the $\text{U(IV)}/\text{U(III)}$ ratio, which was varied by the controlled reduction of U(IV) to U(III) with a zirconium rod immersed in the solution. Measured cell potentials were corrected, using the Nernst Equation, to values versus a unit mole fraction $\text{Ni}/\text{Ni(II)}$ electrode (a hypothetical Ni electrode in 100% NiF_2 salt). The corrected potentials were plotted versus a $\log [\text{U(IV)}/\text{U(III)}]$ ratio. The slope of a line through these points agreed well with the theoretical slope predicted by the Nernst Equation. The standard potential of the $\text{U(IV)}/\text{U(III)}$ ratio, corresponding to a $\text{U(IV)}/\text{U(III)}$ ratio of 1, was found to be 1.48 ± 0.01 V at 500°C .

Different potentiometry sensors depend on the requirements and salt systems. In potentiometry measurements, the results based on the Nernst Equations are the chemical activities, which are not always directly proportional to their concentrations. To get the concentration, the activity coefficient at the relevant concentration needs to be known, or the measurements need to be calibrated by chemical analysis, for example by spectroscopy.

Depending on the requirements, potentiometry can also be used to measure the concentrations of some constituents in the molten salts. As described above, the potentiometry required a well-defined RE with constant potential and a good indicator electrode that responds to the concentration of the constituent of interest.

Figure 56 shows the experimental set up using the Gd/GdCl₃ RE with saturated GdCl₃ salt in the LiCl-KCl eutectic salt and the Gd metal as the indicator electrode, based on the OCP (Gd vs. Gd/GdCl₃ RE) and the molar fraction of the GdCl₃ in the LiCl-KCl salt, the activity coefficient of the GdCl₃ can be evaluated [67]. The activity coefficient is different if using two different standard states, including a liquid or solid standard state. But this difference will not affect the concentration measurement if the standard state selected is internally consistent.

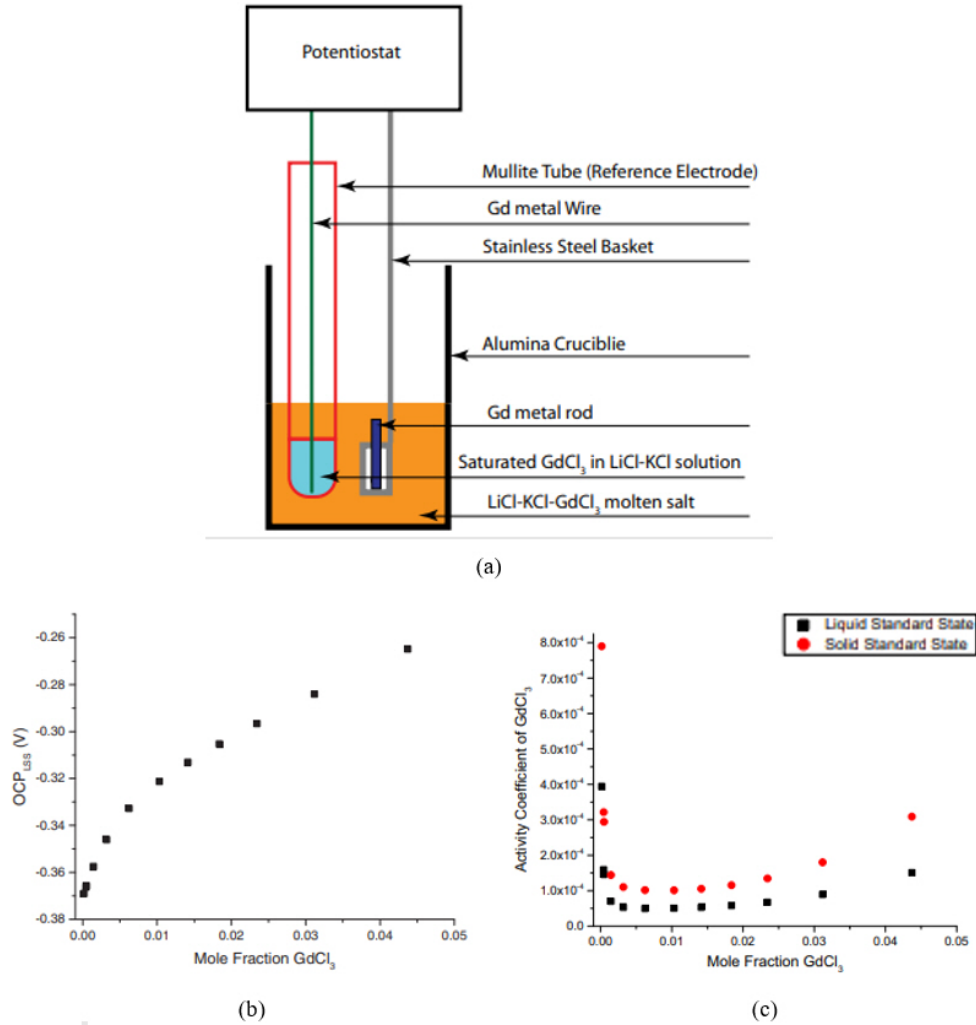


Figure 56. GdCl₃ activity coefficient measurements by EMF. (a) Experimental measurement setup, (b) measured OCP versus molar fraction of GdCl₃ in LiCl-KCl eutectic salt, and (c) activity coefficient of GdCl₃ versus molar fraction of GdCl₃ in LiCl-KCl eutectic salt. Different standard states were used for the two graphs, resulting in different y-axis scales [67].

As described above, the OCP between a WE and a dynamic RE was also used for measuring the RedOx potential in molten salt. For example, the Be (metal) was deposited to a W electrode in a FLiBe salt, and a Be/BeF₂ RE was formed; then the OCP between the WE and Be/BeF₂ was measured to determine the RedOx potential. Usually when the salt is more corrosive, the RedOx potential (WE vs. Be/BeF₂ DRE) will be more positive. Figure 57 shows the potentials measured by the DRE that are prepared using different deposition currents versus a nickel wire electrode in a FLiNaK salt containing 0.1 mol% NiF₂ at 773 K.

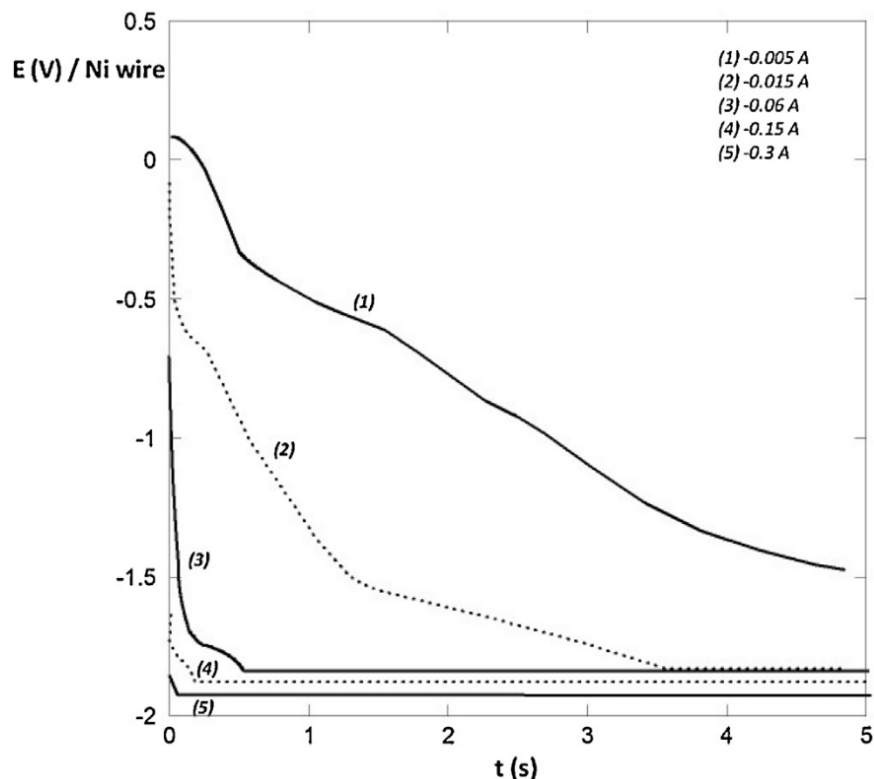


Figure 57. Chronopotentiograms recorded during DRE generation on tungsten electrode against Ni wire for various applied currents in FLiNaK containing 0.1 mol% of NiF₂ at 773 K [68].

Figure 58 shows a RedOx potential sensor assembly based on DRE, and Figure 59 shows the potential difference between a Be-deposited Mo electrode (or Be/BeF₂ dynamic RE) versus a Mo indicator electrode of different FLiBe salts at 500°C. Note that usually the RedOx potential was reported as the potential difference between the indicator electrode and the RE. In general, the RedOx potential is lower or more negative for salts that are purer. With this context, the weighted average RedOx potential (Mo versus Be/BeF₂ RE) is 1.706 ± 0.002 V for the FLiBe salt prepared from LiF and BeF₂ salts and 1.432 ± 0.002 V for the purified MSRE coolant salt by Be reduction and filtration [69]. This demonstrated that the purification process decreased the RedOx potential of the FLiBe salt.

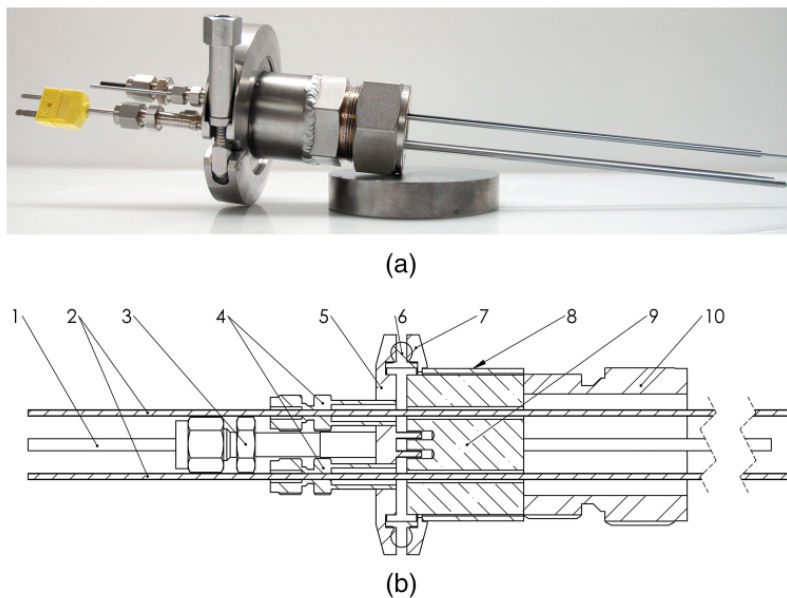


Figure 58. A RedOx potential sensor based on dynamic reference electrode for RedOx potential measurement [69].

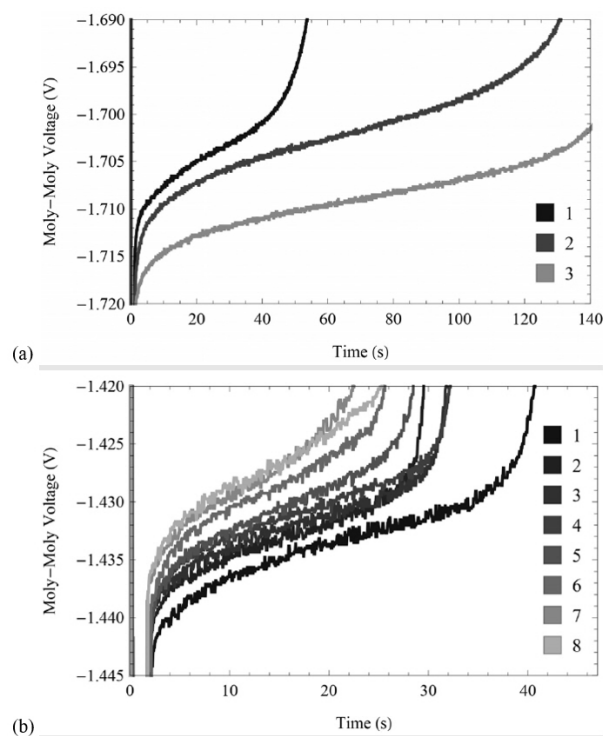


Figure 59. Potential difference between a Be-deposited Mo electrode (or Be/BeF₂ dynamic RE) versus a Mo indicator electrode of different FLiBe salts at 500°C. (a) FLiBe salt produced at University of Wisconsin from LiF and BeF₂ salts, three successive measurements. (b) FLiBe MSRE coolant salt processed by beryllium reduction and filtration, eight successive measurements [69].

TAKEAWAYS: Potentiometry is the measurement of the potential between two electrodes under open circuit potential conditions, which means that no electrical current is passing between the two

electrodes. For example, the resulting potential can be used to determine the presence and concentration of select metal cations in the salt. Potentiometric techniques can be coupled with voltammetry techniques to provide even more information.

5.4. Some Practical Considerations of Potentiometry for MSR Applications

In potentiometry, only two electrodes are involved and there is no current passing through the electrodes. There are several ways to measure the OCP. The two-electrode configuration can be used, and many instruments with a high internal impedance (e.g., potentiostats or some advanced digital voltmeters) can be used to measure and record the OCP data. Sometimes the circuit resistance at high temperature is quite high for REs using ceramic containers that serve as ionic conductive membranes. If the internal impedance of the instrument for OCP measurement is not high enough, some measurement errors may be introduced [70]. In most cases for low impedance RE containers, the common commercially available digital multimeters meet the requirements for OCP measurements.

The OCP is basically the potential difference between the RE and indicator electrode. Both potentials need to be stable during the measurement. In a simplistic interpretation, the potential of the RE is assumed constant. In actuality, particularly for long-term measurement, some potential drifting of the RE may occur, and this will result in measurement errors. Therefore, a periodic calibration of the potentiometric measurements is recommended.

The usefulness of DREs is application specific. Not every molten salt system is suitable for DRE measurements. For example, if the salt contains UF_4 and UF_3 , as is the case in MSR operation, the deposition layer can react with the salt and cause the reduction of U^{4+} to U^{3+} , which may be undesired. The DRE may be suitable for quick RedOx potential measurements for the FLiBe coolant salt used in FHRs.

There is no standard or universally accepted RE for molten salts. The selection of RE is mainly at the discretion of the researcher and based on their specific application. Typically, the NiF_2/Ni RE in different container materials such as graphite and boron nitride have been used for potentiometry measurements in fluoride salt systems. The AgCl/Ag RE and the recently reported NiCl_2/Ni RE in different container materials are used for chloride salt systems. In addition, REs like the $\text{Gd}/\text{saturated GdCl}_3$ RE were also used for some specific potentiometry measurements and applications. This suggests that some other similar metal/saturated-metal-chloride REs may be developed for some similar applications, but their performance needs to be evaluated and demonstrated before being considered for MSR applications. Because the potential of the QREs is not fixed and is dependent on the salt chemistry, the QREs usually are not good RE candidates for potentiometry measurements, in either fluoride or chloride salt systems, particularly if quantitative results are desired. The lack of a standard practice makes it challenging to compare the RedOx potential measurement results of different researchers, even under similar salt compositions and/or test conditions.

TAKEAWAYS: Potentiometry methods are very sensitive to the surface conditions of the electrodes involved. Measurements with voltmeters with high internal impedance are preferred as not to perturb the potential being measured. Certain configurations are susceptible to drift over time, introducing errors into the measurements.

5.5. Voltammetry Techniques for Molten Salts

Voltammetry techniques are widely used for electrochemistry studies of molten salt systems. Many developments have been reported in the molten salt electrochemistry field [71, 72, 73, 74, 75, 76]. For example, the pyroprocessing of spent nuclear oxides fuels involves the electrochemical reduction of uranium oxide to uranium metal in a LiCl-Li₂O salt and the electrorefining of the uranium in LiCl-KCl-UCl₃ salt [77, 78, 79, 80]. Voltammetry is also extensively used in understanding and determining the basic thermodynamic and kinetic properties of salt constituents in a bulk molten salt electrolyte.

In an experimental setup for voltammetry measurements, usually a three-electrode configuration is used (WE, RE, and CE) in conjunction with a potentiostat to (1) control the voltage between the WE and RE and record the current between the WE and CE, or (2) control the current between the WE and CE and record the voltage of the WE versus the RE. In some advanced potentiostats, the voltage between the CE and RE is also measured and recorded, though the potential at the CE (versus RE) is typically not explicitly of interest. In most cases, researchers are interested in the electrochemical processes occurring at the WE. Figure 60 shows a schematic three-electrode configuration for molten salts (the computer for controlling the potentiostat was not shown). All three electrodes are connected to the potentiostat. Most potentiostats have two sense leads which are used to measure or control the potentials between the WE and RE. The surface area for the CE is usually higher than that of the WE, so that the performance of the CE does not in any way interfere with the electrochemical measurement. In most advanced potentiostats, many electrochemical measurement techniques are available in the instrument's software.

There are different voltammetry techniques depending on how the voltage or current is controlled. Voltage and current cannot be controlled simultaneously during measurement sequences. Either voltage is controlled and amperage is the responding signal, or current is controlled and voltage is the responding signal. The relationship between the voltage and current, as well as the time, are recorded during the voltammetry measurement sequence. For monitoring the salt chemistry for MSRs, typical voltammetry measurements include LSV, CV, SWV, and PV, as detailed below, and involve controlling the voltage and measuring the current.

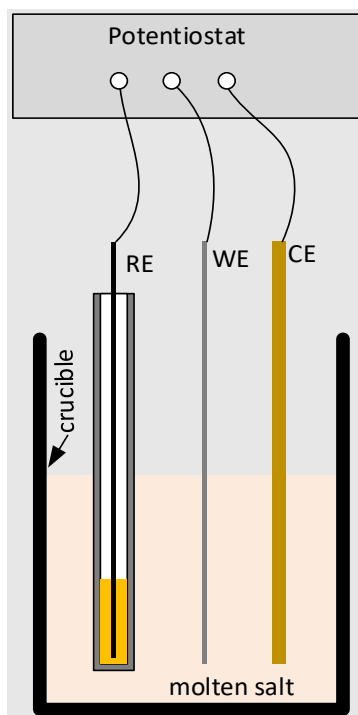


Figure 60. A schematic three-electrode system for molten salt electrochemistry measurements.

5.5.1. Linear Sweep Voltammetry

Linear Sweep Voltammetry (LSV) is a basic voltammetry technique. In LSV, the WE potential versus RE is swept linearly from a starting value to a final value, and the current is measured as a function of time during this sweep. The rate of sweeping, or the slope of the potential vs. time, is also called the scan rate. The results that are typically plotted are the relationships of the current versus the voltage, which are called voltammograms. During the WE potential sweep process, some species in the molten salt may undergo oxidation or reduction processes depending on their reduction potentials and chemical activities in the salts. The currents associated with these oxidation and reduction processes are accurately measured by the potentiostat. The current is a type of rate (a rate of electrical charge transfer associated with chemical reaction) and the current is thus a representation of the process kinetics when oxidation or reduction are present during the sweep. In many cases, the process may have mass transfer limited kinetics for the chemical reaction, in which case the current represents a rate that is partially or fully representative of mass transfer kinetics.

The MSRE used the LSV technique for RedOx potential measurements of its fuel salts. In the MSRE, an apparatus was designed to permit the LSV measurements in a sample of fuel salt [66]. The principle of the LSV for determining the RedOx potential, as schematically shown in Figure 61 [44], is $E_{eq} = E_{1/2} + \frac{2.3RT}{nF} \log \frac{U(IV)}{U(III)}$. E_{eq} is the potential of working electrode versus a reference electrode, which can either be a QRE or thermodynamic RE. E_{eq} is measured when the electrodes are in equilibrium with the salt and the value of E_{eq} is determined by the U(IV)/U(III) ratio. The polarographic half-wave potential, $E_{1/2}$, is approximately equal to standard electrode potential E° , which is the potential when the U(IV)/U(III) ratio is 1 at electrode surface. For LSV at a stationary electrode, when the RedOx couple is reversible, both oxidized and reduced forms are soluble in the salt, which is the case for the U(IV)/U(III) couple, and the concentration of the reduced form in the bulk is lower than the oxidized form (in MSRE, the concentration of the reduced species U(III) was significantly lower than the concentration of the oxidized species U(IV) in fuel salt), the half-wave potential, $E_{1/2}$, corresponds to the potential on a voltammogram at which the current is equal to 85% of the peak current [48,81]. The U(IV)/U(III) ratio determined by the LSV was verified by traditional chemical analysis by spectrophotometry, and the experimental results have good agreement as seen in Table 6 [66].

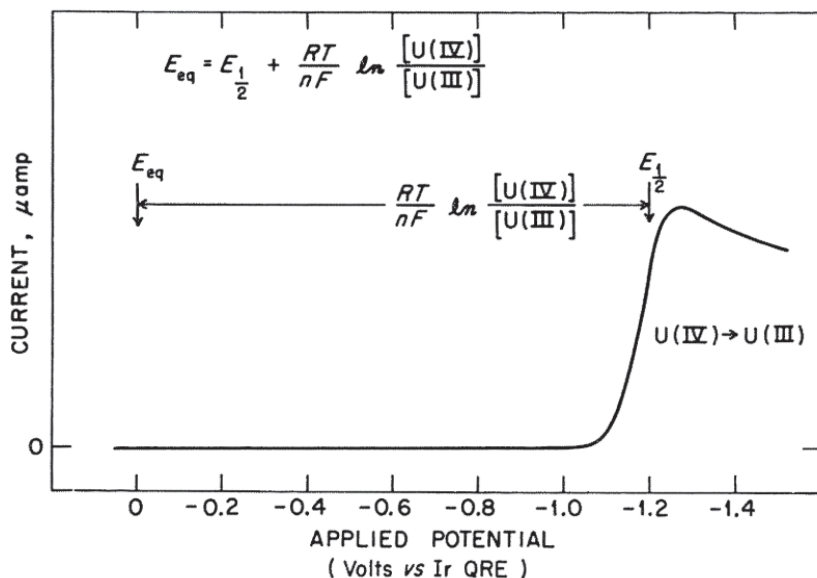


Figure 61. Electrochemical analysis of U(IV)/U(III) used in the MSRE [44].

Table 6. Determination of U(IV)/U(III) ratio in the MSRE [66]

U(IV)/U(III) ratio*	U (mole%)	E _{1/2} vs. equilibrium potential (V)	
		Calculated	Measured
140	~0.3	-0.329	-0.345
29.3	~0.3	-0.225	-0.225
6.7	~0.3	-0.127	-0.130
10	~0.15	-0.153	-0.175
* Determined from spectral measurements			

5.5.2. Cyclic Voltammetry

Cyclic voltammetry (CV) is a widely used technique for measuring the thermodynamic properties of molten salts. During a CV measurement, the WE potential relative to the RE is scanned at a specific rate in terms of millivolts per second (mV s^{-1}) in a positive or negative direction. But unlike in linear sweep voltammetry (where the experiment stops after the final WE potential is reached), during a CV the WE potential is scanned in the reverse direction to return to the initial potential. The two half-cycles create a complete cycle that can be repeated multiple times. From these measurements, the current at the WE is monitored and plotted versus the applied voltage (that is, the WE potential), and the plot is referred to as a cyclic voltammogram. A cyclic voltammogram, also known as a CV curve, can provide information about RedOx potentials and electrochemical reaction rates. For instance, if the electron transfer at the WE surface is fast and the current is limited by the diffusion of analyte species to the electrode surface, then the peak current will be proportional to the square root of the scan rate, as described by the Randles–Sevcik Equation $I_p = 0.4463nFAC(\frac{nvFD}{RT})^{1/2}$, for a reversible soluble-soluble system, such as U(III)/U(IV). From this equation, the diffusion coefficient of the constituent of interest can be determined. To apply CV to the salt chemistry monitoring in MSRs, the concentration of the constituent of interest can be determined after establishing a calibration curve of current versus concentration.

Figure 62 (a) illustrates a typical CV on a glassy carbon WE showing the reduction/oxidation waves of Ni(II)/Ni in FLiNaK with 0.81 mol% NiF₂ at 773 K, and (b) shows CVs at various scan rates (0.050 to 0.300 V s^{-1}). Based on the CV results, the diffusivity of Ni(II) in molten FLiNaK at $T = 748$ to 823 K was determined in the range of 3.27×10^{-7} to $3.04 \times 10^{-6} \text{ cm}^2\text{s}^{-1}$ [82]. The knowledge of the diffusion coefficients is essential for salt chemistry monitoring. In theory, based on the above Randles–Sevcik Equation, if the diffusion coefficient and other CV parameters (e.g., the scan rate and electrode area) are known, then the bulk concentration of the molten salt can be determined by the peak current. In reviewing the electrochemistry of actinides and fission products in molten salts, Zhang summarized the diffusion coefficient of different ions in molten LiCl-KCl salt at 723 K, as shown in Table 7 [73].

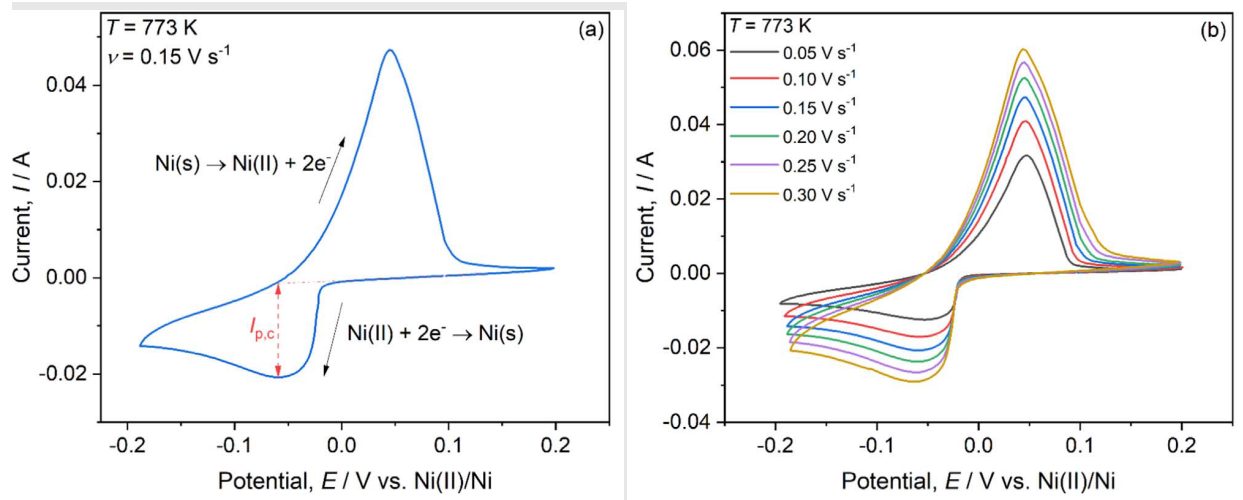


Figure 62. (a) Example CV on a glassy C WE ($A = 0.228 \text{ cm}^2$) indicating the reduction/oxidation waves of Ni(II)/Ni in FLiNaK + 0.81 mol% NiF₂ at 773 K and 0.15 V s^{-1} , (b) CVs at various scan rates (0.050 – 0.300 V s^{-1}) [82].

Table 7. Summary of ion diffusion in molten LiCl-KCl at 723 K [73]

Ions	Coefficient (cm^2/s) @ 723 K
Th ⁴⁺	2.45186×10^{-5}
U ³⁺	1.01929×10^{-5}
U ⁴⁺	6.82658×10^{-6}
Np ³⁺	1.86749×10^{-5}
Np ⁴⁺	2.04971×10^{-5}
Pu ³⁺	7.93686×10^{-6}
Am ³⁺	2.04439×10^{-6}
Am ²⁺	1.00648×10^{-6}
Cm ³⁺	9.28993×10^{-6}
Y ³⁺	9.14377×10^{-6}
Zr ⁴⁺	6.76492×10^{-8}
Zr ²⁺	1.46434×10^{-7}
La ³⁺	1.46636×10^{-5}
Ce ³⁺	9.98503×10^{-6}
Sm ³⁺	7.85288×10^{-6}
Eu ³⁺	1.59995×10^{-5}
Gd ³⁺	5.31242×10^{-6}

The ability for long-term measurements of CVs is desirable for MSRs. Figure 63 shows many CV results over long-term measurement (100 days) in the LiCl-KCl-UCl₃ salt (with various amounts of fission products) for uranium electrorefining at Idaho National Laboratory (INL). The CV results show that, for the LiCl-KCl-UCl₃ salt with a stable composition, the CV measurements appear consistent during the long-term service test [83]. Figure 64 shows the relationship between the peak current densities in CVs and the concentration of corrosion products Cr²⁺ and Fe²⁺ in molten MgCl₂-KCl-NaCl mixture. The

measured Cr^{2+} and Fe^{2+} peaks by CV were consistent with previous estimates of their known concentrations under similar conditions [84].

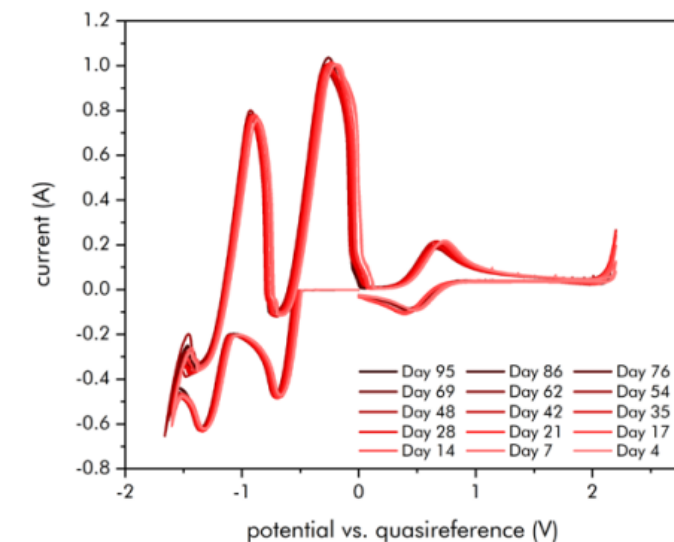


Figure 63. CV results of the LiCl-KCl-UCl_3 salt used for uranium electrorefining at INL for over 100-hour service life test. Scan rate: 500mV s^{-1} , temperature: 500°C , the day number in the plot indicates the immersion times [83].

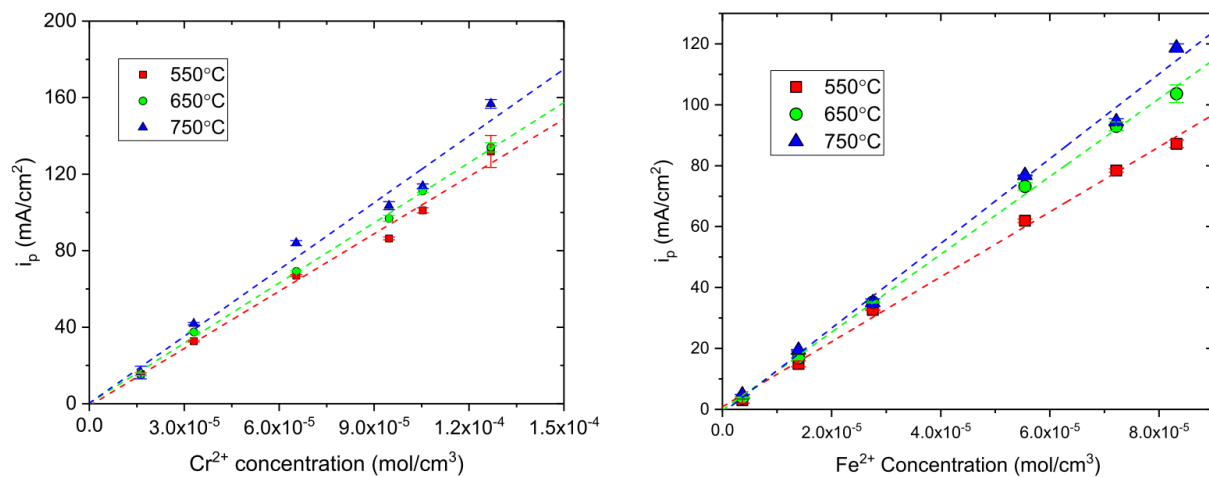


Figure 64. The peak current density obtained from $\text{MgCl}_2\text{-KCl-NaCl}$ salt with various Cr^{2+} and Fe^{2+} concentration 550 , 650 , and 750°C [84].

5.5.3. Square Wave Voltammetry

In square wave voltammetry (SWV), a square wave potential pulse waveform is superimposed on a staircase step to generate a voltametric sweep, as shown in Figure 65. The potential as the base of the peaks increases at a rate linear with time. Typically, the sampling technique of SWV considers the current response during the latter half of the potential pulse. The current is measured two times near the end of the forward and reverse potentials, respectively, and the results of SWV is the relationship between the applied WE potential and the forward and reverse current difference. With this approach, the charging current or non-Faradaic current is minimized, and thus the SWV can provide higher sensitivity measurements than the typical CV. It is therefore likely to be used to measure low concentrations of constituents in molten salts, such as oxide and moisture, which should be maintained as low as achievable in MSRs.

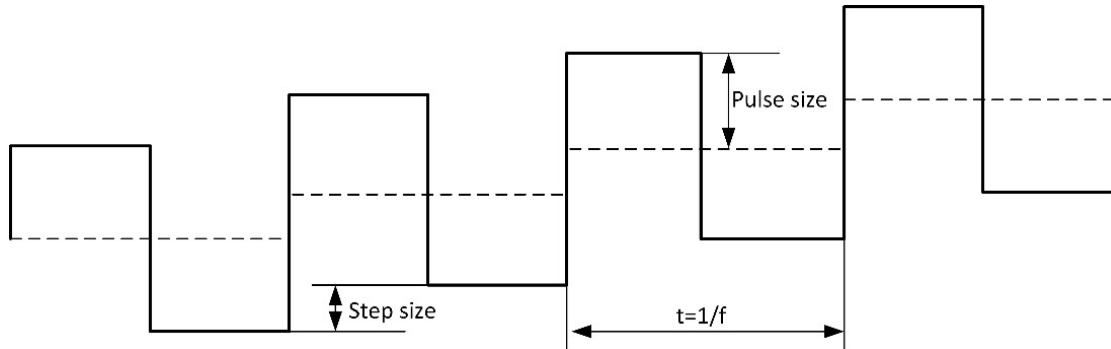


Figure 65. Schematic of square wave voltammetry showing how voltage varies with time [85].

Figure 66 shows the SWV results for the LiCl-Li₂O salt system with various Li₂O contents (0 to 0.77 wt%) [86]. It appears that the lowest measured oxide concentration in this study (600 ppm) was detected by SWV. In another study, 300 ppm oxide in a NaCl-KCl-Na₂O system was detected by SWV. Some unpublished work at INL shows that as low as 50 ppm oxide impurity in a LiCl-KCl salt at 500°C has been detected by SWV. Figure 68 shows the SWV of LiCl-KCl salt with different concentration (5 to 1015 ppm) of Li₂O at 600°C. In addition to measurement of low concentration oxide impurity in salt, the SWV can also be used for salt chemistry monitoring purposes. Figure 69 shows the SWV of LiCl-KCl-UCl₃ salt for electrorefining, which exhibits high sensitivity of SWV [83].

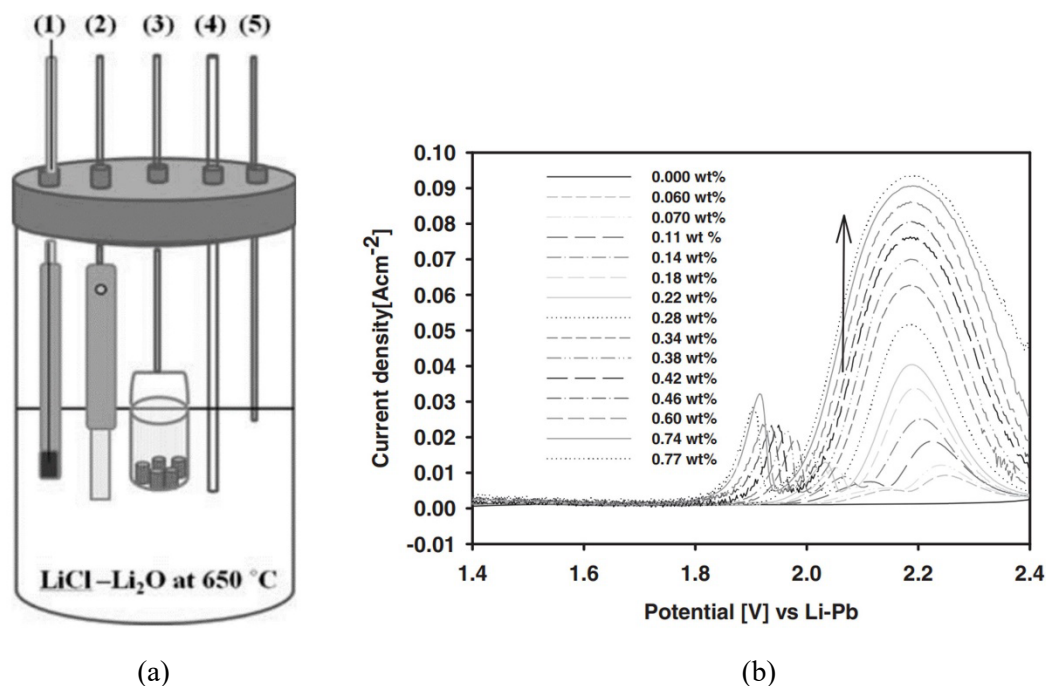


Figure 66. (a) Schematic diagram of the electrochemical cell. (1) Li-Pb reference electrode, (2) Pt anode for electrolytic reduction, (3) cathode basket containing UO₂ pellets, (4) Ni rod (CE for SWV), (5) Pt wire (WE for SWV). (b) SWV at various Li₂O concentrations (0.00 ~ 0.77 wt%) with 20 Hz frequency, 19.95 mV amplitude, 2.55 mV step height. Experimental conditions: Pt wire (ϕ 0.5 mm) WE, Ni rod (ϕ 3 mm) CE, Li-Pb reference electrode in molten LiCl [86].

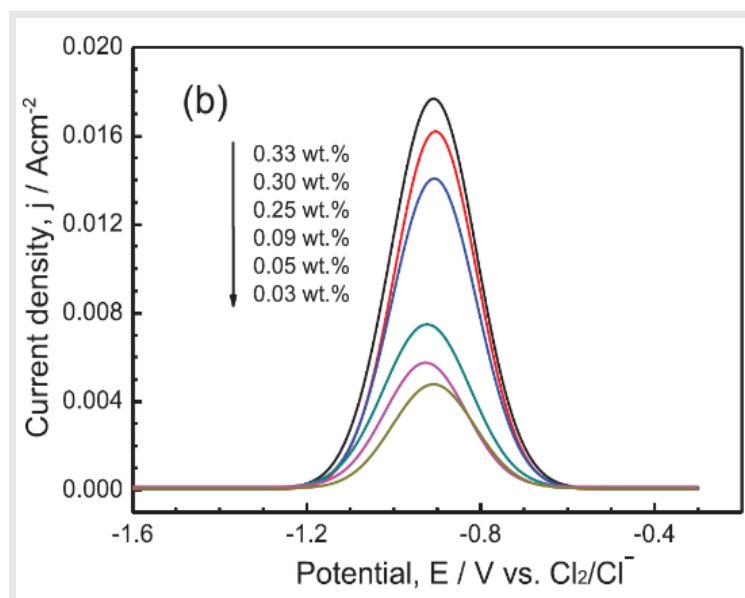


Figure 67. SWV of the NaCl-KCl-Na₂O system for various oxide contents. WE: Gold. Auxiliary electrode: graphite rod. The frequency was 36 Hz and the square wave amplitude was 20 mV [87].

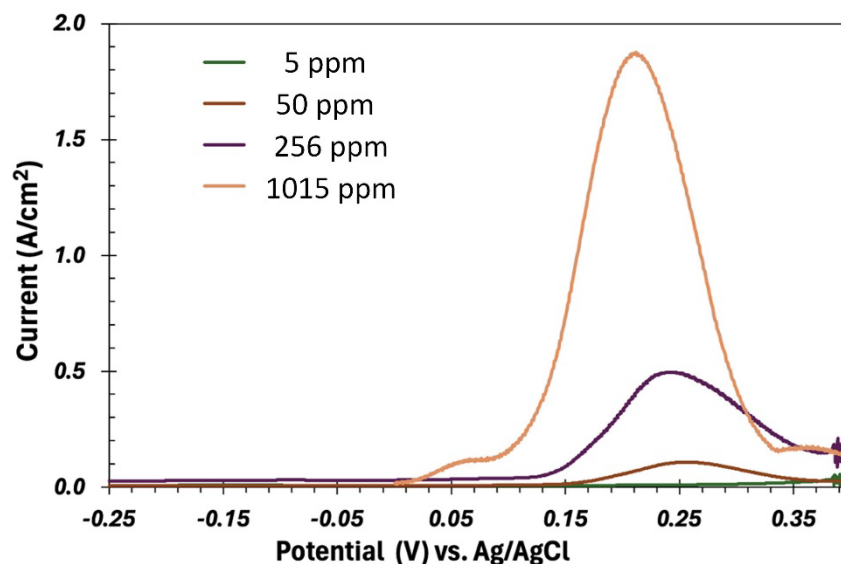


Figure 68. SWV of LiCl-KCl salt with different concentration of Li_2O at 600°C . WE: Au, CE: glassy carbon, RE: Ag/AgCl, frequency: 20 Hz [88].

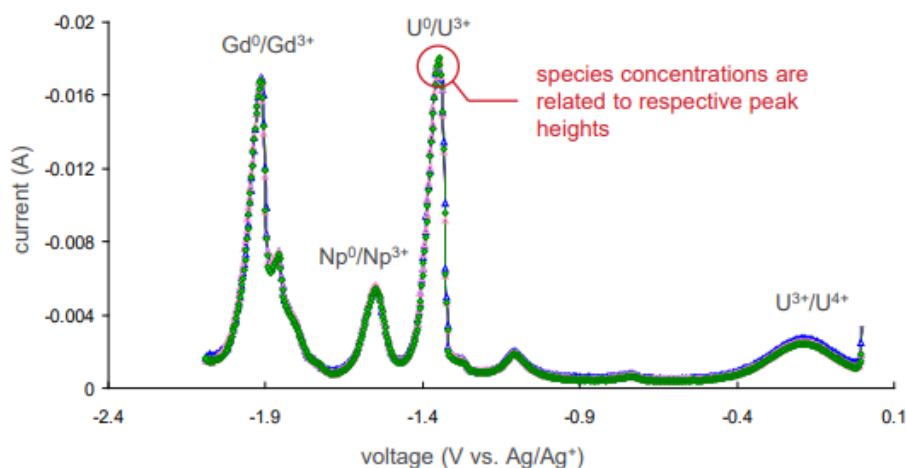


Figure 69. SWV of LiCl-KCl- UCl_3 salt for uranium electrorefining [83].

5.5.4. Normal Pulse Voltammetry

Normal Pulse Voltammetry (NPV) is a potentiostatic method where the voltage applied at the WE is in the form of a series of pulses with increasing amplitude. After each pulse, the WE potentials return to the initial values. Figure 70 shows how the voltage is applied to the WE during the NPV experiment. As shown in the figure, the current is measured near the end of each pulse, which minimizes the background non-faradic current (or charging current) and, thus, increases the measurement sensitivity. The NPV can be used for chemical monitoring, as well as to illustrate the thermodynamic and kinetic properties of molten salts for MSRs.

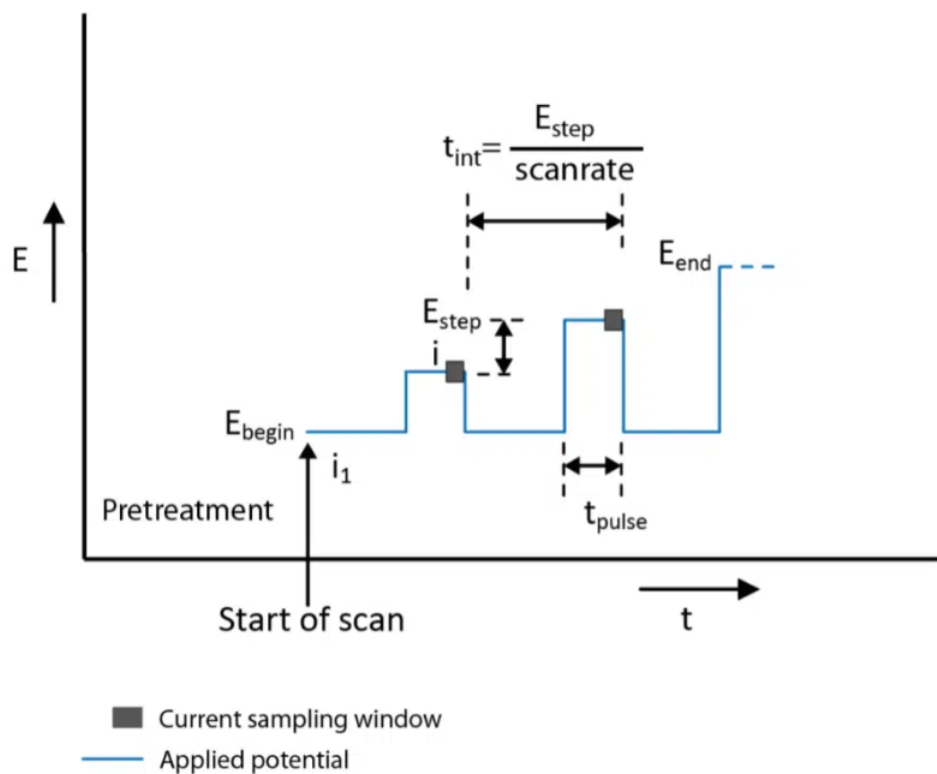


Figure 70. Potential applied during NPV [89].

Figure 71 shows the measured current versus voltage during an NPV test of a LiCl-KCl eutectic salt containing 1.5 wt% MgCl_2 and 5 wt% UCl_3 . Figure 72 shows the correlation of UCl_3 concentration (wt%) and diffusion current density in LiCl-KCl eutectic salts containing various amounts of UCl_3 (1 to 9 wt%) at 773 K.

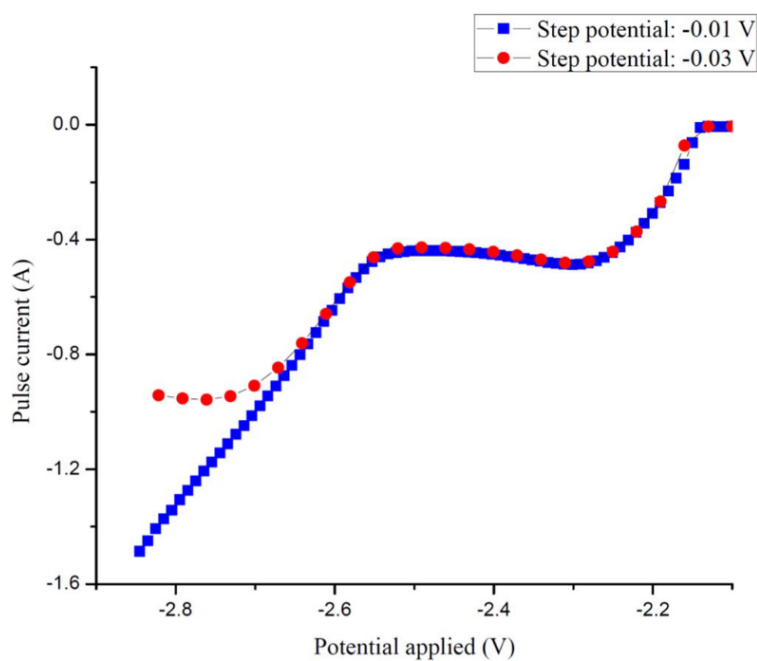


Figure 71. NPV of LiCl-KCl eutectic salt containing 1.5 wt% MgCl_2 and 5 wt% UCl_3 [90].

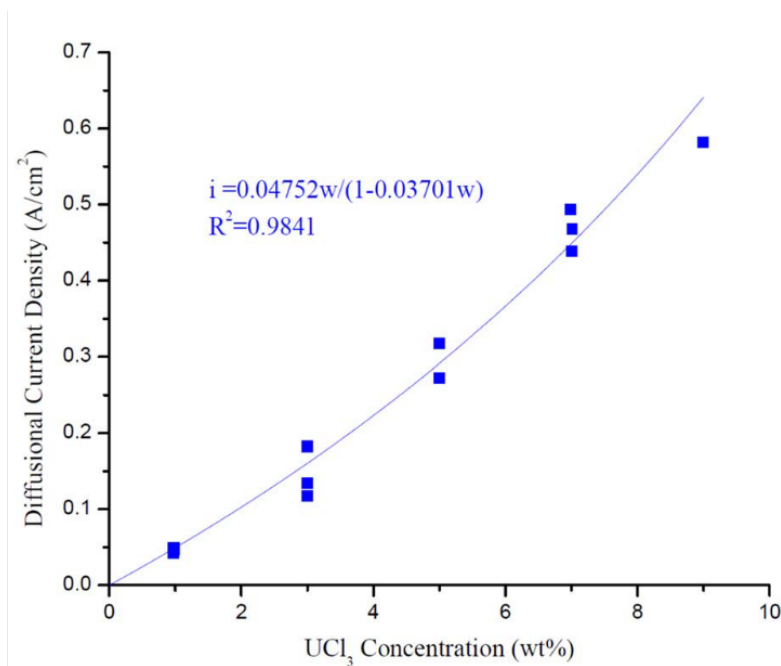


Figure 72. UCl₃ concentration and diffusion current density correlation in LiCl-KCl eutectic at 773 K [90].

5.5.5. Differential Pulse Voltammetry

Differential pulse voltammetry (DPV) is a type of electrochemical method where a series of small regular voltage pulses of the same amplitude are superimposed on the potential sweeping in the linear sweep voltammetry or staircase voltammetry measurements. DPV differs from NPV, which features a series of voltage pulses of increasing amplitudes and the potential returns to the initial value. Figure 73 schematically shows the potential applied to the WE during DPV. The current immediately before each pulse is measured, which minimizes the background non-Faradic current (or charging current) and increases the measurement sensitivity. A peak-shape current versus potential plot is expected in DPV, and the height of the current peak is typically proportional to the concentration of the constituent in the molten salt being tested, making DPV useful for MSR salt chemistry monitoring purposes.

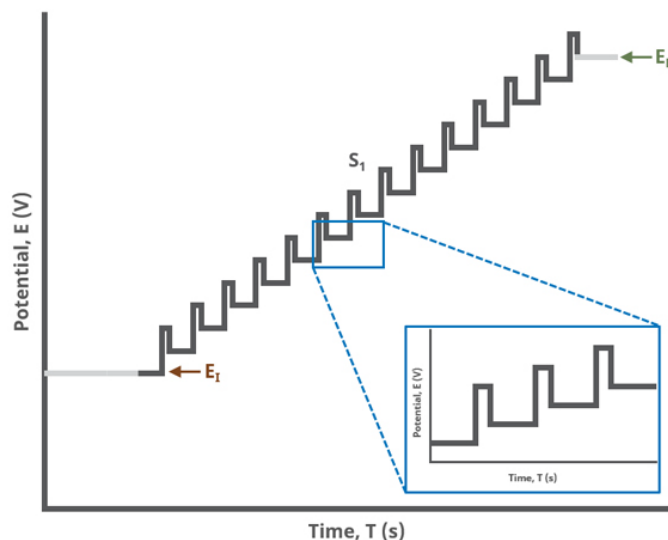


Figure 73. Schematic potential waveform of differential pulse voltammetry [91].

Figure 74 shows a DPV for a fluoride salt melt of $\text{ZrF}_4\text{--BaF}_2\text{--LaF}_3\text{--AlF}_3\text{--NaF}$ (53–20–4–3–20 wt%) (i.e., ZBLAN20) containing 7 ppm of CuF_2 and 6 ppm of FeF_2 at 550°C . Two well-separated, distinct current peaks can be seen for trace amounts of CuF_2 and FeF_2 , suggesting the high sensitivity of the DPV technique.

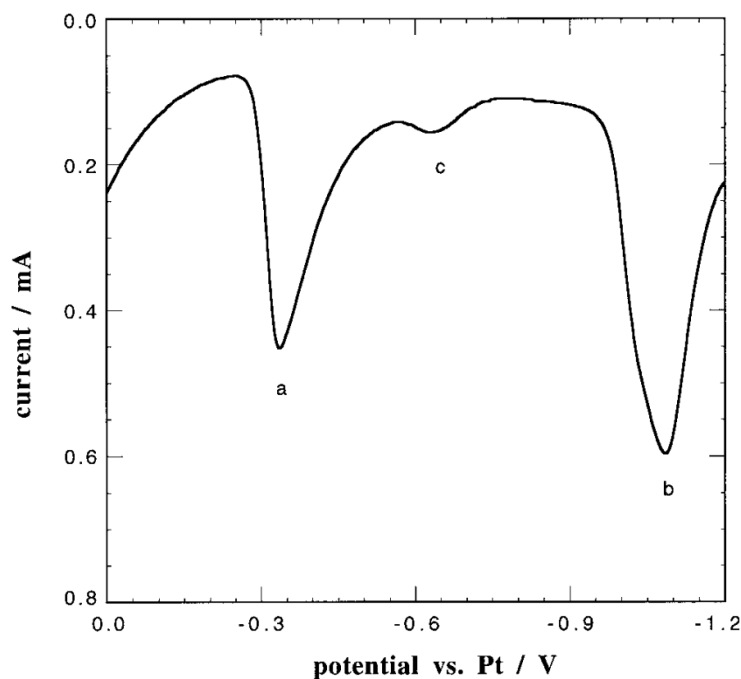


Figure 74. DPV for a ZBLAN20 fluoride melt doped with 7 ppm of CuF_2 (peak a) and 6 ppm of FeF_2 (peak b) at 550°C . Pulse amplitude is 100mV, pulse time is 20 ms, and rest time is 200 ms with a potential scan rate of 20 mV s^{-1} [92].

5.6. Some Practical Considerations of Voltammetry Techniques for MSR Applications

The measurement electrodes should be compatible with the molten salt systems being tested for long periods of time (such as three months or longer) and should not alloy with the deposited metals during the voltammetry measurements.

Electrochemical methods are typically used to study thermodynamic and kinetic properties of salt components of low concentrations in a bulk salt system. These methods have not been shown to be suitable for measuring high concentrations of UCl_3 in fuel salts for fast reactors. For electrochemical measurement of high concentrations, one option is that the salt can be sampled and diluted for analysis.

The turbulent flow of salt in MSRs may introduce some measurement challenges. The sensor may need to be placed in locations where the flow is relatively slow, or more robust electrochemical sensors can be developed to mitigate the salt turbulence effect on the measurement and sensor lifetime. For MSRs, some locations may need to be designed/engineered for voltammetry and potentiometry sensor measurements.

Although the MSRE researchers did test electrochemical sensors in a laboratory salt loop, sensors in the MSRE were not tested in real time during reactor operation. During reactor operation, samples were taken from the pump bowl and transferred to a location for measurements. In-line measurement locations should be included in MSR design and engineering.

Electrodes that perform well in one salt system may not perform as satisfactorily in other salt systems. The performance of electrodes also depends on the selected parameters in voltammetry experiments. For example, it was reported that Pt (generally inert and compatible with most molten fluoride and chloride salt) was not performing well in $\text{NaCl-CaCl}_2\text{-CeCl}_3$ salt systems, as shown in Table 8 [93]. Ag quasi-reference electrode was also found to be unreliable in MSR systems.

Table 8. Performance of electrodes in electrochemistry study of NaCl-CaCl_2 (5 wt%) CeCl_3 salt at 823 K with a scan rate of 500 mV s^{-1} [93].

Type	Stability	Repeatability	Longevity
Ag(I)/Ag RE	Very good	Very good	Very good
Pt (1.00 mm) QRE	Poor	Poor	Poor
Pt (0.25 mm) QRE	Poor	Poor	Poor
Ag (1.00 mm) QRE	Fair	Fair	Fair
Ag (0.25 mm) QRE	Poor	Poor	Poor
W (1.5 mm) QRE	Good	Very good	Very good

5.7. Salt Purification by Electrochemical Methods

It is possible that electrochemical methods can be used for salt purification to remove dissolved oxide and moisture in FLiNaK, FLiBe, and NaCl-CaCl₂ salts. The main purification mechanism is oxidation of the oxide ion (O²⁻) under anodic polarization, which will liberate the oxygen as O₂(g). The purification of molten salt by electrochemical methods has been used for preparing salts for oxide electroreduction studies. For example, CaCl₂ is very hygroscopic and usually contains significant moisture. In CaCl₂-based molten salt electrochemistry experiments, a voltage of 2.5 to 2.7 V was used applied and successfully removed moisture and some RedOx-active impurities [94, 95]. At this voltage level, the CaCl₂ salt remained stable at high-temperature and the moisture in the salt was removed. The removal of the moisture can be seen from the response of current versus time while the voltage was applied. When the current was very low or was just the background current, most of the moisture was considered removed. Different techniques have been reported in literature for salt purification. Figure 75 shows a schematic of molten salt purification achieved by removing moisture electrochemically with the assistance of hydrogen gas. Figure 76 shows an electrochemical process to remove the impurity MgOH⁺ in chloride salt using a Mg anode and W cathode. In this case, the Mg is oxidized to Mg²⁺ so the Mg²⁺ can react with oxide in the salt and form MgO solids. These insoluble solids can be further removed with techniques like filtration.

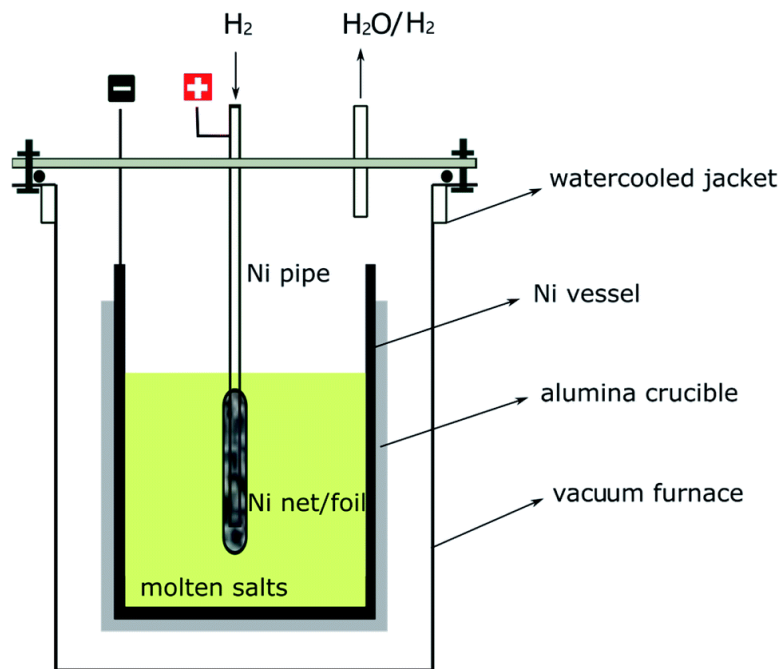


Figure 75. Purification setups for the fluoride or chloride molten salts [96].

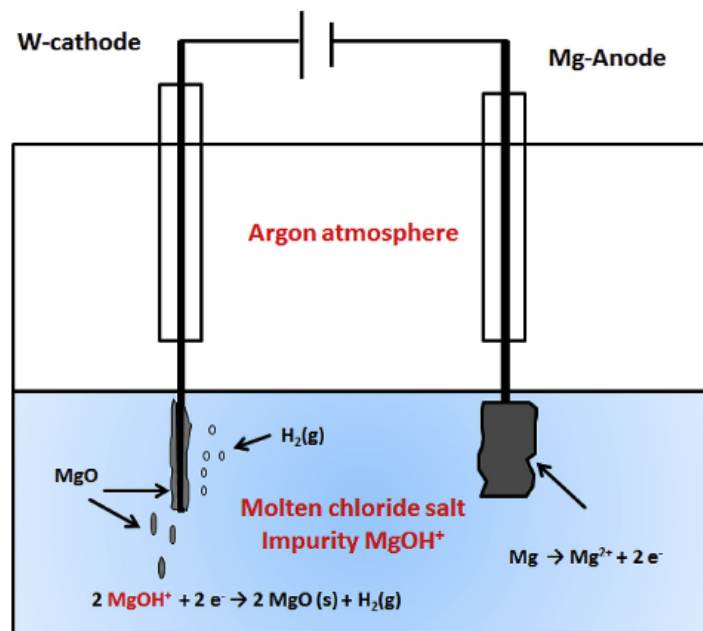


Figure 76. Schematic of assumed reactions and observed phenomena on the W-cathode and Mg-anode in electrochemical purification using pulsed over-potential [97].

Salt purification by electrochemical methods can be used for salts like FLiBe and LiCl-KCl salts that do not contain UF₄, UF₃ or UCl₄, UCl₃ components. As U³⁺ may be easier to oxidize than the oxide ion, moisture, and other impurities, processes for purification by electrochemistry likely would not be used on salt that contains fuel constituents like uranium chloride or fluorides. However, this electrochemical purification method may be readily suitable for the salt coolant used in solid fuel MSRs, such as FHRs, as those salt coolants do not include uranium.

Takeaways: Voltammetry techniques are widely used in understanding and determining the basic thermodynamic and kinetic properties of salt constituents in a bulk molten salt electrolyte. Usually a three-electrode configuration (WE, RE, and CE) is used in conjunction with a potentiostat which is controlled by a computer to perform voltammetry experiments. For monitoring the salt chemistry for MSRs, typical voltammetry measurements include LSV, CV, SWV, and PV.

6. ASSESSMENT OF ELECTROCHEMICAL METHODS FOR APPLICATION TO MOLTEN SALT REACTORS

Some electrochemical technologies can be used for multiple purposes. For example, depending on the experimental setup and design, potentiometry can be used for impurity and RedOx potential measurements, as well as measurement of specific constituents in the salt. Voltammetry with various potential ramping modes is also a capable analytical technique that can be used for different purposes. Note that the general principles and theory of these electrochemical methods are well established, but for each molten salt measurement application, the requirements, accuracy, and sensitivity may be different. The assessment for these electrochemical methods is therefore organized with respect to specific applications, mainly impurity monitoring, salt composition monitoring, and RedOx potential monitoring. It should be noted that this assessment provides an overview of the state-of-technology and does not consider regulatory aspects of the use of electrochemical monitoring technology in a license application.

The assessment of electrochemical methods for MSRs is informed by knowledge and input from the series of ORNL reports on the MSRE, literature review of peer-reviewed journals documenting the status of electrochemical methods, and publicly available reports from international nuclear organizations and national laboratories.

Based on MSRE experiences, the RedOx potential measurement of fuel salt for the MSRE is the most technically mature of the electrochemical applications. Section 5 detailed the status of different electrochemical methods for applications to MSRs. In this chapter, we summarize the maturity of techniques for MSR application purposes, including impurity and RedOx potential monitoring, and present suggestions for further information and data needs to support eventual deployment in MSRs.

6.1. Impurities Monitoring

6.1.1. Dissolved oxygen ion monitoring

Currently available literature on oxide measurements by potentiometry is focused on molten salts that contain significant amounts of oxide, such as the LiCl-Li₂O salts used for the electrochemical reduction of oxide fuels to metals. These measurement techniques are not directly relevant for MSRs applications, which are expected to contain very small amounts of oxide impurity (ppm levels) in the fuel and coolant salts. However, the same theory of operation is expected to work for MSR fuel and coolant salts, but more development and demonstration work is needed.

Square wave voltammetry (SWV) has been demonstrated to test for the presence of oxide impurities at levels of 300 ppm or higher in literature. The authors have unpublished data showing that the SWV is able to test for the presence of oxide anions at concentrations of 50 ppm in a LiCl-KCl salt system at 500°C. It may be possible that the SWV potentially can be used for monitoring the very low concentration (ppm level) oxide ion impurity in fuel salts for MSRs, but more development and demonstration is needed to assess this possibility.

6.1.2. Moisture monitoring

Moisture effects on corrosion are widely known and reported in literature. There is also research on electrochemistry studies of molten salts containing moisture, but they are typically qualitative and not applicable to quantitative moisture monitoring in molten salts.

The voltammetry techniques may be able to qualitatively detect the presence of moisture and the extent of the moisture in the salt based on the current behavior measured during voltammetry and comparing with the same types of measurements using ultra-dry salts.

Although moisture measurements by electrochemical methods are qualitative (because the moisture in fuel salts for MSR's should be strictly controlled and is typically in very low concentrations), qualitative measurements can give very useful information. If problematic levels of moisture are detected in the salt, the moisture must be reduced and could also be indicative of unexpected material entering the reactor system. The voltammetry technique for moisture monitoring is currently a qualitative analytical method and is in early stages of development.

6.2. Salt Composition Monitoring (U, Pu)

With the operation of salt-fueled MSR's, the compositions are quite complicated, with many fission products generated in the fuel salt over time. It is difficult to monitor all or most elements in the salt. The elements of most interest are U, Pu, and rare earths, which are closely related to the fuel salt chemistry. Corrosion products from reactions with structural materials can also be a good indicator of RedOx potential or salt chemistry, so monitoring of such corrosion products can also be considered an aspect of salt composition monitoring.

6.3. RedOx Potential Monitoring (U^{4+}/U^{3+} Ratio)

RedOx potential in fluoride salt fuel is determined by the UF_4/UF_3 ratio. Though there is literature in general on the RedOx potential control of chloride salt (with some assuming that a small amount of UCl_4 is in the salt as a RedOx buffer), this has not been demonstrated in literature and there are no demonstration experiments to show the UCl_4 presence in the fuel salt for chloride-salt-fueled MSR's. To evaluate the electrochemical methods applications to MSR's, it is assumed that a small amount of UCl_4 will be maintained in the fuel salt and the UCl_4/UCl_3 ratio can be used as a RedOx potential indicator. Because of the limited information on the RedOx potential control in a chloride salt reactor system, it is possible that there are various options for controlling the RedOx chemistry of the fuel salt. For example, salt can be maintained in a way that the UCl_4 formation due to the oxidizing fission process can be minimized or eliminated by contacting a reactive metal (or reducing agent). In this case, how the RedOx potential is determined is unknown and the measured RedOx potential is an overall value of the mixed potential. It has been suggested that a RedOx potential buffer similar to UCl_4/UCl_3 ratio could be used for chloride MSR's, but further research is suggested to confirm the utility of that concept.

6.4. Quality Assurance and Engineering Considerations

Quality assurance is meant to ensure that the electrochemical measurements being made are accurate and precise. Engineering considerations are meant to ensure that the electrochemical measurements are robust and maintainable.

6.4.1. Quality Assurance

According to Title 10 of the Code of Federal Regulations, Appendix B to Part 50—Quality Assurance Criteria for Nuclear Power Plants and Fuel Reprocessing Plants, quality assurance “comprises all those planned and systematic actions necessary to provide adequate confidence that a structure, system, or component will perform satisfactorily in service. Quality assurance includes quality control, which comprises those quality assurance actions related to the physical characteristics of a material, structure, component, or system which provide a means to control the quality of the material, structure, component, or system to predetermined requirements.”

For MSR electrochemical measurement methods, the goal of quality assurance is to evaluate and ensure that the results generated are accurate and reliable. The results of electrochemical methods are typically voltage (for potentiometry) or current (for voltammetry under potential control mode), which are determined by salt chemistry and selected operating parameters. The voltage and current results of electrochemical methods need to be converted, based on the scientific mechanism of the selected measurement technique, to salt chemistry data such as RedOx potential, concentrations of chemical constituents, and salt composition.

In addition to salt chemistry, there are other multiple factors related to the electrochemical methods that affect the measurement of salt chemistry by electrochemical methods. For example, in potentiometry, the stability of the RE, and the surface condition of the indicator electrode are extremely important. The drifting of the RE over long-term in-line monitoring needs to be understood and some quality assurance approach needs to be developed to address possible RE drifting issues. Other corrosion or fission products may be deposited on the surface of the indicator electrode, affecting potentiometric measurement. Even a small amount of OCP variation introduced by RE drifting and surface condition changes of electrodes could result in a high uncertainty in measurements intended to give information about salt chemistry.

For voltammetry, the current measured is affected by the surface area of the WE, so it is necessary to control and accurately know its surface area or relative surface area. This is challenging considering the mixing characteristics of the fuel salt in the pumped salt flow. Accurate knowledge and characterization of experiment parameters are needed for high confidence in electrochemical data.

To ensure the validity of the electrochemical sensor monitoring for MSRs, the authors suggest considering:

1. Multiple sensors—such as two potentiometry and two voltammetry sensors—can be designed and installed. They can be compared with each other to show whether measurement results are consistent. Consistent data can be expected, within acceptable uncertainties. Multiple sensor measurements also must be validated or calibrated using other chemical analytical techniques, like spectroscopy. Therefore, when one sensor is removed for maintenance or replacement, there would still be another working sensor to provide high-fidelity in-line measurements until the second sensor is replaced.
2. Cross calibration of sensor data. For example, the RedOx potential determined by potentiometry should be similar to that measured by voltammetry. If there are discrepancies greater than expected or desired, inspections must be conducted to ensure sensors are functioning properly.
3. Periodic calibration and validation of the sensor data by spectroscopy. Spectrophotometry can be used to quantify the U(IV)/U(III) ratio. ICP-MS can measure the isotopic composition of the salt samples. The chemical analysis results can help determine whether the results of electrochemical methods are accurate and exhibit acceptable uncertainty ranges.
4. Sensors should be replaced or maintained without prohibitive difficulties. During long-time operation, the function of the sensor may deteriorate. If it is determined that replacement or maintenance is needed, the replacement or maintenance should be performable sufficiently easily.

Note that these recommendations for quality assurance are based on the understanding and experiences of the authors. Since there are no operational experiences or demonstrations of the sensors, and the design of the MSR and associated components are complicated and may evolve with time, new options may be made available to better ensure quality assurance.

6.4.2. Engineering Aspects

Generally, a stable salt melt is desired for potentiometry and voltammetry sensors. In an MSR, however, the salt melt flows at quite high speeds. This should be considered in the sensor design. For example, the electrode should have enough stiffness so it will not bend in the flow of salt. Additionally, the salt level of the fuel salt may change over time and will change the surface area of the WE in contact of the fuel salt, which will affect the measurements. We make the following recommendations for sensors' locations:

1. In the MSRE, the pump bowl has multiple functions for adding salt and taking salt samples. Some ports may be designed for installation of these sensors.
2. Side-engineered port specifically designed for the sensor is shown in Figure 77. The salt flow in the side tube can be controlled so that it is reasonably slow and stable. This figure is only a simple schematic, so some variations may be designed. For example, a large portion in the side tube similar to the sump tank of the molten salt loop can be used for multiple sensors.
3. For MSRs that are accompanied by a chemical processing plant, the sensor can be installed in some salt tanks for chemical processing. The salt in the chemical processing plant is expected to be much more stable in terms of turbulence and flow rate. The detailed requirements of the sensors are specific to the chemical processing plant.

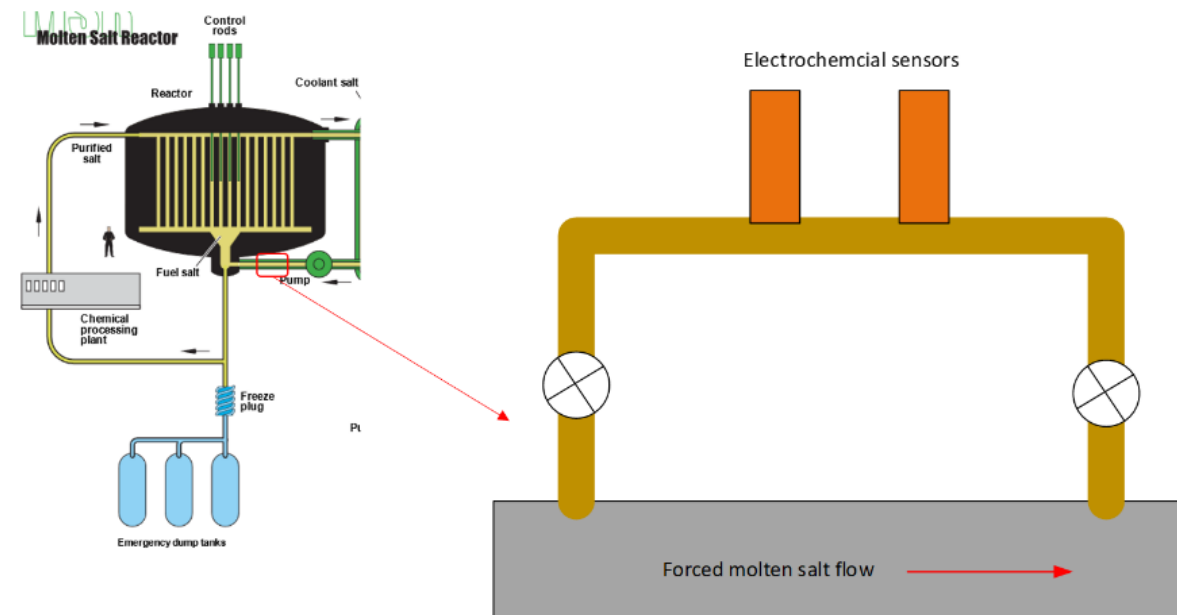


Figure 77. A schematic side tube for electrochemical sensors monitoring.

7. CONSIDERATIONS FOR ADDITIONAL RESEARCH AND DEVELOPMENT ACTIVITIES

RedOx potential monitoring and control have been demonstrated in the MSRE, which used a fluoride fuel salt, and significant amounts of operational data and experience were gained from the experiment. However, the same level of experience does not exist for chloride MSRs, and the available data and methodology for RedOx potential monitoring and control is very limited. One basic background for RedOx potential in fluoride salt is the stability issue of UF_3 , which means a higher UF_4/UF_3 ratio is needed. UCl_3 is expected to be more stable than UF_3 , and UCl_4 is known to be very corrosive compared to UCl_3 .

The understanding of the disproportionation of UCl_3 , particularly in relevant reactor operating conditions, is quite limited and the results in literature seem inconsistent. For example, in vaporization studies of pure UCl_3 salt in a temperature range of 600 to 1000°C, UCl_4 was found in the vapor phase along with UCl_3 . The presence of UCl_4 was most likely due to the disproportionation of UCl_3 via $4\text{UCl}_3 = 3\text{UCl}_4 + \text{U}$ [98]. Thermodynamic studies show that UCl_3 in NaCl- UCl_3 salt is susceptible to disproportionation, and a U(IV)/U(III) ratio of 0.003 to 5% was suggested to avoid the disproportionation of UCl_3 and mitigate the material corrosion in chloride salt reactors [99]. However, in a KCl salt with 11mol% UCl_3 that was molten at 800°C, no evidence of disproportionation of UCl_3 was observed, and it was believed that the disproportionation of UCl_3 is not significant [100]. It is possible that the UCl_3 disproportionation is temperature and concentration dependent. Moltex proposed using Zr metal in contact with chloride salts to ensure the salt is noncorrosive to materials; the rationale for the proposal is the high stability of UCl_3 in a fuel salt [101]. Therefore, more research and development are needed to understand the disproportionation of UCl_3 in relevant reactor operating conditions. If UCl_3 does undergo disproportionation, the U(IV)/U(III) ratio may be used for monitoring the redox potential. Still, more robust experimental data are needed to determine the range of U(IV)/U(III) both to prevent the UCl_3 disproportionation and to mitigate the corrosion of materials in the reactor.

If UCl_3 is stable and the disproportionation does not occur or is negligible under the reactor operating conditions, the salt chemistry management strategy is expected to be different. UCl_4 is a strong oxidizer and its concentration in fuel salt should be minimized to the extent practical or achievable. The RedOx potential measured by potentiometry likely has an overall result of mixed potential. The OCP data are difficult to quantify. A RedOx buffer is desired that is sensitive to RedOx potential changes, similar to the $\text{U}^{4+}/\text{U}^{3+}$ ratio in fluoride salt systems. If no RedOx buffer is used, in general, when the salt chemistry is more corrosive, the RedOx potential will be more positive. But this information is only qualitative and cannot give the needed quantitative RedOx information on the fuel salt because the measured values are a result of mixed potentials. That said, qualitative information is still helpful. If the measured RedOx potential shows some changes, it means something has changed in the fuel salt, which can inform the need for more accurate chemistry analysis.

The REs for fluorides that typically use graphite and boron nitride containers and these materials have not been demonstrated for long-term operation. An RE with a boron nitride container was reported to be stable for 12 days [48]. A reliable RE for long-term operation in fluoride salts is desirable. In chloride salts, although REs have shown good performance for extended periods of time, more work is still needed to exhibit no drifting during long-term monitoring of more extreme MSR environments.

In-line or in-process measurements are preferable. Voltammetry and potentiometry may offer convenient in-line measurements (or in-process measurements). Like the RedOx potential measurements, concentration of corrosion products and low concentration (less than 10 ppm) oxides and moisture impurity can also be monitored using in-line electrochemical methods, and these measurements can be designed to be operated in an automated manner. The in-line measurement can provide information that cannot be achieved by discrete sampling methods.

The lifetime of the cables for electrochemical methods under high radiation environments near the reactor vessel is unknown. Research and development work on the radiation effect on the cables for long-term operation would be beneficial.

Although this report is focused on MSR electrochemical measurement methods, it is worth noting that other techniques may be used in combination with electrochemical methods. For example, spectroscopy is able to give different information on the speciation of salt chemistry and there have been reports on combining spectroscopy and electrochemistry, called spectroelectrochemistry. Because spectroscopy requires collection of salt samples and special containers that provide transparent windows for the light source, there are inherent challenges for industrial applications particularly for long-term in-line monitoring purposes. There is active research on spectroelectrochemistry for molten salts and recent journal articles have been published. Further research would help move the laboratory research to engineering demonstration or deployment in MSRs.

8. SUMMARY

In the simplest real-world applications, the chemical conditions that develop within a molten salt system can be complex. The salt system is never completely isolated. Some MSR applications subject the salt to multiple extreme conditions that occur at the same time. These include high temperatures, contact with materials of construction, contact with gas phases, exposure to different sources of contamination, temperature gradients, pressure gradients, high-velocity flows, cavitation effects, entrainment of solids, exposure to radiation, composition changes due to fission processes, and exposure to tritium even when it is produced in a different location of the reactor.

In salt-fueled MSRs, many fission products are dissolved in molten salts. The noble metal fission products can accumulate in the molten salt or deposit on the materials of construction. Noble gas fission products are recovered in the cover gas by gas sparging. The fission process is mildly oxidizing toward the dissolved uranium (e.g., U^{4+}) in the fuel salt, which gradually makes the fuel salt more oxidizing if no measures are in place to prevent the fuel salt from becoming more oxidizing. The salt composition and RedOx potential of the fuel salt is therefore constantly changing, though slowly. Increasing the oxidizing condition of the fuel salt accelerates structural material corrosion by the fuel salt at high temperatures. The introduction of oxide and moisture can also lead to changes of RedOx potential and possibly cause other issues. Salt chemistry in an MSR is critical for reliable operation and minimization of structural material corrosion by fuel salt and therefore should be maintained in acceptable ranges.

The last two decades have seen significant advancements in molten salt electrochemistry. Electrochemical methods are moving toward providing quantitative measurement of select impurities and RedOx potentials, with potentiometry and LSV having been demonstrated even as early as in the MSRE. In-line salt chemistry monitoring is likely to be seen in the development and deployment of test or commercial MSRs, and electrochemical methods seem to have suitability for this purpose.

This report provides an overview of RedOx potentials of molten salts relevant to MSRs and the status of multiple electrochemical methods for potential MSR applications. The report also provides an overview of electrochemical measurement technologies for molten salts. Electrochemical sensor technologies for molten salts seek to observe at least one of the various signatures related to the chemical conditions of the molten salts. Generally, the purpose of the electrochemical sensor is to determine whether the chemical conditions of the molten salt are within acceptable limits of operation. The chemical conditions of concern may be related to corrosion but may also be related to aspects such as core neutronics, fissile inventory, and phase behaviors. It is important to understand what signature the sensor is designed to observe, and what that signature says about the condition of the salt.

9. ACKNOWLEDGMENTS

This work was sponsored financially by NRC. The review and comments of Steven Herrmann and Prabhat Tripathy are greatly appreciated.

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Appendix A

Dynamic Behavior of Halide Systems

Dynamic Behavior of the Cl, Cr, Fe, Ni System

Building on the concepts of equilibrium thermodynamics and ideal behaviors presented earlier in this report, the dynamic composition behavior of the Cl, Cr, Fe, Ni system at 500°C is shown in Figure 78. This system was selected because these metals are alloying elements in common stainless steels. In systems where precipitation occurs, the precipitating compound could be formed continuously, and such precipitation behavior is outside of the scope of this example. The model begins with initial amounts of Cr (1.0 kmol), Fe (0.9 kmol), and Ni (0.8 kmol) metals, in contact with Ar(g) (1.0 kmol) to which $\text{Cl}_2(\text{g})$ is incrementally added (up to 4.0 kmol). The figure shows that $\text{Cl}_2(\text{g})$ first reacts with chromium to form CrCl_2 . When the chromium is consumed, $\text{Cl}_2(\text{g})$ then reacts with iron to form FeCl_2 . Finally, when the iron is consumed, the $\text{Cl}_2(\text{g})$ then reacts with nickel to form NiCl_2 . Only after the three metals are consumed is $\text{Cl}_2(\text{g})$ liberated into the system.

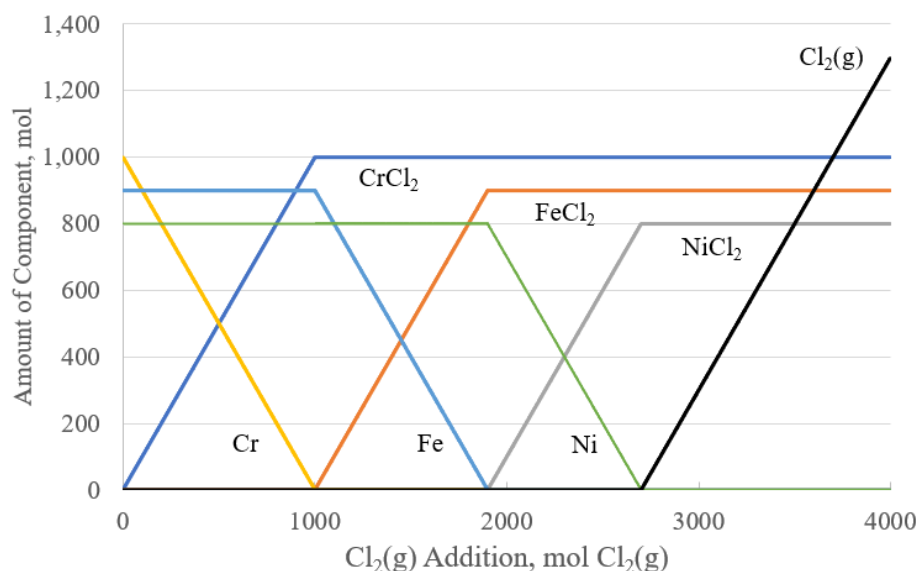


Figure 78. Equilibrium composition as a function of $\text{Cl}_2(\text{g})$ addition to the Cl, Cr, Fe, Ni system at 500°C (Version 1).

Figure 78 by may give the impression that the metals Cr, Fe, and Ni are completely consumed before $\text{Cl}_2(\text{g})$ is present. And that FeCl_2 is not formed until the chromium is consumed, and that NiCl_2 is not formed until the chromium and iron are consumed. However, there are small quantities of these species in the calculation that are not readily observable with the scale chosen for the ordinate of Figure 78. The same data are shown in Figure 79 with a logarithmic ordinate. This figure shows that CrCl_2 , FeCl_2 , NiCl_2 , and $\text{Cl}_2(\text{g})$ exist from the beginning, albeit at very low amounts, which is a common feature of the mathematics of equilibrium thermodynamic behaviors of multicomponent systems. All of thermodynamics is a mathematical model used to predict the behaviors of chemical systems. The calculation of extremely small chemical potentials is mathematically congruent and thermodynamically sound, which results in the calculations of extremely small concentrations and amounts.

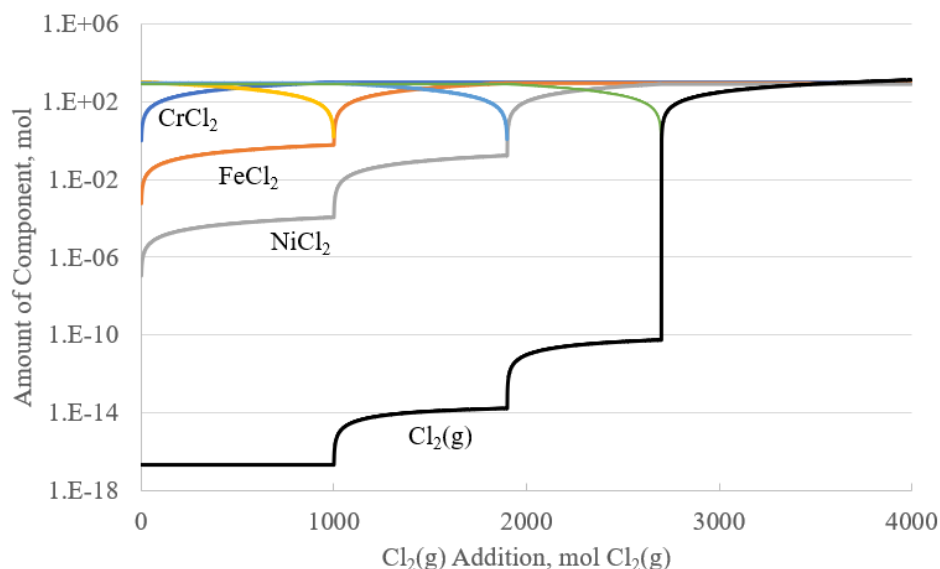


Figure 79. Equilibrium composition as a function of $\text{Cl}_2(\text{g})$ addition to the Cl, Cr, Fe, Ni system at 500°C (Version 2).

In this example, the Cl chemical potential is taken as the partial pressure of $\text{Cl}_2(\text{g})$ in the gas phase, which is a mixture of $\text{Cl}_2(\text{g})$ and $\text{Ar}(\text{g})$ at 1 bar total pressure. The equilibrium composition as a function of the Cl chemical potential is shown in Figure 80. This figure is consistent with Figure 78 and Figure 79, as $\text{Cl}_2(\text{g})$ is added to the system, the Cl chemical potential increases, and Cr, Fe, and Ni are incrementally chlorinated.

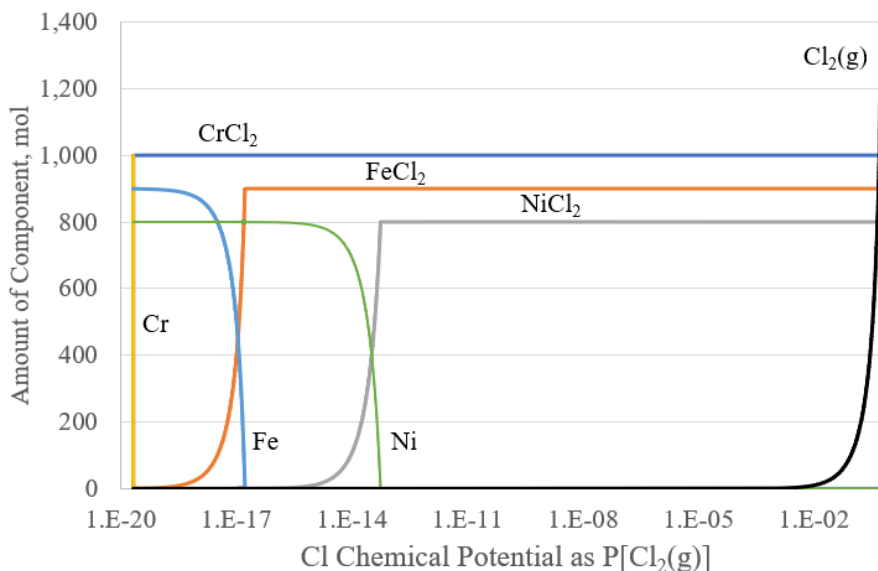


Figure 80. Equilibrium composition as a function of the Cl chemical potential of the Cl, Cr, Fe, Ni system at 500°C.

The Cl chemical potential as a function of the amount of $\text{Cl}_2(\text{g})$ added to the system is shown in Figure 81. The plateaus in the figure correspond to the phase transition of the metals to metal chlorides. The Cl chemical potential is buffered during the phase transitions. For example, the Cl chemical potential is relatively stable until most of the chromium is chlorinated. Then the Cl chemical potential increases to the next plateau where iron is chlorinated. Then the Cl chemical potential increases to the next plateau where nickel is chlorinated. Finally, the Cl chemical potential increases toward 1 as it appears in appreciable quantities in the gas phase. In Figure 81, the Cl chemical potential is buffered three times. As chromium is chlorinated to CrCl_2 , as iron is chlorinated to FeCl_2 , and finally as nickel is chlorinated to NiCl_2 .

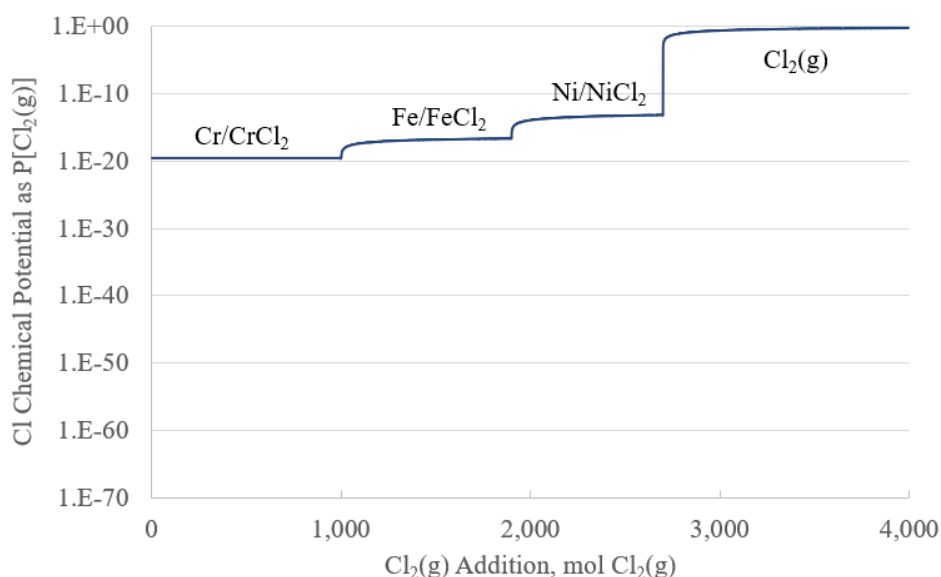


Figure 81. Cl chemical potential as a function of $\text{Cl}_2(\text{g})$ addition to the Cl, Cr, Fe, Ni system at 500°C . Unit of pressure: bar.

TAKEAWAYS: Under equilibrium thermodynamic conditions, the chlorine chemical potential is buffered during the phase transitions between metals and metal chlorides. Generally, the more electropositive metals are chlorinated preferentially. Thermodynamic models often predict the presence of very small quantities of reactants and products. These small quantities can appear as concentrations that are well below the ability of analytical chemistry methods to detect.

Dynamic Behavior of the F, Cr, Fe, Ni System

In a direct comparison to the previous example, the dynamic behavior of the F, Cr, Fe, Ni system at 500°C is shown in Figure 82. Stainless-steel alloys have been proposed for use in some MSR designs, the following comparison shows the differences in behaviors between chloride and fluoride systems. The model begins with initial amounts of Cr (1.0 kmol), Fe (0.9 kmol), and Ni (0.8 kmol) metals, in contact with $\text{Ar}(\text{g})$ (1.0 kmol) to which $\text{F}_2(\text{g})$ is incrementally added (up to 4.0 kmol). The figure shows that $\text{F}_2(\text{g})$ reacts with chromium, iron, and nickel in the same sequence as before. Only after the three metals are consumed is $\text{F}_2(\text{g})$ liberated into the system. Stoichiometrically, Figure 82 looks exactly like Figure 78.

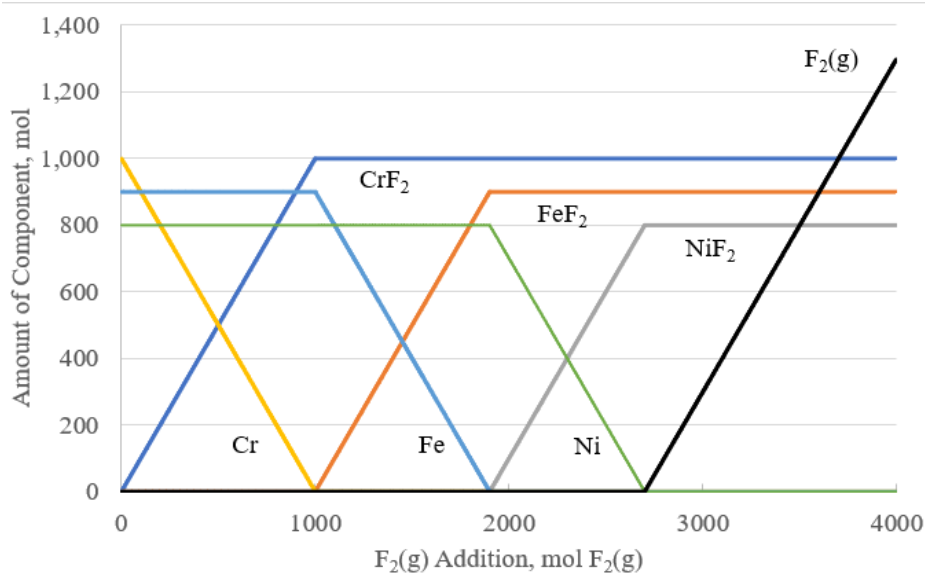


Figure 82. Equilibrium composition as a function of $F_2(g)$ addition to the F, Cr, Fe, Ni system at 500°C (Version 1).

The same data are shown in Figure 83 with a logarithmic ordinate. This figure shows that CrF_2 , FeF_2 , NiF_2 , and $F_2(g)$ exist from the beginning, albeit at very low amounts.

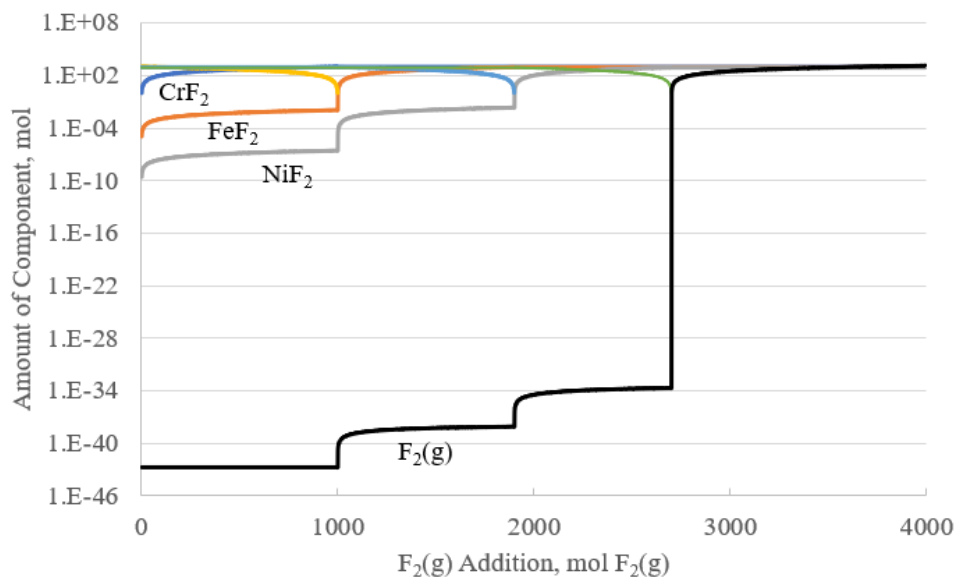


Figure 83. Equilibrium composition as a function of $F_2(g)$ addition to the F, Cr, Fe, Ni system at 500°C (Version 2).

The equilibrium composition as a function of the F chemical potential is shown in Figure 84. This figure is consistent with Figure 82 and Figure 83, as $F_2(g)$ is added to the system, the F chemical potential increases, and Cr, Fe, and Ni are incrementally fluorinated.

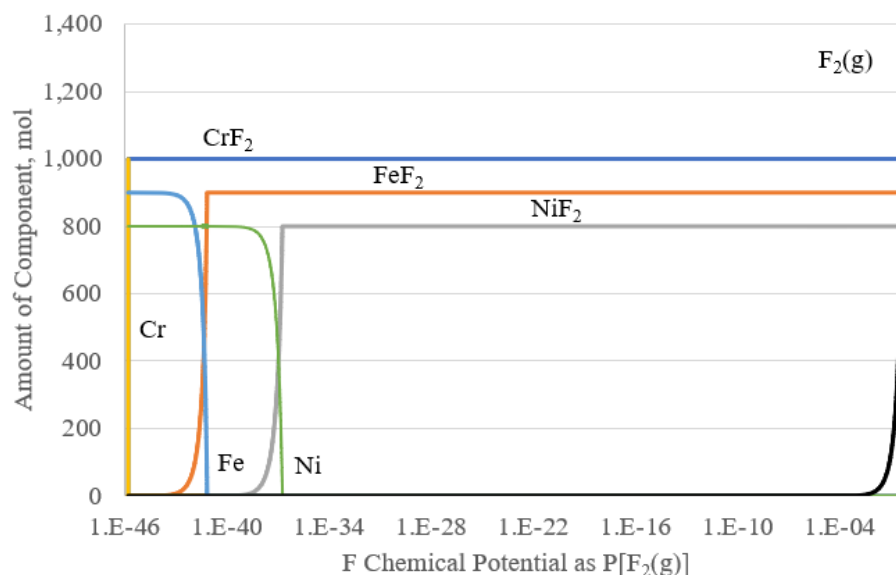


Figure 84. Equilibrium composition as a function of the F chemical potential of the F, Cr, Fe, Ni system at 500°C.

The F chemical potential as a function of the amount of $F_2(g)$ added to the system is shown in Figure 85. The plateaus in the figure correspond to the phase transition of the metals to metal fluorides. The F chemical potential is buffered during the phase transitions. For example, the F chemical potential is relatively stable until most of the chromium is fluorinated. Then the F chemical potential increases to the next plateau where iron is fluorinated. Then the F chemical potential increases to the next plateau where nickel is fluorinated. Finally, the F chemical potential increases toward 1 as it appears in appreciable quantities in the gas phase.

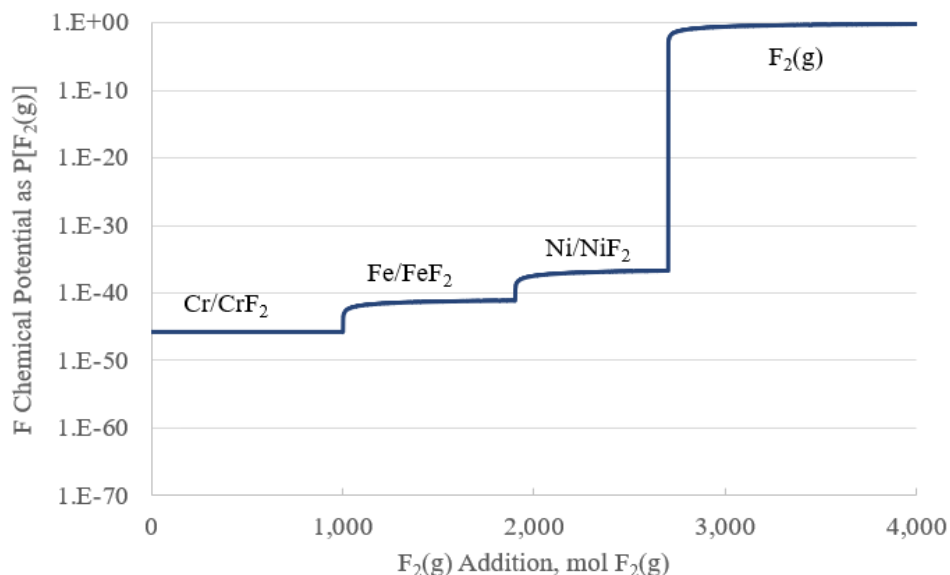


Figure 85. F chemical potential as a function of $F_2(g)$ addition to the F, Cr, Fe, Ni system at 500°C. Unit of pressure: bar.

TAKEAWAYS: Under equilibrium thermodynamic conditions, the fluorine chemical potential is buffered during the phase transitions between metals and metal fluorides. The data from Figure 81 and Figure 85 are shown together in Figure 86. The fluorine chemical potentials needed for fluorination are much lower than the chlorine chemical potentials needed for chlorination.

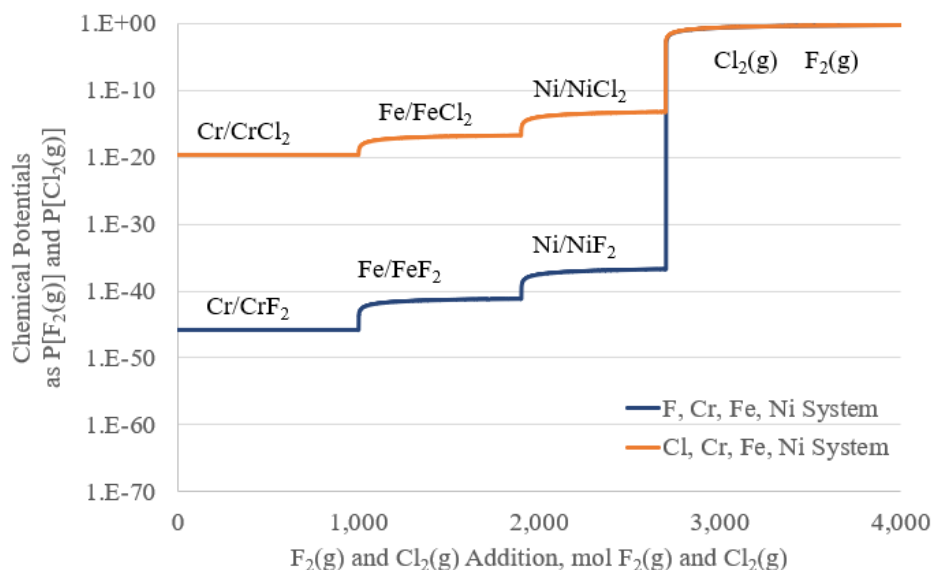


Figure 86. Cl and F chemical potential as functions of $\text{Cl}_2(\text{g})$ and $\text{F}_2(\text{g})$ additions to the Cl, Cr, Fe, Ni, and F, Cr, Fe, Ni systems, respectively, at 500°C . Unit of pressure: bar.

Dynamic Behavior of the Cl, U System

The dynamic behavior of the Cl, U system at 500°C is shown in Figure 87. Uranium chloride is potentially a fissile component for a chloride MSR. The model begins with an initial amount of U (1.0 kmol) metal, in contact with $\text{Ar}(\text{g})$ (1.0 kmol) to which $\text{Cl}_2(\text{g})$ is incrementally added (up to 3.1 kmol). The figure shows that $\text{Cl}_2(\text{g})$ first reacts with uranium to form UCl_3 . However, once the bulk of the uranium is converted to UCl_3 , additional $\text{Cl}_2(\text{g})$ converts the UCl_3 to UCl_4 . Only after the bulk of the uranium is converted to UCl_4 is $\text{Cl}_2(\text{g})$ liberated into the system.

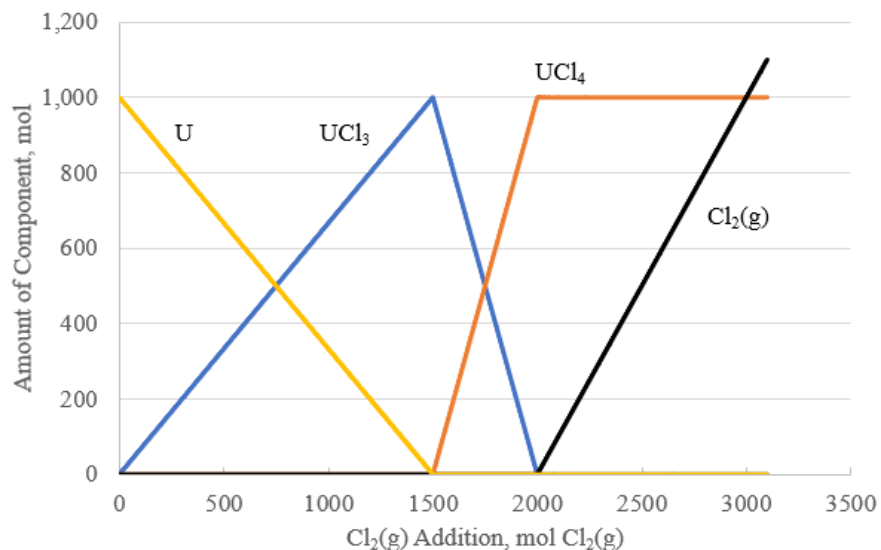


Figure 87. Equilibrium composition as a function of $\text{Cl}_2(\text{g})$ addition to the Cl, U system at 500°C (Version 1).

The same data as above are shown in Figure 88 with a logarithmic ordinate. This figure shows that UCl_3 , UCl_4 , and $\text{Cl}_2(\text{g})$ exist from the beginning, albeit at very low amounts.

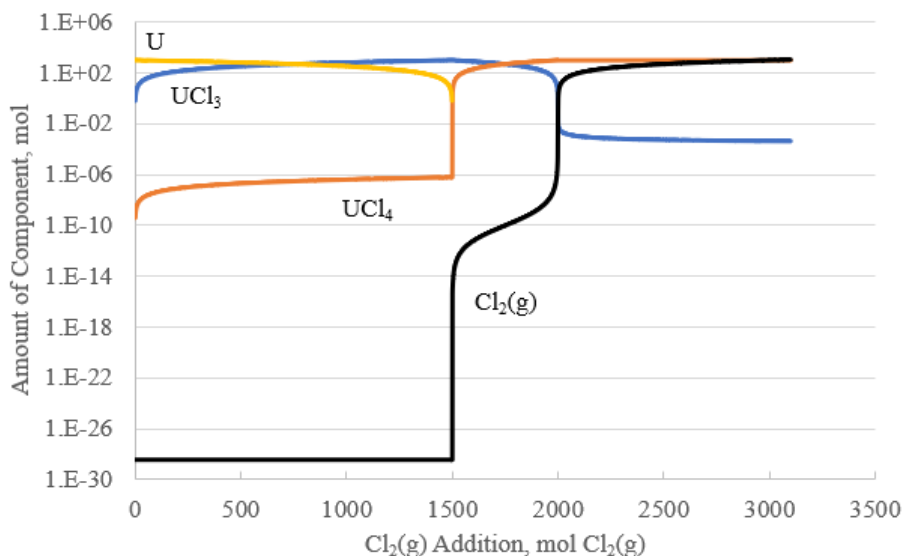


Figure 88. Equilibrium composition as a function of $\text{Cl}_2(\text{g})$ addition to the Cl, U system at 500°C (Version 2).

The equilibrium composition as a function of the Cl chemical potential is shown in Figure 89. This figure is consistent with Figure 87 and Figure 88, as $\text{Cl}_2(\text{g})$ is added to the system, the Cl chemical potential increases, and U is chlorinated to UCl_3 , and UCl_3 is chlorinated to UCl_4 .

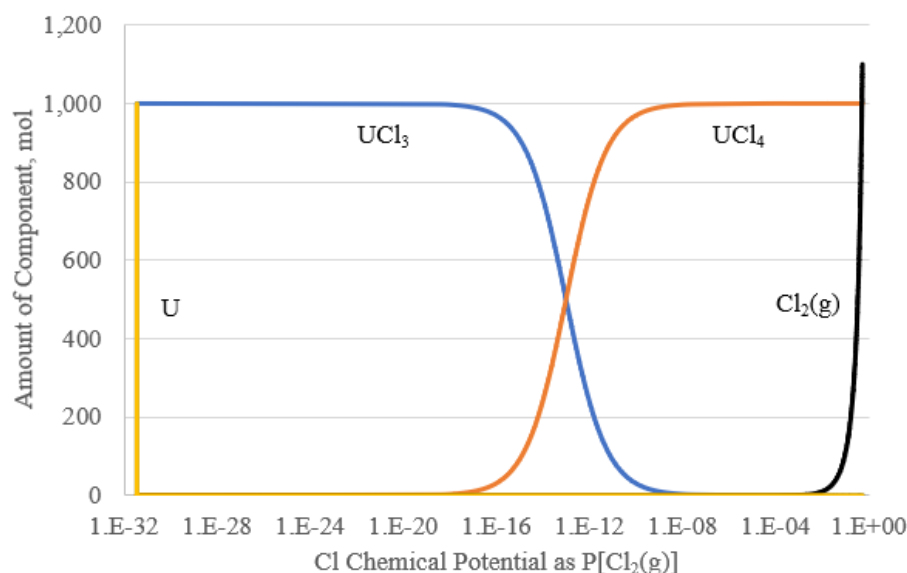


Figure 89. Equilibrium composition as a function of the Cl chemical potential of the Cl, U system at 500°C.

The Cl chemical potential as a function of the amount of $\text{Cl}_2(\text{g})$ added to the system is shown in Figure 90. The plateaus in the figure correspond to the phase transition of uranium metal to UCl_3 , and UCl_3 to UCl_4 . The Cl chemical potential is buffered during the phase transitions. For example, the Cl chemical potential is relatively stable until most of the uranium is chlorinated. Then the Cl chemical potential increases to the next plateau where UCl_3 is chlorinated to UCl_4 . Finally, the Cl chemical potential increases toward 1 as it appears in appreciable quantities in the gas phase.

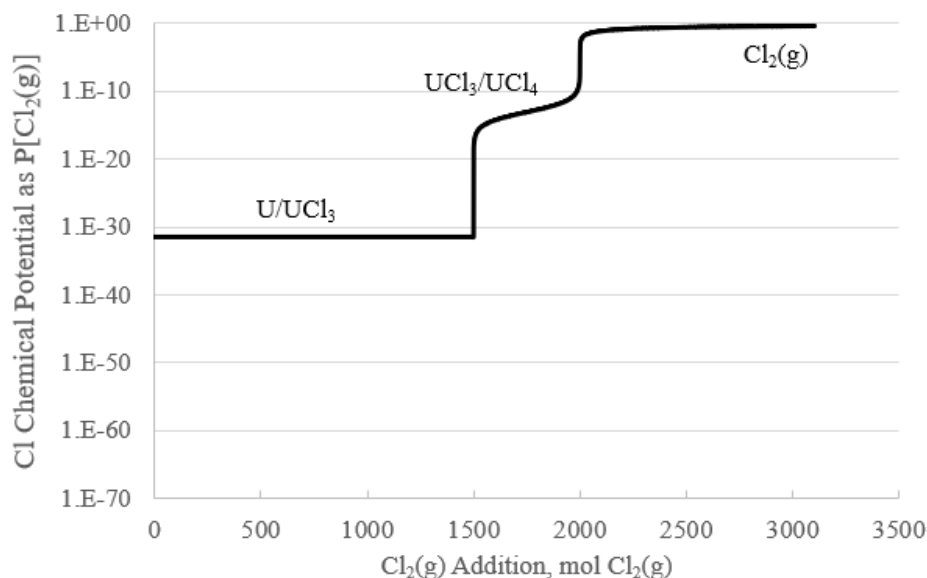


Figure 90. Cl chemical potential as a function of $\text{Cl}_2(\text{g})$ addition to the Cl, U system at 500°C. Unit of pressure: bar.

TAKEAWAYS: Like the phase transitions from metals to metal chlorides, the chlorine chemical potential is buffered during the phase transition of chlorides to higher valence chlorides like UCl_3 to UCl_4 . For all practical purposes, uranium metal is not stable in the presence of UCl_4 . Therefore, when UCl_3 and UCl_4 are present in appreciable quantities, the chlorine chemical potential is buffered and virtually no uranium metal is present.

Dynamic Behavior of the Cl, Na, U System

This system is a modification to the previous system by the addition of sodium. A mixture of sodium chloride and uranium chloride is potentially a fuel salt for a chloride MSR. The dynamic behavior of the Cl, Na, U system at 500°C is shown in Figure 91. The model begins with an initial amount of Na (1.32 kmol) and U (0.68 kmol) metals, in contact with Ar(g) (1.0 kmol) to which $\text{Cl}_2(\text{g})$ is incrementally added (up to 3.1 kmol). This ratio of Na:U was selected to give the eutectic composition of the NaCl-UCl_3 system. The figure shows that $\text{Cl}_2(\text{g})$ first reacts with sodium to form NaCl . Then once the bulk of the sodium is converted to NaCl , $\text{Cl}_2(\text{g})$ reacts with the uranium to form UCl_3 . And once the bulk of the uranium is converted to UCl_3 , additional $\text{Cl}_2(\text{g})$ converts the UCl_3 to UCl_4 . Only after the bulk of the uranium is converted to UCl_4 is $\text{Cl}_2(\text{g})$ liberated into the system.

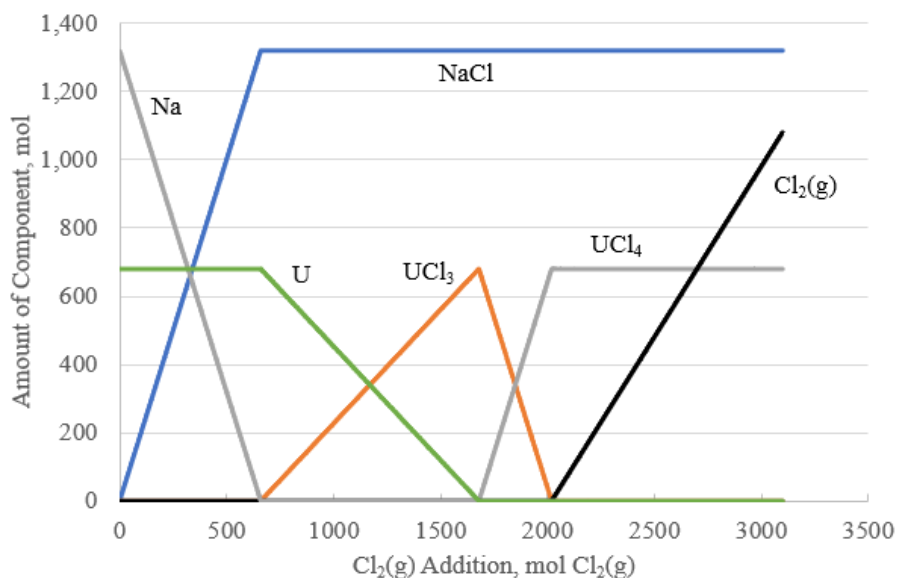


Figure 91. Equilibrium composition as a function of $\text{Cl}_2(\text{g})$ addition to the Cl, Na, U system at 500°C (Version 1).

The same data as above are shown in Figure 92 with a logarithmic ordinate. This figure shows that UCl_3 , UCl_4 , and $\text{Cl}_2(\text{g})$ exist from the beginning, albeit at very low amounts.

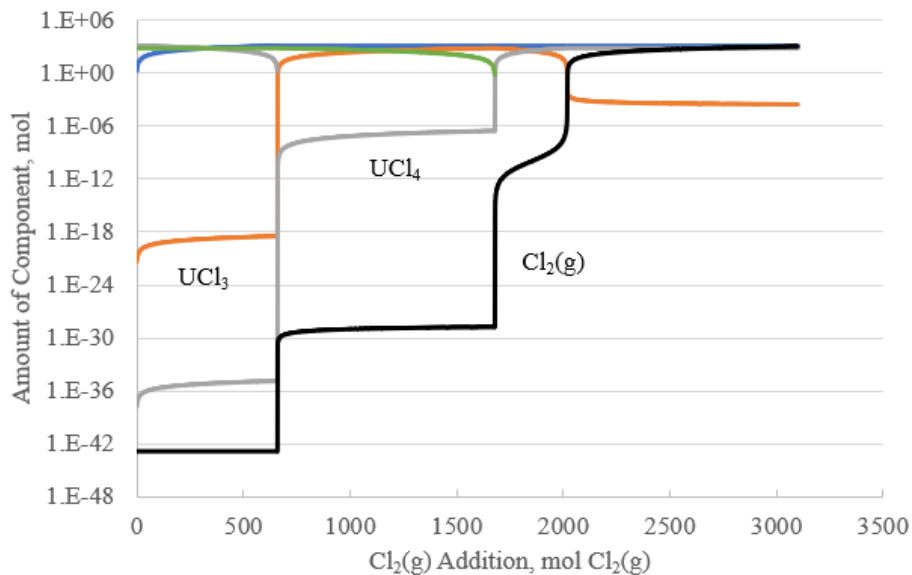


Figure 92. Equilibrium composition as a function of $\text{Cl}_2(\text{g})$ addition to the Cl, Na, U system at 500°C (Version 2).

The equilibrium composition as a function of the Cl chemical potential is shown in Figure 93. This figure is consistent with Figure 78, Figure 91 and Figure 92, as $\text{Cl}_2(\text{g})$ is added to the system, the Cl chemical potential increases, and Na is chlorinated to NaCl, U is chlorinated to UCl_3 , and UCl_3 is chlorinated to UCl_4 .

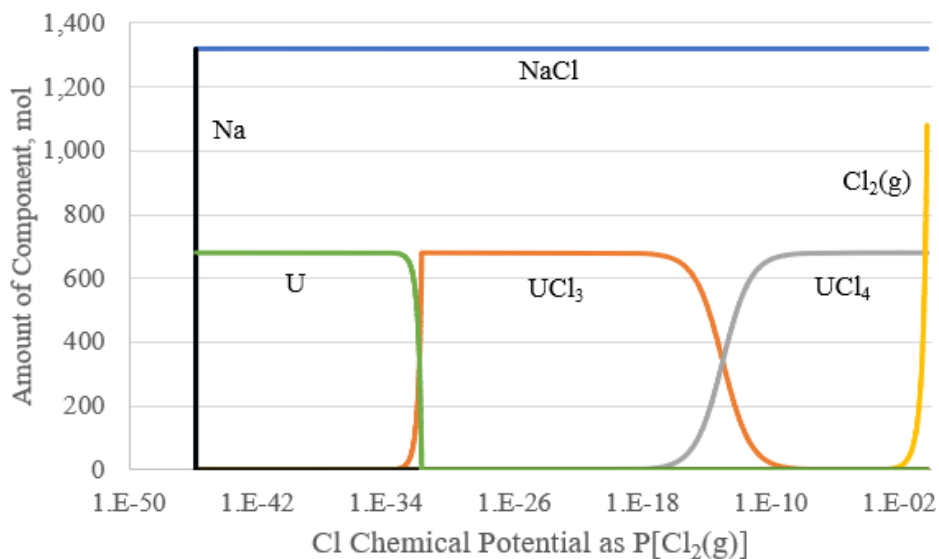


Figure 93. Equilibrium composition as a function of the Cl chemical potential of the Cl, Na, U system at 500°C .

The Cl chemical potential as a function of the amount of $\text{Cl}_2(\text{g})$ added to the system is shown in Figure 94. The plateaus in the figure correspond to the phase transition of sodium to NaCl , uranium to UCl_3 , and UCl_3 to UCl_4 . The Cl chemical potential is buffered during the phase transitions. For example, the Cl chemical potential is relatively stable until most of the sodium is chlorinated. Then the Cl chemical potential increases to the next plateau where uranium is chlorinated. Then the Cl chemical potential increases to the next plateau where UCl_3 is chlorinated to UCl_4 . Finally, the Cl chemical potential increases toward 1 as it appears in appreciable quantities in the gas phase.

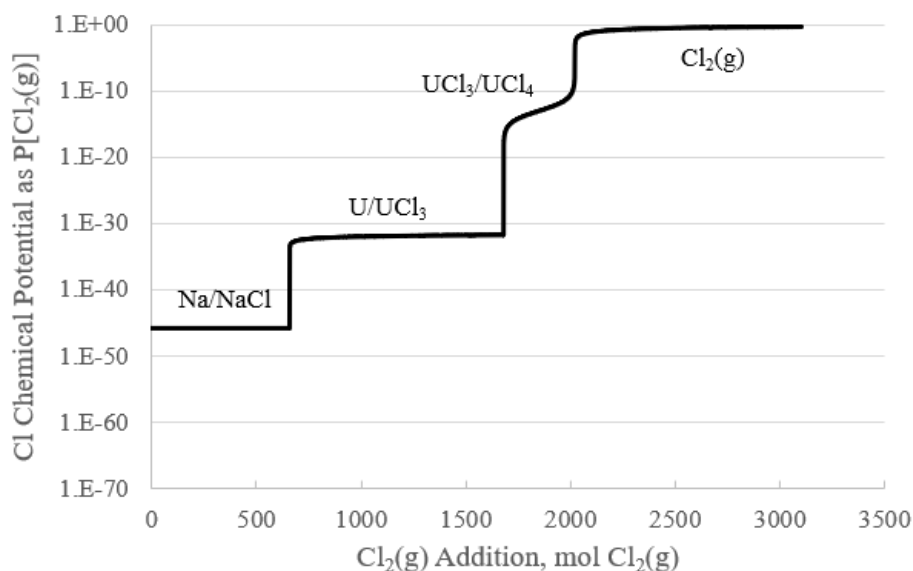


Figure 94. Cl chemical potential as a function of $\text{Cl}_2(\text{g})$ addition to the Cl, Na, U system at 500°C . Unit of pressure: bar.

TAKEAWAYS: The data from Figure 90 and Figure 94 are shown together in Figure 95. The introduction of sodium into the system had little effect on the chlorine chemical potentials associated with the U to UCl_3 , and the UCl_3 to UCl_4 phase transitions.

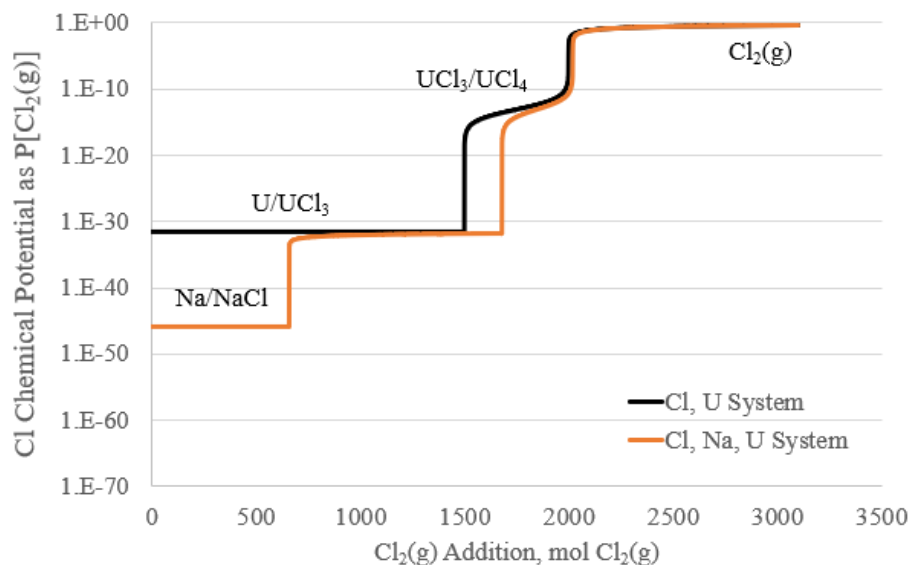


Figure 95. Cl chemical potential as a function of $\text{Cl}_2(\text{g})$ addition to the Cl, U, and Cl, Na, U systems at 500°C . Unit of pressure: bar.

Dynamic Behavior of the F, U System

The dynamic behavior of the F, U system at 500°C is shown in Figure 96. Uranium fluoride is potentially a fissile component for a fluoride MSR. The model begins with an initial amount of U (1.0 kmol) metal, in contact with Ar(g) (1.0 kmol) to which $F_2(g)$ is incrementally added (up to 3.1 kmol). The figure shows that $F_2(g)$ first reacts with uranium to form UF_3 . However, once the bulk of the uranium is converted to UF_3 , additional $F_2(g)$ converts the UF_3 to UF_4 . Only after the bulk of the uranium is converted to UF_4 is $F_2(g)$ liberated into the system.

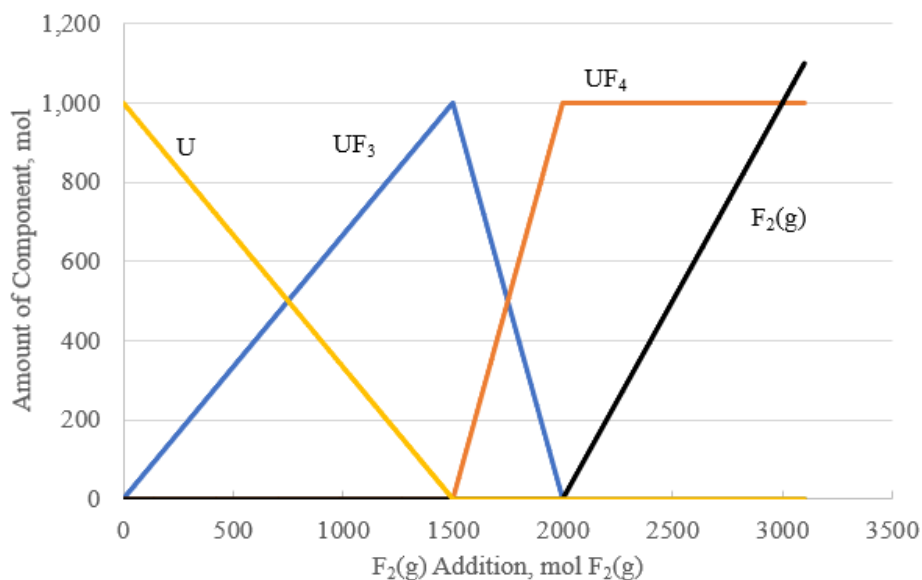


Figure 96. Equilibrium composition as a function of $F_2(g)$ addition to the F, U system at 500°C (Version 1).

The same data as above are shown in Figure 97 with a logarithmic ordinate. This figure shows that UF_3 , UF_4 , and $F_2(g)$ exist from the beginning, albeit at very low amounts.

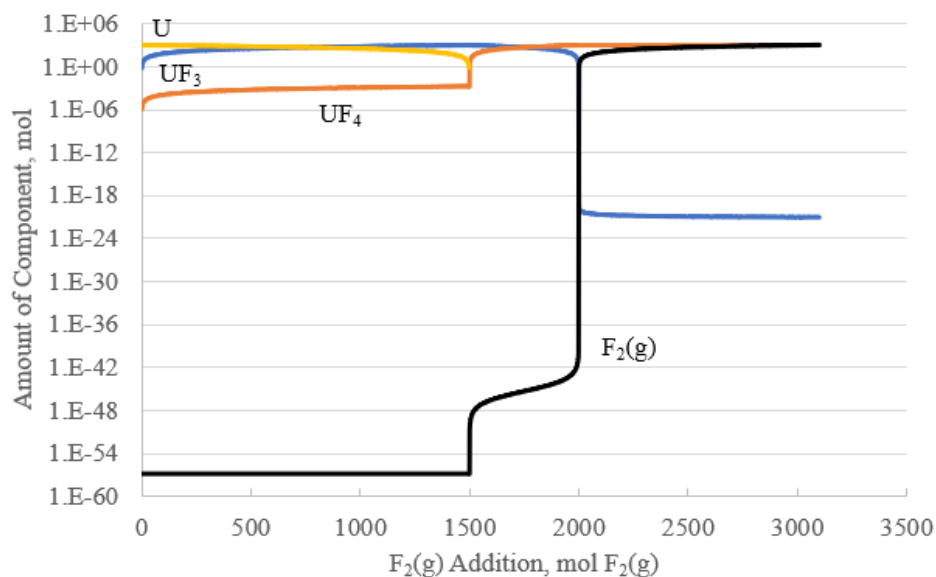


Figure 97. Equilibrium composition as a function of $F_2(g)$ addition to the F, U system at 500°C (Version 2).

The equilibrium composition as a function of the F chemical potential is shown in Figure 98. This figure is consistent with Figure 78, Figure 96 and Figure 97, as $F_2(g)$ is added to the system, the F chemical potential increases, and U is fluorinated to UF_3 , and UF_3 is fluorinated to UF_4 .

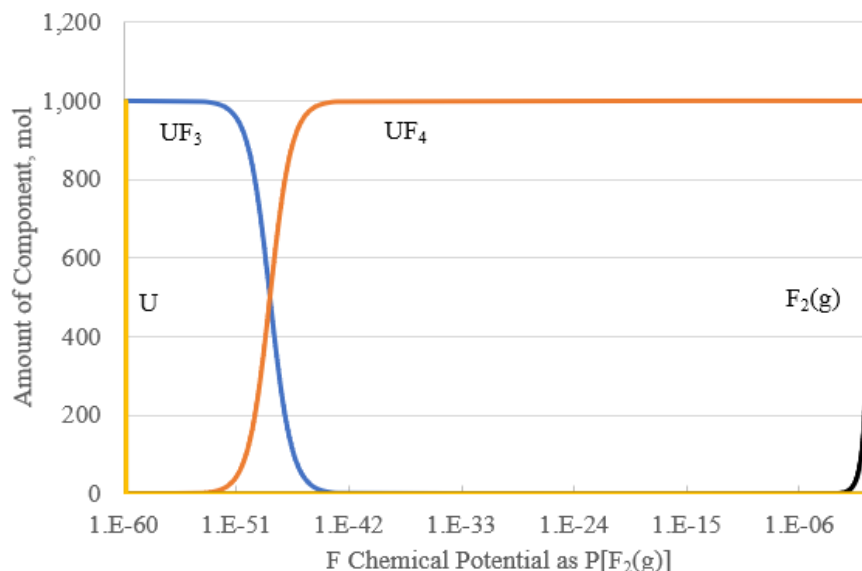


Figure 98. Equilibrium composition as a function of the F chemical potential of the F, U system at 500°C.

The F chemical potential as a function of the amount of $F_2(g)$ added to the system is shown in Figure 99. The plateaus in the figure correspond to the phase transition of uranium metal to UF_3 , and UF_3 to UF_4 . The F chemical potential is buffered during the phase transitions. For example, the F chemical potential is relatively stable until most of the uranium is fluorinated. Then the F chemical potential increases to the next plateau where UF_3 is fluorinated to UF_4 . Finally, the F chemical potential increases toward 1 as it appears in appreciable quantities in the gas phase.

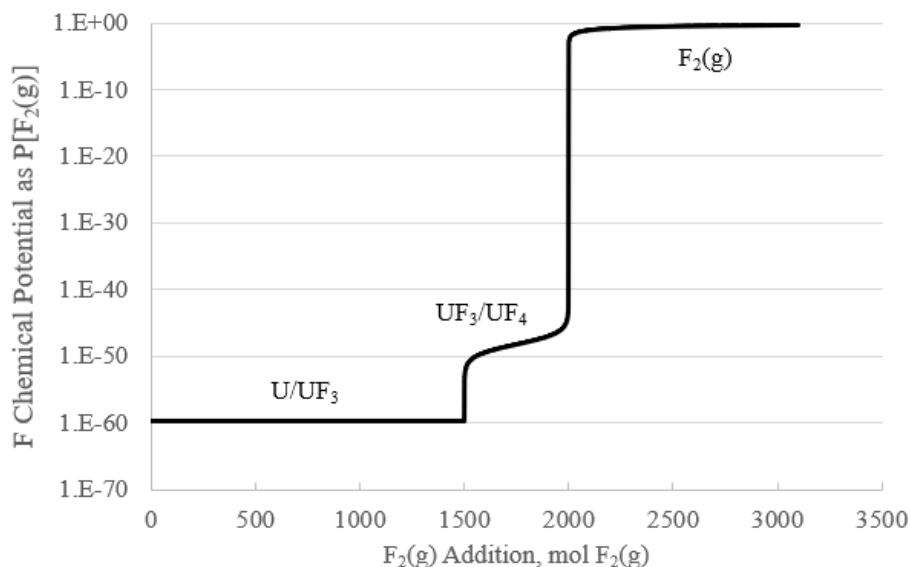


Figure 99. F chemical potential as a function of $F_2(g)$ addition to the F, U system at 500°C. Unit of pressure: bar.

TAKEAWAYS: Like the phase transitions from metals to metal fluorides, the fluorine chemical potential is buffered during the phase transition of fluorides to higher valence fluorides like UF_3 to UF_4 . The data from Figure 90 and Figure 99 are shown together in Figure 100. The fluorine chemical potential needed for fluorination are much lower than the chlorine chemical potentials needed for chlorination.

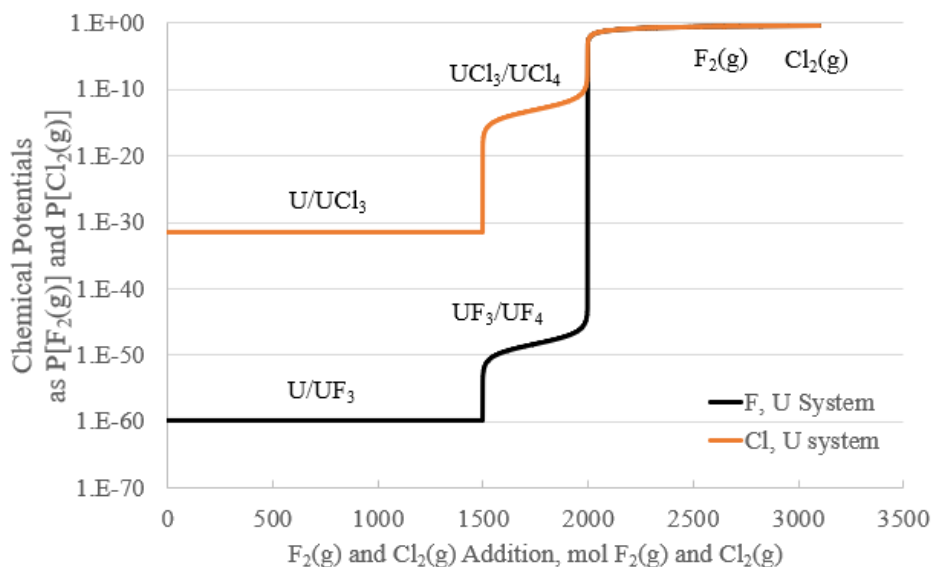


Figure 100. Cl and F chemical potential as functions of $\text{Cl}_2(\text{g})$ and $\text{F}_2(\text{g})$ additions to the Cl, U, and F, U systems, respectively, at 500°C . Unit of pressure: bar.

Dynamic Behavior of the F, Li, Be, U, Zr System (Version 1)

The Oak Ridge National Laboratory (ORNL) MSRE used a nominal fuel salt composition of 65 LiF , 29.2 BeF_2 , 0.8 UF_4 , and 5 ZrF_4 (mol%). The model begins with these four metals, in these ratios to each other, and incrementally adds $\text{F}_2(\text{g})$. The dynamic behavior of the F, Li, Be, U, Zr system at 500°C is shown in Figure 101. Initial amounts of Li (6.5 kmol), Be (2.92 kmol), U (0.08 kmol), and Zr (0.5 kmol) metals, are in contact with $\text{Ar}(\text{g})$ (1.0 kmol) to which $\text{F}_2(\text{g})$ is incrementally added (up to 10.0 kmol). The figure shows that $\text{F}_2(\text{g})$ first reacts with lithium to form LiF . However, once the bulk of the lithium is converted to LiF , additional $\text{F}_2(\text{g})$ converts the beryllium to BeF_2 . Following this, the sequence continues with the conversions of uranium to UF_3 , zirconium to ZrF_4 , and UF_3 to UF_4 . Only after the bulk of the uranium is converted to UF_4 is $\text{F}_2(\text{g})$ liberated into the system.

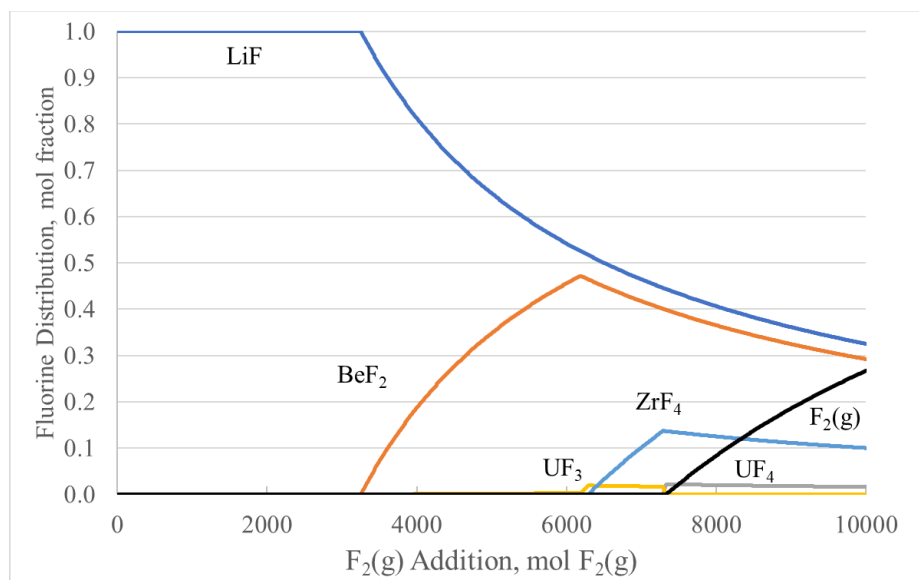


Figure 101. Equilibrium fluorine distribution as a function of $F_2(g)$ addition to the F, Li, Be, U, Zr system at 500°C (Version 1).

The same data as above are shown in Figure 102 with a logarithmic ordinate. This figure shows that fluorine is distributed to BeF_2 , UF_3 , UF_4 , ZrF_4 and $F_2(g)$ from the beginning, albeit at very low amounts.

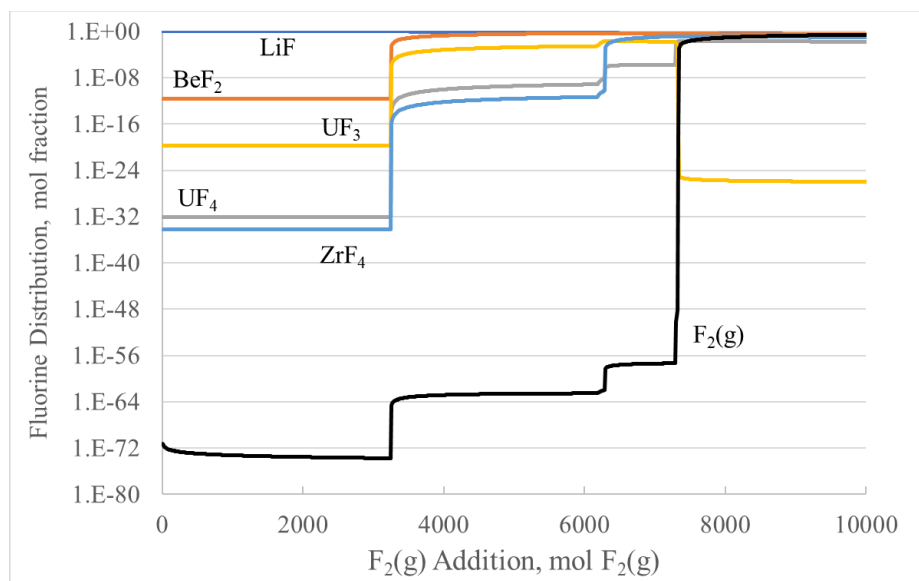


Figure 102. Equilibrium fluorine distribution as a function of $F_2(g)$ addition to the F, Li, Be, U, Zr system at 500°C (Version 2).

The F chemical potential as a function of the amount of $F_2(g)$ added to the system is shown in Figure 103. The plateaus in the figure correspond to the phase transitions as labeled. The F chemical potential is buffered during the phase transitions. For example, the F chemical potential is relatively stable until most of the lithium is fluorinated. Then the F chemical potential increases to the next plateau where beryllium is fluorinated. Then the F chemical potential increases to the next plateau where uranium and zirconium are fluorinated. Finally, the F chemical potential increases toward 1 as it appears in appreciable quantities in the gas phase.

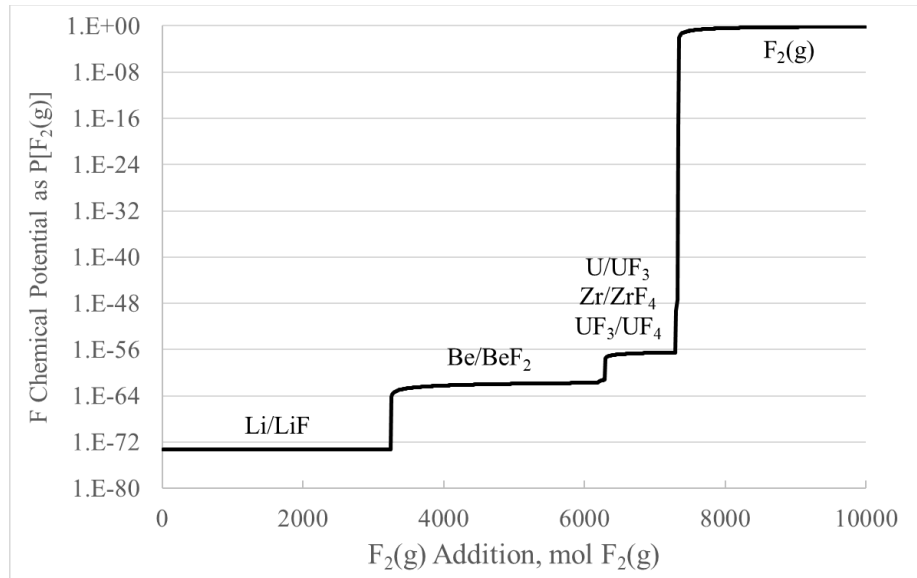


Figure 103. F chemical potential as a function of $F_2(g)$ addition to the F, Li, Be, U, Zr system at 500°C.

In this example, the individual phase transitions of U to UF_3 , Zr to ZrF_4 , and UF_3 to UF_4 are, at this scale, overlapping in Figure 103 causing the details of the transitions to be obscured. This model treated the individual metals as unique phases, meaning the chemical potentials of Li, Be, U, and Zr were each unity throughout the process. The gas phase consisted of $Ar(g)$ and $F_2(g)$, meaning that the chemical potential of $F_2(g)$ was its partial pressure in the gas phase. The salt phase consisted of LiF, BeF_2 , UF_3 , UF_4 , and ZrF_4 , meaning that the chemical potentials of these species were equal to their mole fractions in the salt phase.

Calculations showed that UF_6 was a minor contributor to this example and was therefore not included in the analysis. However, UF_6 is important to fluoride salt MSR. In operation of the MSRE, the fuel salt was converted from ^{235}U to ^{233}U . This was done by stripping the ^{235}U from the salt via UF_6 volatility.

TAKEAWAYS: As the concentrations of salt species are diminished, their individual contributions to the buffering of the fluorine chemical potential are obscured but remain important.

Dynamic Behavior of the F, Li, Be, U, Zr System (Version 2)

This model begins by adding beryllium metal to the fuel salt composition described above, which is 65 LiF, 29.2 BeF_2 , 0.8 UF_4 , and 5 ZrF_4 (mol percent). The addition of beryllium reduces the F chemical potential by the following reactions.



The standard state Gibbs free energy change for Equation (117) to Equation (120) are -906.5, -191.5, -80.0, and -20.7 kJ (mol Be)⁻¹, respectively.

Initial amounts of LiF (6.5 kmol), BeF₂ (2.92 kmol), UF₄ (0.08 kmol), and ZrF₄ (0.5 kmol) metals, are in contact with Ar(g) (1.0 kmol) and F₂(g) (0.1 kmol) and Be is incrementally added (up to 2.0 kmol). The redistribution of fluorine by the addition of beryllium is shown in Figure 104. The Be first consumes the fluorine from F₂(g), then UF₄, then ZrF₄, then finally UF₃ as predicted by Equation (117) to Equation (120).

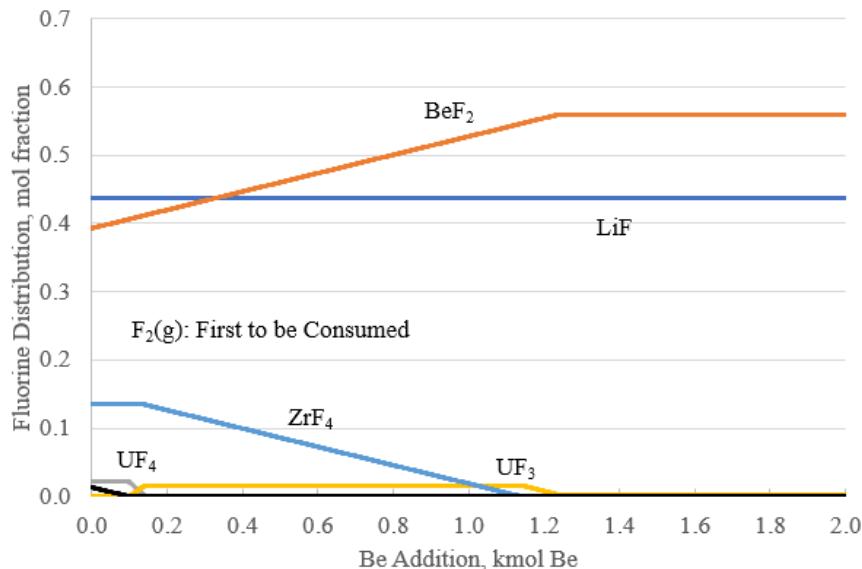


Figure 104. Fluorine distribution as a function of Be addition to the F, Li, Be, U, Zr system at 500°C.

The liberations of Zr, U, and Be by the addition of beryllium are shown in Figure 105. And the F chemical potential as a function of the addition of beryllium is shown in Figure 106. The addition of beryllium reduces the F chemical potential.

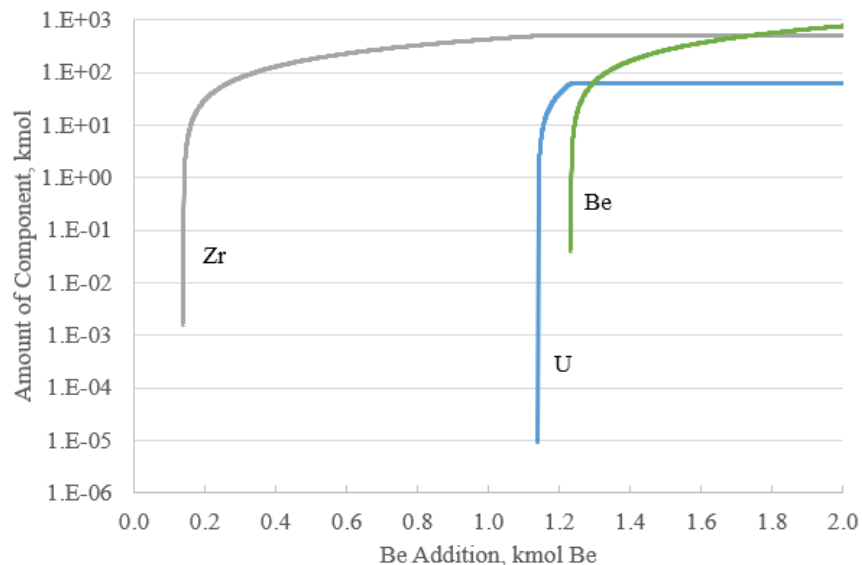


Figure 105. Liberations of Zr, U, and Be as a function of Be addition to the F, Li, Be, U, Zr system at 500°C.

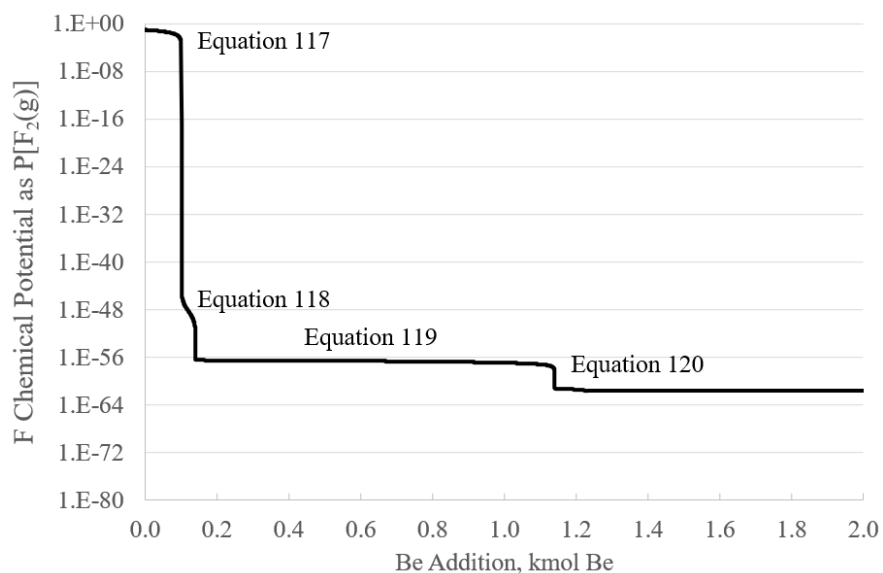


Figure 106. F chemical potential as a function of Be addition to the F, Li, Be, U, Zr system at 500°C. Unit of pressure: bar.

TAKEAWAYS: Adding beryllium metal, which is a highly electropositive reductant, decreased the fluorine chemical potential. However, if it is desired to have ZrF_4 in the salt, then the fluorine chemical potential must be maintained above a certain threshold, which in this example is shown in Figure 106 for Equation 119.

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